Dobrushin uniqueness for the four-dimensional SU(2) Wilson lattice gauge measure for $\beta < 2/9$, with interval/Lean certificates

Version 1.6

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Abstract

We write out Dobrushin's uniqueness criterion, in its Wasserstein (W_1) form, for the $SU(2)$ Wilson lattice gauge measure on $\mathbb{Z}^4$, and compute the single-link influence coefficient *exactly*, uniformly in the field, for two metrics on $SU(2) \cong S^3$. For the Euclidean (chordal) metric of $\mathbb{R}^4$, the metric used by Levin (1980), we show that the one-site coefficient in a field of strength κ equals $\mathbb{E}_\kappa[x_\perp^2] = I_2(\kappa)/(\kappa I_1(\kappa))$, the second moment of a coordinate orthogonal to the field, *for every* $\kappa \geq 0$ (in v1.4: for $\kappa \leq 4/3$). For $\kappa > 0$ the maximum over directions is attained exactly for the directions orthogonal to the field, oblique directions are strictly smaller (with explicit two-sided bounds), and the coefficient is strictly decreasing in κ, hence maximal at zero field, where it equals $1/4$. Hence each of the 18 neighbouring links has influence exactly $\beta/4$, the Dobrushin sum is $9\beta/2$, and the (DLR) Gibbs measure exists and is unique for $\beta < 2/9 \approx 0.2222$, in the normalisation $\beta \sum_p (1 - \frac{1}{2} \operatorname{Re} \operatorname{Tr} U_p)$. The proof of the identity is analytic: it bounds the covariance by the L^1 norm of an explicit divergence-free vector field on the ball with prescribed normal flux, bounds that norm slice by slice, and closes with an exact sum-of-squares identity and a Chebyshev association argument. It is new in this version, it uses no computer, and it is *not* formalised in Lean. The interval-arithmetic proof of v1.4 for $\kappa \leq 4/3$, with its Lean-checked final rational step, is kept in Appendix A as a second proof on that range. The bound $D_{\mathrm{ch}}(\kappa) \leq 1/4$ for all fields, hence the maximum-at-zero-field property conjectured by Levin (Remark 3.7) for $N = 4$, and the exactness of the influence coefficient $\beta/4$ were proved before by Bhattacharjee and Bhattacharya [17]; accordingly, $2/9$ is exactly Levin's published bound transferred to the Wilson link, $2\sqrt{5}/45 \approx 0.0994$, with his factor $\sqrt{5}$ replaced by 1. For the geodesic metric (the main result of v1.2 of this note) the exact coefficient is $8\beta/(9\pi)$, uniformly for $\kappa \leq 3\pi/8$, which gives the weaker range $\beta < \pi/16 \approx 0.196$ by the same method. Each threshold is sharp for its criterion. A remark records that Wu's theorem (2006) turns these coefficients into a Poincaré inequality for $\beta < 2/9$, uniform in the volume and the boundary condition; this is a corollary, not a new method. The single-site machinery is not new: Levin (1980, Euclidean metric) and Royer (1986, geodesic metric) treat rotator models, and Levin's many-body version already covers the Wilson model; none of the cited papers mentions gauge theories. For comparison, the explicit range of Shen–Zhu–Zhu is $\beta < 1/12$, the total-variation criterion (Gross 1979, Simon 1979) gives $\beta < 1/18$ transferred,

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and global Bakry–Émery on link variables, without gauge fixing, cannot pass $\beta = 1/8$ (Proposition 22); with a gauge fixing by a field of straight lines, the Bakry–Émery method reaches $\beta_W < 4/29 \approx 0.138$ in [18, Theorem A]. As an application of the chordal coefficient (Section 7), Dobrushin's comparison for the slab σ-models of Durhuus and Fröhlich gives Wilson's area law for $\beta < 2/3$, uniformly on tori, with free boundary conditions and for every Gibbs measure, with the string-tension bound $\sigma \geq \operatorname{arcosh}(2/\beta - 2)$. This range is not new: Borgs and Seiler (1983; Appendix B of the preprint version) proved confinement for $\beta < 12/17$ by a transfer-matrix argument, and we claim no priority on any threshold, only an independent proof with explicit constants. Apart from this application, no statement on decay of correlations or log-Sobolev inequality is claimed. *Prior public statement.* The uniqueness range $\beta < 2/9$ for the four-dimensional $SU(2)$ model, with exponential decay and a lattice mass gap, the all-field bound $D_{\mathrm{ch}}(\kappa) \leq 1/4$ in the form $W_1(\nu_b, \nu_{b'}) \leq \frac{1}{4}|b - b'|$ on S^3, and the exactness of the influence coefficient were stated publicly, with proofs, in the preprint [17] of 18 October 2026 (Theorems E, F and 7 and Proposition 13 of the PDF cited in our bibliography; the metadata of the Zenodo record 10.5281/zenodo.23449141, version 2.0.0, created on 18 October 2026 at 08:35:38 UTC, describe the same results; the archive of that record was not opened). This note was prepared independently and makes *no priority claim* on them. The series of H. T. Rajabu [19, 20] (Zenodo, from 27 October 2026, 13:09:49 UTC, i.e. about ten hours before v1.4 of this note) states, with computer-assisted proofs that we have not checked: the chordal one-link bound $\|D_v\|_{\mathrm{ch}} \leq q(|s|)\,|v| \leq \frac{1}{4}|v|$ for $|s| \leq 4$, with equality for $v \perp s$ (Part II, Theorem 7.6); uniqueness for the Wilson specification on all links of $\mathbb{Z}^4$ for $\beta_W \leq 0.393$ (Part II, Corollary 5.2) and $\beta_W \leq 0.419$ (preprint of 27 October). In version 2.2, Rajabu states that the Wilson specification on $\mathbb{Z}^4$ has exactly one Gibbs measure for $0 < \beta_W \leq 0.511$ (Theorem 9.1(i)), with exponential clustering of gauge-invariant local chordal-Lipschitz observables and a gap of the physical transfer matrix for $0 < \beta_W \leq 0.51$ (Theorem 9.1(ii)–(iii)); he also states uniqueness in every half-space for every boundary condition for $0 < \beta_W \leq 0.419$ (Section 11). Part II states Wilson's area law on tori for $\beta_W < 2/3$ (Corollary 7.7(iii)). These stated uniqueness ranges go beyond $2/9$, and the area-law range $2/3$ is stated there before §7 of this note; we make *no priority claim* on them either (§8). This note's own contributions are: the exact value $\mathbb{E}_\kappa[x_\perp^2] = I_2/(\kappa I_1)$ of the supremum over directions for every $\kappa \geq 0$, its maximisers, the two-sided estimate for oblique directions and the strict monotonicity in κ, with an analytic proof; the exact geodesic coefficient $8\beta/(9\pi)$ for $\kappa \leq 3\pi/8$; two interval-arithmetic certificates, each recertified by an independent implementation (Appendix A and §5.2), with Lean-checked final steps; an explicit transfer of the published constants of Levin, Royer and Shen–Zhu–Zhu with their conventions; and the area-law application of Section 7 (an independent proof by Dobrushin's comparison, valid for every Gibbs measure and with explicit constants; the range itself is not new, see above).

Status labels. Every claim carries one label: [proved] (complete argument written here or in the cited source); [exact] (identity or inequality verified in exact rational arithmetic or by symbolic computation; computer-checked, not Lean); [interval-certified] (rigorous modulo trust in the interval library and in the reductions, which are proved on paper but not formalised: those of Appendix A and §5.1, and, for the cross-checks of §4.9, the definitions of §4.4); [Lean-checked] (kernel-checked in Lean 4, axioms `propext`, `Classical.choice`, `Quot.sound` only; this label never covers more than the Lean statements quoted verbatim in §A.3 and §5.2, which are rational inequalities, conditional implications and threshold arithmetic; *the analytic proof of §4 is not formalised*); [read] (statement read in the original source, location given); [not checked] (cited from a source of which only the metadata or the abstract were seen, or a combination of published results not verified against the sources); [conjecture]. The label [numerical] marks floating-point evidence (including ball-arithmetic evaluations without a rigorous quadrature bound) which is *not* part of any proof.

1 Statement and conventions

Let E be the set of (positively oriented) links of $\mathbb{Z}^4$ and $\mathcal{P}$ the set of unoriented plaquettes. A configuration is $U = (U_\ell)_{\ell \in E} \in SU(2)^E$; U_p is the ordered product around p (its real trace does not depend on the starting point or orientation, since $\operatorname{Re}\operatorname{Tr} W = \operatorname{Re}\operatorname{Tr} W^{-1}$ on $SU(2)$). For a finite $\Lambda \subset E$ and boundary condition $\bar{U}$, the Wilson specification is

$$\mu_\Lambda^{\bar{U}}(dU_\Lambda) \;\propto\; \exp\Big(-\beta \sum_{p \cap \Lambda \neq \emptyset} \big(1 - \tfrac{1}{2}\operatorname{Re}\operatorname{Tr} U_p\big)\Big) \prod_{\ell \in \Lambda} dU_\ell, \tag{1}$$

with dU the normalised Haar measure. Throughout, $\beta = \beta_W$ is the *Wilson* coupling, and the bare coupling g is *defined* by $\beta_W = 2N/g^2$, i.e. $\beta_W = 4/g^2$ for $SU(2)$; the coupling g_0 of Osterwalder–Seiler is a different normalisation (Table 1). A Gibbs measure is a probability measure on $SU(2)^E$ satisfying the DLR equations for (1). We identify $SU(2)$ with the unit sphere $S^3 \subset \mathbb{R}^4$ (unit quaternions) and use two metrics on it: the *chordal* (Euclidean) distance $\rho_{\mathrm{ch}}(x, y) = |x - y|$ of $\mathbb{R}^4$, of diameter 2, and the *geodesic* (great-circle) distance ρ, of diameter π. They generate the same topology and Borel σ-algebra.

convention	weight	relation to β_W (for $SU(2)$)
Wilson (this note)	$\exp(-\beta_W \sum_p (1 - \frac{1}{N}\operatorname{Re}\operatorname{Tr} U_p))$	—
$\frac{\beta}{N}\operatorname{Re}\operatorname{Tr}$	$\exp(\frac{\beta}{N}\sum_p \operatorname{Re}\operatorname{Tr} U_p)$	$\beta_W = \beta$
raw trace	$\exp(\beta \sum_p \operatorname{Re}\operatorname{Tr} U_p)$	$\beta_W = 2\beta$
't Hooft (SZZ, CNS)	$\exp(N\beta \sum_p \operatorname{Re}\operatorname{Tr} U_p)$	$\beta_W = N^2\beta = 4\beta$
Osterwalder–Seiler	$\exp(\frac{1}{2g_0^2}\sum_P \operatorname{Re}\chi(g_P))$	$\beta_W = 1/g_0^2$ (hypothesis*), i.e. $g_0 = g/2$

Table 1: Conventions. The 't Hooft line is [2, (1.2)] (see §8); the Osterwalder–Seiler line is [3, (2.1)–(2.2), p. 442], read on the page image ([read]). *The conversion $\beta_W = 1/g_0^2$ is a *hypothesis*: it holds if χ is the fundamental character and each geometric plaquette is counted once in $\sum_P$. On p. 442, χ is "a character on $\mathcal{G}$" and plaquettes are "closed loops $uvxy$"; neither condition is stated there. The conversion is used only to translate thresholds, never in a proof.

Theorem 1 (Main result)**.** *For $0 \le \beta < 2/9 \approx 0.2222$ there is exactly one Gibbs measure for the $SU(2)$ Wilson specification (1) on $\mathbb{Z}^4$. Equivalently: $g^2 > 18$ with g defined by $\beta_W = 2N/g^2 = 4/g^2$; in the 't Hooft normalisation of [2], $\beta_{\mathrm{tH}} < 1/18 \approx 0.0556$; in the normalisation of [3] (under the hypothesis of Table 1), $g_0^2 > 9/2$.*

Status: existence [proved] (Proposition 3); uniqueness [read] (Dobrushin 1970, Thm 4) + [proved] (§3, chordal metric) + Theorem 6, which is [proved] by an analytic argument valid for all fields (§4; not formalised in Lean). For $\kappa \le 4/3$, which is all that $\beta < 2/9$ needs, a second proof is [interval-certified], with its last rational step [Lean-checked] (Appendix A). The threshold arithmetic is [Lean-checked] (`somme_dobrushin_lt_one`, `kappa_max_lt`).

Theorem 2 (Geodesic criterion; the main result of v1.2)**.** *With the geodesic metric ρ, Dobrushin's sum for (1) equals $16\beta/\pi$ for $0 \le \beta < \pi/16 \approx 0.19635$; so the geodesic W_1 criterion gives uniqueness for $\beta < \pi/16$ and fails at $\beta = \pi/16$.*

Status: [proved] (§3) + Theorem 17, which is [proved] for $\kappa = 0$ and [interval-certified] for $\kappa > 0$, with its last rational step [Lean-checked] (§5.2). The uniqueness statement for $\beta < \pi/16$ is contained in Theorem 1; what Theorem 2 adds is the exact geodesic coefficient $8\beta/(9\pi)$ and its field-uniformity.

Each threshold is *sharp for its criterion*: with single-site conditionals, the chordal sum equals $9\beta/2$ and the geodesic sum $16\beta/\pi$ exactly (Proposition 5), so the chordal criterion fails at $\beta = 2/9$ and the geodesic one at $\beta = \pi/16$. Nothing is claimed about other metrics, block or multi-site versions of the criterion, or the true uniqueness region.

An application: the area law (§7). The chordal coefficient is not specific to uniqueness. Applied to the three-dimensional σ-models into which Durhuus and Fröhlich [24] decompose the four-dimensional theory, it gives, by Dobrushin's comparison, Wilson's area law for $\beta < 2/3$, uniformly on tori, with free boundary conditions and for every Gibbs measure, with a string tension at least $\operatorname{arcosh}(2/\beta - 2)$ (Theorem 7.1). The range is not new: Borgs and Seiler [25] proved confinement for $\beta < 12/17$ by a different method (§7.2), and no priority is claimed for any threshold.

2 Dobrushin's theorem and its hypotheses

We use the original Russian article [1] (Math-Net.Ru scan `tvp1856`; the quotations below were read on the rendered page images; English wording is ours).

Setting (§1, p. 469). T is a countable set; the random variables ξ_t take values in a measurable space $(X, \mathfrak{B})$. Dobrushin's standing assumption is that all probability measures on finite products of $(X, \mathfrak{B})$ are perfect ("все вероятностные меры на прямых произведениях конечного числа экземпляров измеримого пространства $(X, \mathfrak{B})$ совершенны"), and for this it suffices (verbatim) that "$\mathfrak{B}$ было σ-алгеброй всех борелевских подмножеств для некоторой метрики, превращающей X в полное сепарабельное метрическое пространство". Given an integer function ν on T,

$$G_{t_0} = \{t : \nu(t) \le \nu(t_0),\ t \ne t_0,\ t \in T\}, \qquad t_0 \in T \qquad [1,\ (1.2)],$$

and a field has conditional distributions $\{p_{t_0}\}$ if $\mathbf{P}\{\xi_{t_0} \in B \mid \xi_t, t \in G_{t_0}\} = p_{t_0}(B \mid x_t, t \in G_{t_0})$ [1, (1.3)].

Wasserstein distance (§3, p. 479). "Назовем *расстоянием Васерштейна* (см. [12]) число $R(P,Q) = \inf \mathrm{M}\rho(\eta,\zeta)$, (3.1) где нижняя грань берется по всем парам случайных величин $\eta, \zeta \ldots$ таким, что η имеет распределение P и ζ — распределение Q". This is W_1 defined by couplings, for the metric ρ of X (any metric making X complete and separable).

Theorem 4 (§5, p. 488), verbatim. "Для того чтобы существовало не более, чем одно, случайное поле $\{\xi_t, t \in T\}$ с заданными условными распределениями $\{p_{t_0}(\cdot|\cdot), t_0 \in T\}$, для которого при некоторых $x_0 \in X$ и $C < \infty$ $\mathrm{M}\rho(\xi_t, x_0) \le C < \infty$, $t \in T$, (5.1) достаточно, чтобы при некотором $d < 1$ и всех $t_0 \in T$ $\sum_{t \in G_{t_0}} r(t, t_0) \le d < 1$, (5.2) где числа $r(t, t_0)$ ... таковы, что расстояние Васерштейна $R(p_{t_0}(\cdot|x^1_t, t \in G_{t_0}), p_{t_0}(\cdot|x^2_t, t \in G_{t_0})) \le \sum_{t \in G_{t_0}} r(t, t_0)\rho(x^1_t, x^2_t)$ (5.3) ... Условие (5.3) выполнено, в частности, если $r(\bar{t}, t_0) = \sup R(\cdots)/\rho(x^1_{\bar{t}}, x^2_{\bar{t}})$, (5.4) где верхняя грань берется по всем парам условий ..., отличным лишь при $t = \bar{t}$." (Translation: at most one field with the given conditionals and property (5.1) exists provided that, for some $d < 1$ and all t_0, (5.2) holds with numbers r satisfying the Wasserstein bound (5.3); (5.3) holds in particular for the supremum (5.4) over pairs of conditions differing only at $\bar{t}$.)

Theorem 1 (existence, pp. 472–473). Two conditions suffice; on p. 473 Dobrushin writes "Условие 1 заведомо выполнено, если X — бикомпакт. (Можно положить $h(x) \equiv 1$.)" and, in the Markov case (conditionals depending on finitely many x_t, $t \in U_{t_0}$), "условие 2 означает непрерывность $p_{t_0}(\cdot \mid x_t, t \in U_{t_0})$ по переменным x_t в топологии слабой сходимости".

Verification of the hypotheses for (1). Let d be either ρ_{ch} or ρ.

1. $T = E$ is countable; $X = SU(2) \cong S^3$ with d is compact, hence complete and separable; $\mathfrak{B}$ is its Borel σ-algebra (the same for both metrics). [proved]

2. Take $\nu \equiv 0$; then $G_{t_0} = T \setminus \{t_0\}$ (the case Dobrushin himself mentions on p. 488, "В случае $G_{t_0} = \{t : t \neq t_0\}$"). The kernel $p_\ell(\cdot \mid U_{E\setminus\ell})$ is the single-link conditional law (2) below; it has a density continuous in all variables, hence is jointly measurable. Every DLR measure for (1) satisfies [1, (1.3)] with these kernels. So uniqueness in Dobrushin's class implies uniqueness of the Gibbs measure. [proved]

3. (5.1) holds for every field with $x_0 = \mathbf{1}$ and $C = \operatorname{diam} S^3$ (2 for ρ_{ch}, π for ρ). [proved]

4. (5.2)–(5.4): p_ℓ depends only on the 18 neighbouring links (§3), so $r(t,\ell) = 0$ for all other t. With r the supremum (5.4): for $d = \rho_{\mathrm{ch}}$ and every $\beta \geq 0$, $\sum_t r(t,\ell) = 9\beta/2$ (Proposition 5 and Theorem 6); for $\beta < 2/9$ take $d = 9\beta/2 < 1$. For $d = \rho$ and $\beta < \pi/16$, $\sum_t r(t,\ell) = 16\beta/\pi$ (Theorem 17). [proved] given Theorems 6 and 17.

5. Existence: Theorem 1 applies (Condition 1 because X is compact; Condition 2 because the conditional density is continuous in the neighbouring links), but it yields a field with the prescribed *single-site* conditionals. We do not use it: existence of a measure satisfying the DLR equations for *every* finite Λ is proved directly in Proposition 3. [proved]

A second modern statement of the W_1 criterion on complete metric single-spin spaces is [8, Thm 3.5] (p. 8 of arXiv v4) [read].

Proposition 3 (Existence in the DLR sense)**.** *For every $\beta \geq 0$ there is at least one probability measure μ on $SU(2)^E$ such that $\mu\gamma_\Delta = \mu$ for every finite $\Delta \subset E$, where $\gamma_\Delta(\cdot \mid \bar U) = \mu_\Delta^{\bar U} \otimes \delta_{\bar U_{E\setminus\Delta}}$ is the kernel (1).*

Status: [proved].

Proof. Write $H_\Delta(U) = \sum_{p\cap\Delta\neq\emptyset}(1 - \frac{1}{2}\operatorname{Re}\operatorname{Tr} U_p)$, a finite sum of continuous functions of finitely many links, with $0 \leq H_\Delta \leq h_\Delta := 2\,|\{p : p\cap\Delta \neq \emptyset\}|$. Thus

$$\gamma_\Delta f(\bar U) = Z_\Delta(\bar U)^{-1} \int f(U_\Delta, \bar U_{E\setminus\Delta})\, e^{-\beta H_\Delta(U_\Delta, \bar U)}\, dU_\Delta, \qquad Z_\Delta \geq e^{-\beta h_\Delta} > 0.$$

(a) Feller property. If f is continuous on the compact space $SU(2)^E$ (product topology), the integrand is jointly continuous in $(U_\Delta, \bar U)$ and bounded, so $\gamma_\Delta f$ and Z_Δ are continuous by dominated convergence, and $\gamma_\Delta f$ is continuous. *(b) Consistency.* For finite $\Delta \subset \Lambda$ one has $\gamma_\Lambda\gamma_\Delta = \gamma_\Lambda$: the density of $\gamma_\Lambda(\cdot \mid \bar U)$ is $e^{-\beta H_\Lambda}/Z_\Lambda$ and $H_\Lambda = H_\Delta + H'$ with H' independent of U_Δ (the plaquettes meeting Λ but not Δ), so by Fubini integrating $\gamma_\Delta f$ against γ_Λ first in U_Δ reproduces $\gamma_\Lambda f$. *(c) Limit.* Fix $\bar U$ and finite $\Lambda_n \uparrow E$, and let $\mu_n = \gamma_{\Lambda_n}(\cdot \mid \bar U)$. By compactness (Prokhorov, or Riesz–Markov on $C(SU(2)^E)$) a subsequence converges weakly to some μ. For finite Δ and continuous f, (b) gives $\mu_n(\gamma_\Delta f) = \mu_n(f)$ as soon as $\Delta \subset \Lambda_n$; by (a) both sides converge along the subsequence, so $\mu(\gamma_\Delta f) = \mu(f)$ for all continuous f, hence $\mu\gamma_\Delta = \mu$. □

Every such μ satisfies in particular [1, (1.3)] with $G_{t_0} = T \setminus \{t_0\}$ (take $\Delta = \{\ell\}$), which is the direction used for uniqueness in item 2.

3 The single-link conditional law and the influence matrix

Quaternion form. For unit quaternions u, w, $\frac{1}{2}\operatorname{Re}\operatorname{Tr}(uw) = \langle u, \bar{w}\rangle_{\mathbb{R}^4}$. Link ℓ lies in $2(d-1) = 6$ plaquettes; for each such p, $\frac{1}{2}\operatorname{Re}\operatorname{Tr} U_p = \frac{1}{2}\operatorname{Re}\operatorname{Tr}(U_\ell W_p)$ where the *staple* W_p is a product of the three other links of p (or of their inverses; we use $\operatorname{Re}\operatorname{Tr} A = \operatorname{Re}\operatorname{Tr} A^{-1}$ to put U_ℓ first). Hence

$$p_\ell(du \mid \cdot) = q_s(du) \propto e^{\beta\langle u,s\rangle}\, du, \qquad s = \sum_{p\ni\ell} \overline{W_p} \in \mathbb{R}^4, \quad |s| \le 6. \tag{2}$$

Write $\kappa = \beta|s| \in [0, 6\beta]$, $\hat{s} = s/|s|$, and $q_\kappa \propto e^{\kappa\langle u,\hat{s}\rangle}$. In the notation of Royer [10, p. 31], $q_s = \nu(\beta s)$ on S^n with $n = 3$: "For each vector $h \in \mathbb{R}^{n+1}$ let $\nu(h)$ be the probability measure on S^n, the density of which is proportional to $x \to \exp(x.h)$" [read], on the Numdam page image; in Levin's notation [11, (3.11), p. 68] it is $\mu_{\mathbf{h}}$ with $\mathbf{h} = \beta s$ and $N = 4$ [read]. The single-site analysis below is therefore an instance of the setting of Levin [11] and Royer [10] for n-vector (rotator) models; what is specific to the gauge model is the staple structure and the count of neighbours. In fact the whole Wilson model is a 4-vector model with four-body interactions in the sense of Levin [11, Remark 3.6, (3.48), p. 73]: writing $u_1, \dots, u_4 \in S^3 \subset \mathbb{R}^4$ for the links of p (with $u^{-1} = \bar{u}$), $\frac{1}{2}\operatorname{Re}\operatorname{Tr} U_p = \langle u_1 u_2 \bar{u}_3 \bar{u}_4, \mathbf{1}\rangle$ is a multilinear function of (u_1, u_2, u_3, u_4). [proved]

Why 18 neighbours. Each of the 6 plaquettes through ℓ contains 3 further links. Two distinct plaquettes through ℓ span different coordinate planes, or the same plane on opposite sides of ℓ, so they share no link other than ℓ. Hence the $6 \times 3 = 18$ links are pairwise distinct, each shares exactly one plaquette with ℓ, and a change of one of them changes exactly one staple. In general dimension the count is $3 \cdot 2(d-1)$. [proved]

Definition 4. For $\kappa \ge 0$ and a unit vector $v \in \mathbb{R}^4$ put

$$D_{\mathrm{ch}}(\kappa, v) = \sup\{\operatorname{Cov}_{q_\kappa}(f, \langle u, v\rangle) : f \text{ 1-Lipschitz for } \rho_{\mathrm{ch}}\},$$
$$D(\kappa, v) = \sup\{\operatorname{Cov}_{q_\kappa}(f, \langle u, v\rangle) : f \text{ 1-Lipschitz for } \rho\},$$

and $D_{\mathrm{ch}}(\kappa) = \sup_{|v|=1} D_{\mathrm{ch}}(\kappa, v)$, $D(\kappa) = \sup_{|v|=1} D(\kappa, v)$. By rotation invariance about $\hat{s}$, both depend on v only through the angle $\alpha \in [0, \pi]$ between v and $\hat{s}$; replacing f by $-f$ shows that they bound $|\operatorname{Cov}|$.

Proposition 5 (Influence coefficients)**.** *Let $r_{\mathrm{ch}}(y, \ell)$ and $r(y, \ell)$ be Dobrushin's coefficient (5.4) for a neighbour y of ℓ, for the metrics ρ_{ch} and ρ respectively. Then*

$$\beta\, D_{\mathrm{ch}}(0) \;\le\; r_{\mathrm{ch}}(y, \ell) \;\le\; \beta \sup_{0\le\kappa\le 6\beta} D_{\mathrm{ch}}(\kappa), \qquad \beta\, D(0) \;\le\; r(y, \ell) \;\le\; \beta \sup_{0\le\kappa\le 6\beta} D(\kappa).$$

With $D_{\mathrm{ch}}(0) = 1/4$ and Theorem 6, for every $\beta \ge 0$: $r_{\mathrm{ch}}(y, \ell) = \beta/4$ and $\sum_y r_{\mathrm{ch}}(y, \ell) = 18\beta/4 = 9\beta/2$. With $D(0) = 8/(9\pi)$ (Lemma 18) and Theorem 17, for $\beta < \pi/16$: $r(y, \ell) = 8\beta/(9\pi)$ and $\sum_y r(y, \ell) = 16\beta/\pi$.

Status: [proved] (the upper bounds use Kantorovich–Rubinstein duality [8, Thm 10.8], quoted below). The geodesic upper bound is the argument of Royer [10, (3)–(4), p. 31] for $\nu(h)$ on S^n, here with $n = 3$, $K = \beta$ and z the moving staple: verbatim, "$\mathrm{R}(\nu(Kz_1 + \zeta), \nu(Kz_2 + \zeta)) \leqslant |K|(C(n)/(n+1))^{1/2} r(z_1, z_2)$. (3)" and, along a minimal geodesic $z(t)$, "$\left|\frac{d}{dt}\int f(y)\nu(Kz(t) + \zeta, dy)\right| = |K \operatorname{cov}(f(y), y.z'(t))|$ (4)" [read]. The chordal upper bound is the argument of Levin, who writes the distance "as the variation of an expectation along an appropriate path in **h**-space" [11, p. 68], the straight segment [11, (3.35), p. 71] [read]. Royer and Levin then bound the covariance by field-uniform constants; we instead compute $\sup_f$ of the covariance exactly, as a function of $\kappa = |h|$.

Proof. Upper bounds. Let U, U' differ only at y, and let $W = AU_y^{\pm 1}B$ be the unique staple containing y (A, B fixed). The map $U_y \mapsto \overline{AU_y^{\pm 1}B}$ is a composition of left and right multiplications by unit quaternions and of quaternion conjugation, hence the restriction to S^3 of an orthogonal map of $\mathbb{R}^4$; it is an isometry for both ρ_{ch} and ρ. *Chordal:* let $s(t) = s + t(s' - s)$, $t \in [0,1]$, the straight segment between the two values of s; then $|s(t)| \le 6$ by convexity and $|s' - s| = |\overline{W'} - \overline{W}| = \rho_{\mathrm{ch}}(U_y, U_y')$. For f 1-Lipschitz for ρ_{ch},

$$\int f \, d(q_{s'} - q_s) = \int_0^1 \beta \, \mathrm{Cov}_{q_{s(t)}} \left(f, \langle u, s' - s\rangle\right) dt \le \beta \, \rho_{\mathrm{ch}}(U_y, U_y') \sup_{\kappa \le 6\beta} D_{\mathrm{ch}}(\kappa).$$

Geodesic: along a constant-speed geodesic $t \mapsto U_y(t)$, $t \in [0,1]$, from U_y to U_y', the vector $s(t)$ moves on a great circle of the unit sphere translated by the other five staples, with $|\dot s(t)| = \rho(U_y, U_y')$, and $|s(t)| \le 6$. For f 1-Lipschitz for ρ,

$$\int f \, d(q_{s(1)} - q_{s(0)}) = \int_0^1 \beta \, \mathrm{Cov}_{q_{s(t)}} \left(f, \langle u, \dot s(t)\rangle\right) dt \le \beta \, \rho(U_y, U_y') \sup_{\kappa \le 6\beta} D(\kappa).$$

In both cases the supremum over 1-Lipschitz f of the left side is Dobrushin's Wasserstein distance (3.1) for the corresponding metric, by the Kantorovich–Rubinstein duality as stated in [8, Thm 10.8, p. 53] (verbatim): "Let μ and ν be two probability measures on a metric space $(\mathcal{X}, \rho)$ such that $\int_{\mathcal{X}} \rho(x, x_0)\, \mu(dx) < +\infty$ and $\int_{\mathcal{X}} \rho(x, x_0)\, \nu(dx) < +\infty$, for some $x_0 \in \mathcal{X}$. Then the Wasserstein metric equals the Kantorovich-Rubinstein metric, i.e., $\rho_W(\mu, \nu) = \sup_{f \in \mathrm{Lip}_1(\mathcal{X})} \left(\int_{\mathcal{X}} f(x)\, \mu(dx) - \int_{\mathcal{X}} f(x)\, \nu(dx)\right)$" [read], where ρ_W is the infimum over couplings [8, (2.1), p. 6], i.e. Dobrushin's (3.1). The moment condition is trivial on the compact S^3. Dividing by the distance of U_y and U_y' and taking the supremum gives the bounds on (5.4).

Lower bounds. The 18 neighbours are distinct and each staple involves three of them only, so the six staples can be prescribed independently. Given a unit $v \in \mathbb{R}^4$, choose the staple containing y so that $w := \overline{W} \perp v$, and the other five staples (unit vectors) with sum $-w$ (possible since $|-w| = 1 \le 5$); then $s = 0$. Moving U_y along a unit-speed geodesic moves $\overline{W}$ isometrically along a geodesic of S^3 through w; choosing it with initial velocity v (tangent at w since $v \perp w$) gives $\overline{W}(\varepsilon) = \cos\varepsilon\, w + \sin\varepsilon\, v$, so $s(\varepsilon) = \varepsilon v + O(\varepsilon^2)$, $\rho(U_y, U_y(\varepsilon)) = \varepsilon$ and $\rho_{\mathrm{ch}}(U_y, U_y(\varepsilon)) = 2\sin(\varepsilon/2) = \varepsilon + O(\varepsilon^3)$. For fixed 1-Lipschitz f (for either metric), $W_1(q_0, q_{s(\varepsilon)}) \ge \int f\, d(q_{s(\varepsilon)} - q_0) = \beta\varepsilon \, \mathrm{Cov}_{q_0}(f, \langle u, v\rangle) + O(\varepsilon^2)$ (only the trivial direction of the duality is used). Dividing by the distance and letting $\varepsilon \to 0$, then optimising f, gives $r_{\mathrm{ch}}(y, \ell) \ge \beta D_{\mathrm{ch}}(0, v)$ and $r(y, \ell) \ge \beta D(0, v)$; both zero-field values are independent of v (Lemmas 7 and 18). □

4 The chordal one-link identity for all fields

Throughout this section $\hat s = e_1$, $x_1 = \langle u, e_1\rangle$, $x_2 = \langle u, e_2\rangle$ with $e_2 \perp e_1$, $\bar\sigma$ is the normalised Haar measure of S^3, and $dq_\kappa = e^{\kappa x_1}\, d\bar\sigma / \bar Z(\kappa)$ with

$$\bar Z(\kappa) = \mathbb{E}_{\bar\sigma} e^{\kappa x_1} = \sum_{k \ge 0} \frac{\kappa^{2k}}{4^k\, k!\, (k+1)!} = \frac{2 I_1(\kappa)}{\kappa}$$

(the marginal law of x_1 under $\bar\sigma$ is $\frac{2}{\pi}\sqrt{1 - t^2}\, dt$). We write $\mathbb{E}_\kappa$ for $\mathbb{E}_{q_\kappa}$ and $m_1 = \mathbb{E}_\kappa x_1 = \bar Z'/\bar Z$. For $\kappa > 0$ put $R(t) = \sqrt{1 - t^2}$ and $E(t) = e^{\kappa t}/\bar Z(\kappa)$ on $[-1, 1]$. The law of x_1 under q_κ is $d\mu(t) = \frac{2}{\pi} R(t) E(t)\, dt$, and for a function g on $[-1, 1]$ we write $\langle g\rangle := \int g\, d\mu = \mathbb{E}_\kappa[g(x_1)]$ (a bracket with a single argument is this average; the inner product of $\mathbb{R}^4$ always carries a comma). Since q_κ is invariant under the rotations of (x_2, x_3, x_4), which are isometries of ρ_{ch}, $D_{\mathrm{ch}}(\kappa, v)$ depends on v only through α.

Theorem 6 (Chordal one-link identity, all fields)**.** *For every $\kappa \geq 0$,*

$$D_{\mathrm{ch}}(\kappa) = D_{\mathrm{ch}}(\kappa, e_2) = \mathbb{E}_\kappa[x_2^2] = \frac{1}{3}\Big(1 - \frac{\bar{Z}''(\kappa)}{\bar{Z}(\kappa)}\Big) \quad \Big(= \frac{I_2(\kappa)}{\kappa\, I_1(\kappa)} \ \text{ for } \kappa > 0\Big), \qquad D_{\mathrm{ch}}(0) = \tfrac{1}{4}.$$

For $\kappa > 0$ and a unit vector v at angle α to $\hat{s}$,

$$\mathbb{E}_\kappa[x_2^2] - \cos^2\alpha\, \Delta(\kappa) \ \leq\ D_{\mathrm{ch}}(\kappa, v) \ \leq\ \mathbb{E}_\kappa[x_2^2] - \tfrac{1}{2}\cos^2\alpha\, \Delta(\kappa) - \tfrac{1}{6}\cos^2\alpha\, \mathcal{S}(\kappa) \ \leq\ \mathbb{E}_\kappa[x_2^2] - \tfrac{1}{2}\cos^2\alpha\, \Delta(\kappa), \tag{3}$$

where $\Delta(\kappa) = m_1^2 + 4m_1/\kappa - 1 = -\kappa \frac{d}{d\kappa}\mathbb{E}_\kappa[x_2^2] = \frac{\kappa}{3}\operatorname{Cov}_\kappa(x_1^2, x_1) > 0$ and $\mathcal{S}(\kappa) \geq 0$ is the explicit quantity (9). Consequently, for $\kappa > 0$ the maximum over directions is attained exactly for $v \perp \hat{s}$ ($\alpha = \pi/2$), $\kappa \mapsto D_{\mathrm{ch}}(\kappa)$ is strictly decreasing, and

$$\tfrac{1}{4} - D_{\mathrm{ch}}(\kappa) = \frac{4\bar{Z}''(\kappa) - \bar{Z}(\kappa)}{12\,\bar{Z}(\kappa)} \ \geq\ \frac{\kappa^2}{96\,\bar{Z}(\kappa)} > 0 \quad (\kappa > 0), \qquad D_{\mathrm{ch}}(\kappa) = \tfrac{1}{4} - \tfrac{\kappa^2}{96} + \tfrac{\kappa^4}{1536} - \tfrac{\kappa^6}{23040} + \tfrac{13\kappa^8}{4423680} - \cdots.$$

Status: [proved] (§§4.1–4.7); the proof is analytic and uses no computer. The value at $\kappa = 0$ and the lower bound in (3) are Lemma 7; the upper bound is Lemmas 10–12; the identity for $\frac{1}{4} - D_{\mathrm{ch}}$ and the bound by $\kappa^2/(96\bar{Z})$ are [proved] by exact power series; the expansion is [exact] (exact series division). The proof is new in this version. It is *not* formalised in Lean, it was read by one independent reviewer and has not been refereed (§9, item 3), and the numerical and interval cross-checks of §4.9 are not used in it. On $\kappa \leq 4/3$ a second, computer-assisted proof (the argument of v1.4) is in Appendix A.

4.1 Reflection: the value at zero field and the lower bound

Lemma 7 (Reflection)**.** *Let v be a unit vector, $R_v x = x - 2\langle x, v\rangle v$, and q a probability measure on S^3 with $R_{v\#}q = q$. Then $\sup\{\operatorname{Cov}_q(f, \langle x, v\rangle) : f$ 1-Lipschitz for $\rho_{\mathrm{ch}}\} = \mathbb{E}_q\langle x, v\rangle^2$. In particular $D_{\mathrm{ch}}(0, v) = \mathbb{E}_{\bar{\sigma}}\langle x, v\rangle^2 = 1/4$ for every v, and $D_{\mathrm{ch}}(\kappa, e_2) = \mathbb{E}_\kappa[x_2^2]$ for every $\kappa \geq 0$. [proved]*

Proof. Put $u = \langle x, v\rangle$; $u \circ R_v = -u$, so $\mathbb{E}_q u = 0$ and $\operatorname{Cov}_q(f, u) = \mathbb{E}_q[fu] = \frac{1}{2}\mathbb{E}_q[(f(x) - f(R_v x))u]$. Since $|x - R_v x| = 2|u|$, this is $\leq \mathbb{E}_q u^2$, with equality for $f = u$, which is 1-Lipschitz. For $q = \bar{\sigma}$, $\mathbb{E}u^2 = \frac{1}{4}\mathbb{E}|x|^2 = \frac{1}{4}$; for $q = q_\kappa$ and $v = e_2$, R_{e_2} preserves x_1, hence q_κ. □

The case $v \perp \mathbf{h}$ of Lemma 7 is Levin's Lemma 3.2: for $\hat{\mathbf{e}} \perp \mathbf{h}$, "$|\mu_{\mathbf{h}}[(\mathbf{x}\cdot\hat{\mathbf{e}})f]| \leqslant \frac{1}{N}$ (3.12)", proved by the same reflection "$\leqslant \mu_{\mathbf{h}}(x_1^2)$ (3.14)" [11, p. 68] [read]; it is also Lemma 6(ii) of v1 of [17]. The content of Theorem 6 is that, for $N = 4$ and every κ, no other direction does better.

4.2 Moments

Lemma 8 (Elementary identities)**.** *For $\kappa > 0$:*

(i) $\kappa\langle R^2\rangle = 3m_1$. *Hence* $\mathbb{E}_\kappa[x_2^2] = \frac{1}{3}\langle R^2\rangle = m_1/\kappa$, $\langle t^2\rangle = 1 - 3m_1/\kappa$ *and* $\operatorname{Var}_\kappa x_1 = 1 - 3m_1/\kappa - m_1^2$.

(ii) $m_1' = \operatorname{Var}_\kappa x_1$, *i.e.* $m_1' = 1 - 3m_1/\kappa - m_1^2$. *With* $\Delta(\kappa) := \mathbb{E}_\kappa[x_2^2] - \operatorname{Var}_\kappa x_1 = m_1^2 + 4m_1/\kappa - 1$, *one has* $\dfrac{d}{d\kappa}\Big(\dfrac{m_1}{\kappa}\Big) = -\dfrac{\Delta(\kappa)}{\kappa}$.

(iii) $\langle t^3\rangle = m_1 - \dfrac{3}{\kappa} + \dfrac{12m_1}{\kappa^2}$ *and* $\operatorname{Cov}_\kappa(x_1^2, x_1) = \dfrac{3\Delta(\kappa)}{\kappa}$.

(iv) $m_1 = I_2(\kappa)/I_1(\kappa)$, *hence* $\mathbb{E}_\kappa[x_2^2] = \dfrac{I_2(\kappa)}{\kappa I_1(\kappa)} = \dfrac{1}{3}\Big(1 - \dfrac{\bar{Z}''}{\bar{Z}}\Big)$.

[proved]

Proof. (i) $(R^3E)' = RE\,(\kappa R^2 - 3t)$ and R^3E vanishes at ±1, so $0 = \int_{-1}^{1}(R^3E)'dt = \frac{\pi}{2}(\kappa\langle R^2\rangle - 3m_1)$. Given $x_1 = t$, (x_2, x_3, x_4) is uniform on a 2-sphere of radius $R(t)$, so $\mathbb{E}_\kappa[x_j^2 \mid x_1] = R(x_1)^2/3$ for $j = 2, 3, 4$. (ii) $m_1 = (\log \bar{Z})'$, so $m_1' = \bar{Z}''/\bar{Z} - (\bar{Z}'/\bar{Z})^2 = \mathbb{E}_\kappa x_1^2 - m_1^2 = \mathrm{Var}_\kappa\, x_1$, and (i) gives the Riccati form. Then $\Delta = m_1/\kappa - (1 - 3m_1/\kappa - m_1^2) = m_1^2 + 4m_1/\kappa - 1$ and $\kappa m_1' - m_1 = \kappa - 3m_1 - \kappa m_1^2 - m_1 = -\kappa\Delta$, which is $\kappa^2 \frac{d}{d\kappa}(m_1/\kappa)$. (iii) $(tR^3E)' = RE\,(R^2 - 3t^2 + \kappa t R^2)$ gives $0 = \langle R^2\rangle - 3\langle t^2\rangle + \kappa(m_1 - \langle t^3\rangle)$, i.e. $\kappa\langle t^3\rangle = \frac{3m_1}{\kappa} - 3\big(1 - \frac{3m_1}{\kappa}\big) + \kappa m_1$. Then $\mathrm{Cov}_\kappa(x_1^2, x_1) = \langle t^3\rangle - \langle t^2\rangle m_1 = \frac{3}{\kappa}(m_1^2 + \frac{4m_1}{\kappa} - 1)$. (iv) $\bar{Z} = 2I_1/\kappa$ and $I_1' = I_2 + I_1/\kappa$ give $\bar{Z}' = 2I_2/\kappa$, so $m_1 = I_2/I_1$. Finally $\mathbb{E}_\kappa x_1^2 = \bar{Z}''/\bar{Z}$ and $\mathbb{E}_\kappa[x_2^2] = \frac{1}{3}(1 - \mathbb{E}_\kappa x_1^2)$. □

4.3 Divergence-free fields

Lemma 9 (Flux bound)**.** *Let $u = \langle x, v\rangle$ and $w = (u - \mathbb{E}_\kappa u)\, e^{\kappa x_1}/\bar{Z}(\kappa)$ on S^3, so that $\mathrm{Cov}_\kappa(f, u) = \mathbb{E}_{\bar\sigma}[fw]$. Let $V \in C^1(\bar{B}; \mathbb{R}^4)$ on the closed unit ball $\bar{B} \subset \mathbb{R}^4$ with $\operatorname{div} V = 0$ in B and $V(x)\cdot x = w(x)$ on S^3. Then, for every f 1-Lipschitz for ρ_{ch},*

$$\mathrm{Cov}_\kappa(f, u) \le \frac{1}{2\pi^2}\int_B |V(y)|\, dy.$$

[proved]

Proof. McShane's extension $F(y) = \min_{x\in S^3}(f(x) + |y - x|)$ is 1-Lipschitz on $\mathbb{R}^4$ and equals f on S^3 (here the chordal metric is essential: it is the restriction of the Euclidean one); by Rademacher's theorem $|\nabla F| \le 1$ a.e. Since FV is Lipschitz on $\bar{B}$ and $\operatorname{div}(FV) = \nabla F \cdot V$ a.e., the divergence theorem and $|S^3| = 2\pi^2$ give $\mathbb{E}_{\bar\sigma}[fw] = \frac{1}{2\pi^2}\int_{S^3} F\, V\cdot x\, d\sigma = \frac{1}{2\pi^2}\int_B \nabla F\cdot V\, dy \le \frac{1}{2\pi^2}\int_B |V|\, dy$. □

For $\kappa = 0$ the constant field $V = v$ is admissible and gives $|B|/|S^3| = (\pi^2/2)/(2\pi^2) = 1/4$: a second proof of $D_{\mathrm{ch}}(0) \le 1/4$. Applied to $F(y) = \langle v, y\rangle$ (equality case), the same computation gives $\frac{1}{2\pi^2}\int_B \langle v, V\rangle\, dy = \mathbb{E}_{\bar\sigma}[uw] = \mathrm{Var}_\kappa(u)$ for every admissible V.

The explicit field. Let $\kappa > 0$. Write $y = (y_1, z)$ with $z = (0, y_2, y_3, y_4)$ and $R = R(y_1)$. Put $q(t) := (t - m_1)E(t)$, so that $\int_{-1}^{1} q(s)R(s)\, ds = \frac{\pi}{2}(\langle t\rangle - m_1) = 0$, and

$$A(t) = \frac{3}{R(t)^3}\int_t^1 q(s)R(s)\, ds = -\frac{3}{R(t)^3}\int_{-1}^{t} q(s)R(s)\, ds \qquad (-1 < t < 1) \tag{4}$$

(the two expressions agree because $\int_{-1}^{1} qR = 0$). Differentiating $A\,R^3 = 3\int_t^1 qR\, ds$ gives $(AR^3)' = -3qR$, i.e. with $(R^3)' = -3tR$,

$$tA(t) - \tfrac{1}{3}(1 - t^2)A'(t) = q(t) = (t - m_1)\, e^{\kappa t}/\bar{Z}(\kappa). \tag{5}$$

For $v = \cos\alpha\, e_1 + \sin\alpha\, e_2$ put

$$V = \cos\alpha\,[A(y_1)\, e_1 - \tfrac{1}{3}A'(y_1)\, z] + \sin\alpha\, E(y_1)\, e_2. \tag{6}$$

Then $\operatorname{div} V = \cos\alpha\,(A' - A') + \sin\alpha\,\partial_{y_2}E(y_1) = 0$ (z has three components), and on S^3, where $|z|^2 = 1 - x_1^2$, $V\cdot x = \cos\alpha\, q(x_1) + \sin\alpha\, x_2 E(x_1) = (u - \cos\alpha\, m_1)e^{\kappa x_1}/\bar{Z} = w$, because $\mathbb{E}_\kappa u = \cos\alpha\, m_1$ ($\mathbb{E}_\kappa x_2 = 0$ by the symmetry $x_2 \mapsto -x_2$). [proved]

Regularity, for every $\kappa > 0$. Write $t = \cos\vartheta$ and $s = \cos\varphi$. Then $\int_t^1 q(s)R(s)\, ds = \int_0^\vartheta q(\cos\varphi)\sin^2\varphi\, d\varphi = \vartheta^3 G(\vartheta^2)$ with G entire, because $\varphi \mapsto q(\cos\varphi)\sin^2\varphi$ is an even entire function vanishing to second order at 0. As $\sin^3\vartheta = \vartheta^3 H(\vartheta^2)$ with H entire and $H(0) = 1$, the function $A = 3G(\vartheta^2)/H(\vartheta^2)$ is analytic in ϑ^2 near 0, hence analytic in $t = 1 - \vartheta^2/2 + \cdots$ near $t = 1$. Near $t = -1$ the same argument applies to the second expression in (4). In $(-1, 1)$ analyticity is clear. So A is real-analytic on $[-1, 1]$ and $V \in C^\infty(\bar{B})$ for every $\kappa > 0$: the field is admissible in Lemma 9. (The series in κ used in v1.4 for this purpose, convergent for $|\kappa| \le 2$, is needed only for the certificates of Appendix A.) [proved]

4.4 The slice estimate

On $[-1,1]$ put $\mathsf{u} := A/E$ and $\mathsf{w} := t\mathsf{u} - t + m_1$ (functions of t; they are unrelated to the symbols u, w of Lemma 9). They are real-analytic, hence bounded with bounded derivatives. By (5), $\frac{1}{3}R^2A' = tA - q = \mathsf{w}E$, i.e.

$$A = \mathsf{u}E, \qquad R^2A' = 3\mathsf{w}E, \qquad R^2(\mathsf{u}' + \kappa\mathsf{u}) = 3\mathsf{w}, \qquad \mathsf{w} = t\mathsf{u} - t + m_1. \tag{7}$$

Define

$$\Xi(\kappa) := \langle R^2\mathsf{u}^2\rangle + \tfrac{3}{5}\langle \mathsf{w}^2\rangle - \langle R^2\rangle. \tag{8}$$

Lemma 10 (Slice estimate)**.** *For $\kappa > 0$ and every α:* $D_{\mathrm{ch}}(\kappa,\alpha) \le \mathbb{E}_\kappa[x_2^2] + \dfrac{\cos^2\alpha}{6}\,\Xi(\kappa)$.

[proved]

Proof. By Lemma 9 applied to (6), $D_{\mathrm{ch}}(\kappa,\alpha) \le \frac{1}{2\pi^2}\int_B |V|\,dy$. Fix $t = y_1 \in (-1,1)$; the slice $\{y_1 = t\}\cap B$ is the ball $\{|z| \le R(t)\}$ of $\mathbb{R}^3$. Put $a = A'(t)/3$. Then

$$\begin{aligned} |V|^2 &= \cos^2\alpha\, A^2 + (\sin\alpha\, E - \cos\alpha\, a\, y_2)^2 + \cos^2\alpha\, a^2(y_3^2 + y_4^2) \\ &= \cos^2\alpha\, A^2 + \sin^2\alpha\, E^2 - 2\sin\alpha\cos\alpha\, aE\, y_2 + \cos^2\alpha\, a^2|z|^2. \end{aligned}$$

Since $E > 0$, $\sqrt{P} \le (P + E^2)/(2E)$ for every $P \ge 0$. The cross term is odd in y_2 and integrates to 0 over the ball, and $\int_{|z|\le R}|z|^2dz = \frac{3}{5}R^2\cdot\frac{4}{3}\pi R^3$. With $\sin^2\alpha = 1 - \cos^2\alpha$ and $a^2\cdot\frac{3}{5}R^2 = \frac{1}{15}R^2A'^2$,

$$\int_{|z|\le R}|V|\,dz \;\le\; \frac{4}{3}\pi R^3\Big[E + \cos^2\alpha\;\frac{A^2 + \frac{1}{15}R^2A'^2 - E^2}{2E}\Big].$$

Integrating in t (Fubini) and dividing by $2\pi^2$,

$$\frac{1}{2\pi^2}\int_B |V|\,dy \;\le\; \frac{2}{3\pi}\int_{-1}^{1} R^3E\,dt + \frac{\cos^2\alpha}{3\pi}\int_{-1}^{1} R^3\,\frac{A^2 + \frac{1}{15}R^2A'^2 - E^2}{E}\,dt.$$

The first term is $\frac{1}{3}\langle R^2\rangle = \mathbb{E}_\kappa[x_2^2]$ (Lemma 8(i)). In the second, (7) gives $R^3A^2/E = R^3E\mathsf{u}^2$ and $R^5A'^2/(15E) = \frac{3}{5}RE\mathsf{w}^2$, so the integrand is $RE\,(R^2\mathsf{u}^2 + \frac{3}{5}\mathsf{w}^2 - R^2)$, and $\frac{\cos^2\alpha}{3\pi}\int RE(\cdots)\,dt = \frac{\cos^2\alpha}{3\pi}\cdot\frac{\pi}{2}\,\Xi = \frac{\cos^2\alpha}{6}\,\Xi$. □

No positivity of the components of V is assumed: the amplitude $E(y_1) > 0$ of the reflection field (the exact field at $\alpha = \pi/2$) is the reference scale. This is what replaces the function $m(y_1)$ of v1.4 (Appendix A), which degenerates for large κ (§4.9).

4.5 An exact identity

Lemma 11. *For $\kappa > 0$, $\Xi(\kappa) = -3\Delta(\kappa) - \mathcal{S}(\kappa)$, where*

$$\mathcal{S}(\kappa) := \frac{1}{5}\Big\langle 12\Big(\mathsf{w} - \frac{\kappa R^2\mathsf{u}}{4}\Big)^2 + \frac{\kappa^2}{4}R^4\mathsf{u}^2\Big\rangle \;\ge 0. \tag{9}$$

[proved]

Proof. Step 1 (a differential identity). For $\phi \in C^1[-1,1]$ put $\eta = \frac{\kappa}{5}R^4 - tR^2$ and

$$\mathcal{Q}[\phi] = \tfrac{1}{3}\big(R^2\phi' + (\kappa R^2 - 3t)\phi\big)^2 - R^2\phi^2 - \tfrac{1}{15}R^4(5\phi'^2 + 4\kappa\phi\phi' + 2\kappa^2\phi^2).$$

Expanding, the terms in ϕ'^2 cancel ($\frac{1}{3}R^4-\frac{1}{3}R^4$), the coefficient of $\phi\phi'$ is $\frac{2}{3}R^2(\kappa R^2-3t)-\frac{4\kappa}{15}R^4 = 2\eta$, and that of ϕ^2 is $c_0 := \frac{1}{3}(\kappa R^2-3t)^2 - R^2 - \frac{2\kappa^2}{15}R^4 = \frac{\kappa^2}{5}R^4 - 2\kappa t R^2 + 3t^2 - R^2$. Thus $\mathcal{Q}[\phi] = \eta\,(\phi^2)' + c_0\,\phi^2$. Next, $(R^5E)' = RE\,(\kappa R^4 - 5tR^2)$ and $(tR^3E)' = RE\,(R^2 - 3t^2 + \kappa t R^2)$, so $(R\,E\,\eta)' = \frac{\kappa}{5}(R^5E)' - (tR^3E)' = R\,E\,c_0$. Hence

$$R\,E\,\mathcal{Q}[\phi] = (R\,E\,\eta\,\phi^2)'.$$

Since $R\eta = R^3(\frac{\kappa}{5}R^2 - t)$ vanishes at ±1 and ϕ is bounded, $\langle\mathcal{Q}[\phi]\rangle = \frac{2}{\pi}\int_{-1}^{1} RE\,\mathcal{Q}[\phi]\,dt = 0$.

Step 2. Take $\phi = \mathsf{u}$. By (7), $R^2\mathsf{u}' + (\kappa R^2 - 3t)\mathsf{u} = 3\mathsf{w} - 3t\mathsf{u} = 3(m_1 - t)$, so the first term of $\mathcal{Q}[\mathsf{u}]$ is $3(t-m_1)^2$, whose mean is $3\,\mathrm{Var}_\kappa\,x_1$. Thus $\langle R^2\mathsf{u}^2\rangle + \frac{1}{15}\langle R^4(5\mathsf{u}'^2 + 4\kappa\mathsf{u}\mathsf{u}' + 2\kappa^2\mathsf{u}^2)\rangle = 3\,\mathrm{Var}_\kappa\,x_1$. Eliminating $\mathsf{u}' = 3\mathsf{w}/R^2 - \kappa\mathsf{u}$ gives $\frac{1}{15}R^4(5\mathsf{u}'^2 + 4\kappa\mathsf{u}\mathsf{u}' + 2\kappa^2\mathsf{u}^2) = 3\mathsf{w}^2 - \frac{6\kappa}{5}R^2\mathsf{u}\mathsf{w} + \frac{\kappa^2}{5}R^4\mathsf{u}^2$, hence

$$\langle R^2\mathsf{u}^2\rangle + 3\langle\mathsf{w}^2\rangle - \tfrac{6\kappa}{5}\langle R^2\mathsf{u}\mathsf{w}\rangle + \tfrac{\kappa^2}{5}\langle R^4\mathsf{u}^2\rangle = 3\,\mathrm{Var}_\kappa\,x_1. \tag{10}$$

Step 3. By Lemma 8, $\langle R^2\rangle = 3\mathbb{E}_\kappa[x_2^2] = 3(\mathrm{Var}_\kappa\,x_1 + \Delta)$. Inserting this and (10) in (8),

$$\Xi = \langle R^2\mathsf{u}^2\rangle + \tfrac{3}{5}\langle\mathsf{w}^2\rangle - 3\,\mathrm{Var}_\kappa\,x_1 - 3\Delta = -3\Delta - \tfrac{1}{5}\langle 12\mathsf{w}^2 - 6\kappa R^2\mathsf{u}\mathsf{w} + \kappa^2R^4\mathsf{u}^2\rangle,$$

and $12\mathsf{w}^2 - 6\kappa R^2\mathsf{u}\mathsf{w} + \kappa^2R^4\mathsf{u}^2 = 12\big(\mathsf{w} - \frac{\kappa R^2\mathsf{u}}{4}\big)^2 + \frac{\kappa^2}{4}R^4\mathsf{u}^2$. □

4.6 Positivity of Δ

Lemma 12. *$\Delta(\kappa) > 0$ for every $\kappa > 0$; equivalently, $\kappa \mapsto \mathbb{E}_\kappa[x_2^2] = I_2/(\kappa I_1)$ is strictly decreasing.*

[proved]

Proof. By Lemma 8(iii), $\Delta = \frac{\kappa}{3}\,\mathrm{Cov}_\kappa(x_1^2, x_1)$. Let μ_0 be the law of x_1 under $\bar\sigma$ (it is even) and $d\nu = \cosh(\kappa t)\,d\mu_0/\bar Z$, a probability measure since $\bar Z = \int\cosh(\kappa t)\,d\mu_0$. For even g, $\mathbb{E}_\kappa g(x_1) = \nu[g]$; for odd g, $\mathbb{E}_\kappa g(x_1) = \nu[g\tanh\kappa t]$. Hence $\mathrm{Cov}_\kappa(x_1^2, x_1) = \nu[t^3\tanh\kappa t] - \nu[t^2]\,\nu[t\tanh\kappa t] = \mathrm{Cov}_\nu(t^2,\ t\tanh\kappa t)$. Both t^2 and $t\tanh\kappa t$ are strictly increasing functions of $y = t^2$. For T, T' independent with law ν, $\mathrm{Cov}_\nu(t^2, t\tanh\kappa t) = \frac{1}{2}\mathbb{E}\big[(T^2 - T'^2)(T\tanh\kappa T - T'\tanh\kappa T')\big]$; the integrand is ≥ 0, and > 0 on the event $T^2 \neq T'^2$, which has positive probability. □

4.7 Proof of Theorem 6

Lower bound. The function $f = \langle x, v\rangle$ is 1-Lipschitz for ρ_{ch}, so $D_{\mathrm{ch}}(\kappa, v) \geq \mathrm{Cov}_\kappa(f, \langle x, v\rangle) = \mathrm{Var}_\kappa\langle x, v\rangle = \cos^2\alpha\ \mathrm{Var}_\kappa\,x_1 + \sin^2\alpha\,\mathbb{E}_\kappa[x_2^2] = \mathbb{E}_\kappa[x_2^2] - \cos^2\alpha\,\Delta$ (as $\mathrm{Cov}_\kappa(x_1, x_2) = 0$ and $\mathbb{E}_\kappa x_2 = 0$). At $\alpha = \pi/2$, $D_{\mathrm{ch}}(\kappa, e_2) = \mathbb{E}_\kappa[x_2^2]$ is Lemma 7.

Upper bound. Lemmas 10 and 11 give $D_{\mathrm{ch}}(\kappa,\alpha) \leq \mathbb{E}_\kappa[x_2^2] + \frac{\cos^2\alpha}{6}(-3\Delta - \mathcal{S}) = \mathbb{E}_\kappa[x_2^2] - \frac{\cos^2\alpha}{2}\Delta - \frac{\cos^2\alpha}{6}\mathcal{S}$, and $\mathcal{S} \geq 0$. By Lemma 12, $\Delta > 0$, so for $\kappa > 0$ and $\cos\alpha \neq 0$, $D_{\mathrm{ch}}(\kappa,\alpha) < \mathbb{E}_\kappa[x_2^2]$. With the case $\alpha = \pi/2$: $D_{\mathrm{ch}}(\kappa) = \mathbb{E}_\kappa[x_2^2]$, attained exactly at $\alpha = \pi/2$ (the vectors v and $-v$ give the same value, $f \mapsto -f$). By Lemma 8(ii) and Lemma 12, $\frac{d}{d\kappa}\mathbb{E}_\kappa[x_2^2] = -\Delta/\kappa < 0$: D_{ch} is strictly decreasing on $(0,\infty)$, and continuous at 0, where $D_{\mathrm{ch}}(0) = 1/4$ (Lemma 7).

The formulas. $\mathbb{E}_\kappa[x_2^2] = I_2/(\kappa I_1) = \frac{1}{3}(1 - \bar Z''/\bar Z)$ is Lemma 8(iv); hence $\frac{1}{4} - \mathbb{E}_\kappa[x_2^2] = (4\bar Z'' - \bar Z)/(12\bar Z)$. From $z_{2k+2}/z_{2k} = \frac{1}{4(k+1)(k+2)}$ for the coefficients $z_{2k} = 1/(4^k k!(k+1)!)$ of $\bar Z$, the coefficient of κ^{2k} in $4\bar Z'' - \bar Z$ is $\frac{3k}{k+2}z_{2k} \geq 0$, equal to $\frac{1}{8}$ for $k = 1$. Hence $4\bar Z'' - \bar Z \geq \kappa^2/8$ and $\frac{1}{4} - D_{\mathrm{ch}} \geq \kappa^2/(96\bar Z)$. The expansion follows by exact series division. □

Remark 13 (a constant on bounded fields). Since $\bar Z$ is increasing on $[0,\infty)$, $\frac{1}{4} - D_{\mathrm{ch}}(\kappa) \geq \kappa^2/(96\bar Z(K))$ on $[0, K]$. For $K = 4/3$, $\bar Z(4/3) \in [1.2393065416564178344 \pm 3\cdot 10^{-20}]$ gives $1/(96\bar Z(4/3)) \geq 0.0084052$, so $\frac{1}{4} - D_{\mathrm{ch}} \geq 0.0084\,\kappa^2$ on $[0, 4/3]$ [interval-certified].

4.8 Consequences

Corollary 14. *(a) For every $\beta \geq 0$ and every neighbour y of ℓ, the chordal influence coefficient is $r_{\mathrm{ch}}(y,\ell) = \beta/4$ and $\sum_y r_{\mathrm{ch}}(y,\ell) = 9\beta/2$; so the chordal Dobrushin criterion holds if and only if $\beta < 2/9$. (b) $W_1(\nu_b, \nu_{b'}) \leq \frac{1}{4}|b - b'|$ for all $b, b' \in \mathbb{R}^4$ (with $\nu_b \propto e^{\langle x,b\rangle}\bar{\sigma}$ and W_1 for ρ_{ch}), and the constant $\frac{1}{4}$ is best possible. (c) For $N = 4$ (single spin space $S^3 \subset \mathbb{R}^4$), every unit $\hat{\mathbf{c}}$ and every $\mathbf{g} \in \mathbb{R}^4$, the quantity $\sup_{\|f\|_L \leq 1} |\mu_{\mathbf{g}}[(\mathbf{x}\cdot\hat{\mathbf{c}} - \langle \mathbf{x}\cdot\hat{\mathbf{c}}\rangle_{\mathbf{g}})f]|$ of Levin's Remark 3.7 is $D_{\mathrm{ch}}(|\mathbf{g}|, \hat{\mathbf{c}}) \leq D_{\mathrm{ch}}(|\mathbf{g}|) \leq \frac{1}{4}$, with equality at $\mathbf{g} = 0$; so its maximum over $\mathbf{g}$ is attained at $\mathbf{g} = 0$, with value $1/N = 1/4$, and $D_{\mathrm{ch}}(|\mathbf{g}|)$ is strictly decreasing in $|\mathbf{g}|$.*

[proved]

Proof. (a) In Proposition 5, $\beta D_{\mathrm{ch}}(0) \leq r_{\mathrm{ch}}(y,\ell) \leq \beta \sup_{\kappa \leq 6\beta} D_{\mathrm{ch}}(\kappa)$, and the supremum equals $D_{\mathrm{ch}}(0) = \frac{1}{4}$ by Theorem 6; there are 18 neighbours. (b) By the segment formula in the proof of Proposition 5 (here with $s = b$, $\beta = 1$), $\int f\, d(\nu_{b'} - \nu_b) = \int_0^1 \mathrm{Cov}_{\nu_{b+t(b'-b)}}(f, \langle x, b' - b\rangle)\, dt \leq |b' - b| \sup_\kappa D_{\mathrm{ch}}(\kappa) = \frac{1}{4}|b' - b|$ for f 1-Lipschitz, then Kantorovich–Rubinstein duality [8, Thm 10.8]; the constant is attained at $b = 0$ (Lemma 7). (c) The supremum over f is $D_{\mathrm{ch}}(|\mathbf{g}|, \hat{\mathbf{c}})$ by definition, and $D_{\mathrm{ch}}(|\mathbf{g}|, \hat{\mathbf{c}}) \leq D_{\mathrm{ch}}(|\mathbf{g}|) \leq D_{\mathrm{ch}}(0) = \frac{1}{4} = D_{\mathrm{ch}}(0, \hat{\mathbf{c}})$ by Theorem 6 and Lemma 7. □

Parts (a) and (b) are already in [17] (Proposition 13 with Theorem 7; Theorem 7), and so is the non-strict monotonicity of the transverse moment $\kappa \mapsto I_2/(\kappa I_1)$ (Lemma 5(iii)); see §8 for what is new.

4.9 Cross-checks (not part of the proof)

Nothing in this subsection is used in the proof above. All scripts are in the folder `identity_all_kappa/` of the accompanying archive (SHA-256 below).

Artefacts (SHA-256). Paths are relative to the folder `identity_all_kappa/` of the accompanying archive.

file	SHA-256
`scripts/verif_symbolique.py`	`9458229aaf3b2a3d2527b60df9c42c311c7ecbd988e7c5e512b1ebbc42ac12d1`
`scripts/serie_identite_exacte.py`	`be7832a2df16a6a0011b7cf57d4f56de328ee2e266aad893b24c5b0ef0efe980`
`scripts/ident_num_arb.py`	`9302093edba6325cfe484f75a1fc09e96596654d19e718a22effcfd42fe38a98`
`scripts/ident_num_mp.py`	`d87c807e59799631cebfe01220217656d9f447e33b75c1e11a86f96fb09496e7`
`scripts/cert_xi_arb.py`	`623ddee924c1ef6eb31815a8ad2a208bfd339e4f20e8c25edc25232c7cf925ee`
`scripts/cert_xi_mpiv.py`	`097d406a2ef83c733f0de14f36f44bf683b0cf0611b7b81289c4c729884fadd7`
`scripts/cert_delta_boites.py`	`a3a3c1d98d101ce1845585aa6024683b1bedd01485e38ae052e6de52630cdf17`
`scripts/flux_champ_exact.py`	`214fe55874ece9334c33584c8900765fa739b79f17df166d109f7c2ca3c3320d`
`scripts/ot_controle.py`	`7cdc9b22d506182c91a3b8d13e4364e6e4143bfe7525f276bc0a4488a67a5973`
`scripts/d7_limite.py`	`17e3de668fc92d53c75ad573ccf7b7e740b94920a30b4e6d74481151c5fe454e`

Remark 15 (Why v1.4 stopped at $\kappa \leq 4/3$). The one-variable inequality (25) of Appendix A, $\max(\tilde{J}_0, \tilde{J}_1) < \tilde{N}/\bar{Z}^2$, is false for $\kappa > \kappa^* \approx 3.280$, and the hypothesis $m > 0$ of Proposition 24

check	what was done and the outcome	status
differential identity and algebra of §4.5, Lemma 8	SymPy: $RE\,\mathcal{Q}[\phi] = (RE\eta\phi^2)'$ for arbitrary ϕ; the three derivative formulas; the slice identity $R^3(A^2 + \frac{1}{15}R^2A'^2 - E^2)/E = R^3E\,(\mathsf{u}^2 + \frac{1}{15}R^2(\mathsf{u}' + \kappa\mathsf{u})^2 - 1)$; the Riccati equation (to 10^{-51} for I_2/I_1); $\frac{d}{d\kappa}(m_1/\kappa) = -\Delta/\kappa$; $\mathrm{Cov}(x_1^2, x_1) = 3\Delta/\kappa$; the identity $b' - (1 - 3b/\kappa - b^2) = -(s-2)^2/(s\kappa^2)$ for $b = (s-2)/\kappa$, $s = \sqrt{\kappa^2+4}$	[exact]
$\Xi + 3\Delta + \mathcal{S} = 0$ as formal power series in κ	exact fractions, up to κ^{39}, all coefficients vanish; $\Xi = -\frac{3}{32}\kappa^2 + \frac{363}{25600}\kappa^4 - \cdots$, $\Delta = \frac{1}{48}\kappa^2 - \frac{1}{384}\kappa^4 + \frac{1}{3840}\kappa^6 - \cdots$, $\mathcal{S} = \frac{1}{32}\kappa^2 - \frac{163}{25600}\kappa^4 + \cdots$	[exact]
the same identity at fixed κ	Arb, 90 digits, u by Taylor recursion and Gauss–Legendre quadrature (two rules), 20 values of $\kappa \in [0.01, 400]$ and 100 random values in $[0.010, 208]$: residual $\le 3.7 \cdot 10^{-89}$; an independent mpmath code (integral representation of u, no Taylor series), 30 digits, $\kappa \in \{\frac{1}{2}, \frac{4}{3}, 3, 7, 12, 25\}$: residual $\le 6.1 \cdot 10^{-31}$	[numerical]
$\Xi(\kappa) < 0$ on $[4/3, 20]$	two independent interval implementations (Arb, 13 974 boxes; mpmath.iv with exact fractions, 18 267 boxes), continuous covering; worst $\Xi_{\mathrm{hi}}/\langle R^2\rangle$ is $-1.3 \cdot 10^{-4}$ (Arb, near $\kappa = 20$, where the technique reaches its limit) and -0.164 (mpmath.iv). With Lemma 10 this gives $D_{\mathrm{ch}} = \mathbb{E}_\kappa[x_2^2]$ on that range without Lemmas 11–12	[interval-certified]
$\Delta(\kappa) > 0$	Arb, 933 boxes cover $[1/5, 10^4]$: $\Delta > 0 \iff m_1 > b$, where $b(\kappa) = (\sqrt{\kappa^2+4} - 2)/\kappa$ is the positive root of $b^2 + 4b/\kappa = 1$, and both m_1 and b are increasing, so $m_1(\kappa_0) > b(\kappa_1)$ on a box $[\kappa_0, \kappa_1]$ suffices	[interval-certified]
chain $\mathrm{Low} \le F \le B \le \mathbb{E}_\kappa[x_2^2]$	$F = \frac{1}{2\pi^2}\int_B \lvert V\rvert$ for the field (6) by 3D quadrature, $B = \mathbb{E}_\kappa[x_2^2] + \frac{\cos^2\alpha}{6}\Xi$, $\mathrm{Low} = \mathbb{E}_\kappa[x_2^2] - \cos^2\alpha\,\Delta$: holds at 98 grid points and at 180 random (κ, α), $\kappa \in [0.05, 60]$ (no violation; the smallest slack is $2.5 \cdot 10^{-8}$, at angles close to 90° where the three bounds coincide); at $\kappa = 4/3$, $\alpha = 0°, 45°, 90°$: $F = 0.208916, 0.221397, 0.233321$, as in v1.4	[numerical]
direct optimal transport	exact discrete transport (POT), reduction by the symmetry in (x_3, x_4), Richardson extrapolation, 25 values $\kappa \in [0, 50] \times 10$ angles: the value at 90° equals $\mathbb{E}_\kappa[x_2^2]$ to 10^{-7}, the maximum over α is attained at 90° for every κ tested, and $L_{\mathrm{OT}} \le B$ (Table 3)	[numerical]

Table 2: Cross-checks of Theorem 6.

κ	α	L_{OT}	Low	F	B	$\mathbb{E}_\kappa[x_2^2]$
2	0°	0.172701	0.162709	0.172869	0.179396	0.216564
2	40°	0.191871	0.184961	0.191924	0.194753	0.216564
2	90°	0.216564	0.216564	0.216564	0.216564	0.216564
10	0°	0.024799	0.014112	0.025200	0.046808	0.085419
10	40°	0.057718	0.043574	0.057745	0.062761	0.085419
10	90°	0.085419	0.085419	0.085419	0.085419	0.085419
50	0°	0.002252	0.000594	0.002268	0.009855	0.019403
50	40°	0.012568	0.008365	0.012568	0.013800	0.019403
50	90°	0.019403	0.019403	0.019403	0.019403	0.019403

Table 3: Optimal-transport values L_{OT} of $D_{\mathrm{ch}}(\kappa,\alpha)$ against the chain of bounds ([numerical]; $L_{\mathrm{OT}} \le B$ and $\mathrm{Low} \le L_{\mathrm{OT}}$ within $2\cdot 10^{-5}$ in all 250 cases).

κ	$D_{\mathrm{ch}}(\kappa) = I_2/(\kappa I_1)$	$\frac{1}{4} - D_{\mathrm{ch}}$	$\kappa^2/(96\bar{Z})$	Δ
1/2	0.2474359	0.002564143	0.002524	0.00505
1	0.2401937	0.009806276	0.009216	0.01847
4/3	0.2333214	0.01667858	0.01494	0.03007
2	0.2165637	0.03343629	0.02619	0.05385
3	0.1893079	0.06069212	0.03557	0.07977
4	0.1645118	0.08548818	0.03415	0.09107
6	0.127121	0.122879	0.01834	0.09024
10	0.08541853	0.1645815	0.00195	0.07131
20	0.04629939	0.2037006	$9.814\cdot 10^{-7}$	0.04265
50	0.01940306	0.2305969	$2.243\cdot 10^{-18}$	0.01881

Table 4: Values of the exact coefficient ([numerical], mpmath, 30 digits).

fails for $\kappa \gtrsim 6.85$ (high-precision Arb, [numerical]): the ratio $\max(J_0, J_1)/(\mathbb{E}_\kappa[x_2^2] - \mathrm{Var}_\kappa\, x_1)$ equals 0.3154 at $\kappa = 4/3$ and 0.4679 at $\kappa = 2$ (the values 0.32 and 0.47 of v1.4 are recovered), 0.9831 at 3.25 and 1.1318 at 3.5; $\min m$ vanishes at $\kappa = 6.8458$. This is a limit of the second-order estimate of v1.4, not of the result: for large κ, $A \ll E$ in the bulk of the law, so the uniform minorant $m(y_1)$ of $\langle v, V\rangle$ degenerates, while the exact flux of the same field stays well below $\mathbb{E}_\kappa[x_2^2]$ for every $\alpha < \pi/2$ (Table 3). The slice estimate of §4.4 uses $E(y_1)$ as scale instead of $m(y_1)$, and the criterion $\Xi < 0$ holds, with a growing margin, on the whole grid tested ($\Xi/\langle R^2\rangle$ from -0.0078 at $\kappa = 1/4$ to -0.998 at $\kappa = 400$).

Remark 16 (Earlier numerical evidence and the quality of the field). Before Theorem 6 was proved, linear programs for $\sup_f$ (reduction to functions of two coordinates, HiGHS; logs `calc/lp2d_m14.log` and `calc/check1d.log`) gave for $\max_\alpha D_{\mathrm{ch}}(\kappa,\alpha)$ the values 0.2500, 0.2474, 0.2402, 0.2332, 0.2164 at $\kappa = 0$, 0.5, 1, 4/3, 2, the maximum over directions being attained for $v \perp \hat{s}$; the same code reproduces the geodesic $D(0)$ to 0.15% [numerical]. These values agree with $\mathbb{E}_\kappa[x_2^2] = 0.25$, 0.24744, 0.24019, 0.23332, 0.21656 (the discretised programs are neither upper nor lower bounds; the value $0.2164 < 0.21656$ at $\kappa = 2$ is a discretisation error, since $\mathbb{E}_\kappa[x_2^2]$ is an exact lower bound for $D_{\mathrm{ch}}(\kappa)$). The full flux bound of the explicit field (6) is within $1.7\cdot 10^{-4}$ of the linear programs for all directions tested, so the field is close to optimal [numerical].

5 The geodesic one-link bound

This section is the main result of v1.2 of this note, unchanged in substance. It gives the exact field-uniform coefficient for the geodesic metric, hence Theorem 2.

Theorem 17 (One-link bound, geodesic metric)**.** *For all $\kappa \in [0, 59/50]$ and all unit v,*

$$D(\kappa, v) \;\le\; \frac{8}{9\pi} - 0.0085\,\kappa^2,$$

with equality at $\kappa = 0$. Since $6 \cdot \pi/16 = 3\pi/8 < 59/50$, this covers $\kappa \le 6\beta$ for all $\beta < \pi/16$.

Status: $\kappa = 0$ [proved]; $\kappa > 0$ [interval-certified]. Precisely: Lean proves only that the monotone bound `bP` is < 0 in each of the 160 boxes, i.e. $\Phi < 0$ on $[0, K^2]$ *if* the exported rationals bound the coefficients ([Lean-checked], §5.2); the constant 0.0085 and the covering of $\alpha \in [0, \pi]$ by the boxes (box $\mapsto$ α-interval) are [interval-certified], not in Lean. The inequality $3\pi/8 < 59/50$ is [Lean-checked] (`kappa_max_lt`).

Lemma 18 ($\kappa = 0$)**.** *$D(0, v) = 8/(9\pi)$ for every unit v. [proved]*

Proof. Let $\gamma \in [0, \pi]$ be the angle between u and v; under Haar measure its law is $\frac{2}{\pi}\sin^2\gamma\, d\gamma$. *Upper bound:* $\cos\gamma$ is a first eigenfunction of $-\Delta_{S^3}$ (eigenvalue 3), so $\psi = \cos\gamma/3$ solves $-\Delta\psi = \langle u, v\rangle$ and, for 1-Lipschitz f, $\mathrm{Cov}(f, \langle u, v\rangle) = \mathbb{E}[\nabla f \cdot \nabla\psi] \le \mathbb{E}|\nabla\psi| = \frac{1}{3} \cdot \frac{2}{\pi}\int_0^\pi \sin^3\gamma\, d\gamma = \frac{8}{9\pi}$. *Lower bound:* $f = \pi/2 - \gamma$ is 1-Lipschitz and (integration by parts)

$$\mathrm{Cov}(f, \cos\gamma) = -\mathbb{E}[\gamma\cos\gamma] = -\frac{2}{\pi}\int_0^\pi \gamma\cos\gamma\sin^2\gamma\, d\gamma = -\frac{2}{\pi}\Big(-\frac{4}{9}\Big) = \frac{8}{9\pi}.$$

□

5.1 Reduction (proved, not formalised)

Step 1: transport. Let $L = \Delta_{S^3} + \kappa\nabla x \cdot \nabla$ with $x = \langle u, \hat{s}\rangle$; L is symmetric in $L^2(q_\kappa)$, with Dirichlet form $\mathbb{E}_{q_\kappa}[\nabla\varphi\cdot\nabla\psi]$. *Weak solutions (Lax–Milgram).* Let H be the space of $\varphi \in H^1(S^3)$ with $\mathbb{E}_{q_\kappa}\varphi = 0$, with the inner product $\mathbb{E}_{q_\kappa}[\nabla\varphi \cdot \nabla\psi]$; it is a Hilbert space by the Poincaré inequality of Step 3 ($\lambda > 0$). For $g \in L^2(q_\kappa)$ with $\mathbb{E}_{q_\kappa} g = 0$, the functional $\varphi \mapsto \mathbb{E}_{q_\kappa}[\varphi g]$ is bounded on H by $\|g\|_2\lambda^{-1/2}\|\nabla\varphi\|_2$, so by Riesz there is a unique $\psi \in H$ with $\mathbb{E}[\nabla\varphi\cdot\nabla\psi] = \mathbb{E}[\varphi g]$ for all $\varphi \in H^1$ (constants integrate against g to zero); we write $-L\psi = g$ in this weak sense, which is all that is used below. With $g = \langle u, v\rangle - \mathbb{E}_{q_\kappa}\langle u, v\rangle$ and f 1-Lipschitz for ρ (so $f \in H^1$ with $|\nabla f| \le 1$ a.e., by Rademacher's theorem), $\mathrm{Cov}_{q_\kappa}(f, \langle u, v\rangle) = \mathbb{E}[\nabla f \cdot \nabla\psi_v] \le \mathbb{E}|\nabla\psi_v|$, so $D(\kappa, v) \le \mathbb{E}_{q_\kappa}|\nabla\psi_v|$. [proved]

Step 2: reduced ODEs and exact series. Write $v = \cos\alpha\,\hat{s} + \sin\alpha\, e$ with $e \perp \hat{s}$, $y = \langle u, e\rangle$, and seek $\psi_v = \cos\alpha\, g(x) + \sin\alpha\, y\, h(x)$, $p = g'$. Using $\Delta_{S^3} g(x) = (1-x^2)g'' - 3xg'$, $\Delta_{S^3}(yh) = y[(1-x^2)h'' - 5xh' - 3h]$, $|\nabla x|^2 = 1 - x^2$ and $\nabla x \cdot \nabla y = -xy$, the equation reduces to

$$(1-x^2)(p' + \kappa p) - 3xp = -(x - m), \qquad (1-x^2)(h'' + \kappa h') - 5xh' - (3 + \kappa x)h = -1,$$

with $m = \mathbb{E}_{q_\kappa} x$. Expanding $p = \sum_n \kappa^n p_n$, $h = \sum_n \kappa^n h_n$, $m = \sum_n \kappa^n m_n$ gives polynomials p_n, h_n with rational coefficients, computed exactly; p_n, h_n have parity $(-1)^n$ and $m_n = 0$ for n even (asserted in exact arithmetic by the certificate script). For the truncation ψ_N at order N, the residual is $L\psi_N + (\langle u, v\rangle - \cos\alpha\, m) = \cos\alpha\,(m_N - m) + R'$ with $R' = \kappa^{N+1}\big[\cos\alpha(1 - x^2)p_N + \sin\alpha\, y((1-x^2)h_N' - xh_N)\big]$ (identity also checked symbolically in $\mathbb{R}^4$ at rational points of S^3). Since the full residual R has q_κ-mean zero, $|\cos\alpha(m_N - m)| \le \|R'\|_\infty$, hence $\|R\|_\infty \le 2\kappa^{N+1}\rho_N$, where ρ_N is the sum of ℓ^1 norms of the coefficients of the two bracketed polynomials ($\rho_{16} \approx 8.1 \cdot 10^{-10}$). [proved]

Step 3: correction via Poincaré. Let $-L\varphi = R$; then $\psi_v = \psi_N + \varphi$ exactly. The density of q_κ w.r.t. Haar measure has log-oscillation at most 2κ and the spectral gap of $-\Delta_{S^3}$ is 3, so by the Holley–Stroock perturbation argument [9] the Poincaré constant of q_κ satisfies $\lambda \geq 3e^{-2\kappa}$. (Proof: write $dq_\kappa = h\, d\sigma$ with σ the Haar measure, $h = e^{\kappa x}/\tilde{Z}$, so $e^{-\kappa}/\tilde{Z} \leq h \leq e^{\kappa}/\tilde{Z}$; then $\mathrm{Var}_q f \leq \mathbb{E}_q(f - \mathbb{E}_\sigma f)^2 \leq \frac{e^\kappa}{\tilde{Z}} \mathrm{Var}_\sigma f \leq \frac{e^\kappa}{3\tilde{Z}} \mathbb{E}_\sigma |\nabla f|^2 \leq \frac{e^{2\kappa}}{3} \mathbb{E}_q |\nabla f|^2$.) Then $\mathbb{E}|\nabla\varphi|^2 = \mathbb{E}[\varphi R] \leq \|R\|_2 (\mathbb{E}|\nabla\varphi|^2/\lambda)^{1/2}$, whence

$$\mathbb{E}|\nabla\varphi| \leq \|R\|_2/\sqrt{\lambda} \leq \varepsilon_N(\kappa) := 2\kappa^{N+1}\rho_N e^\kappa/\sqrt{3}.$$

So $D(\kappa, v) \leq \mathbb{E}|\nabla\psi_N| + \varepsilon_N(\kappa)$. [proved]

Step 4: AM–GM. With $\varrho = \sin\gamma/3 = |\nabla\psi_0|$ ($\cos\gamma = \langle u, v\rangle$), pointwise $|X| \leq (|X|^2/\varrho + \varrho)/2$, hence $\mathbb{E}|\nabla\psi_N| \leq U_N := \mathbb{E}[(|\nabla\psi_N|^2/\varrho + \varrho)/2]$, with equality at $\kappa = 0$. [proved]

Step 5: polynomial coordinates. The image of the uniform measure on S^3 under $u \mapsto (\langle u, \hat{s}\rangle, \langle u, e\rangle)$ is uniform on the unit disc, and $|\nabla\psi_N|^2$ is a polynomial in (x, y, κ). In the rotated frame $s = \langle u, v\rangle$, $t = \sqrt{1 - s^2}\,\tau$, the singular weight $1/\varrho$ cancels against the Jacobian and

$$U_N(\kappa, \alpha)\, Z(\kappa) = \iint_{[-1,1]^2} \Big[\tfrac{3}{2}|\nabla\psi_N|^2 + \tfrac{1-s^2}{6}\Big] e^{\kappa x}\, ds\, d\tau, \qquad Z(\kappa) = \pi \sum_{k\geq 0} \frac{(\kappa^2/4)^k}{k!\,(k+1)!} = \pi \bar{Z}(\kappa).$$

At $\kappa = 0$ the integral is $8/9$ and $Z(0) = \pi$, recovering $8/(9\pi)$. The coefficient of κ^j is the integral of a polynomial of degree $\leq j + 2$ in (s, τ): the terms odd in τ integrate to zero (also exactly on symmetric Gauss nodes), and in the remaining terms $\sqrt{1 - s^2}$ occurs only squared. [proved]

Step 6: exact factorisation of κ^2. Put $\Delta_{\mathrm{g}}(\kappa, \alpha) := (U_N + \varepsilon_N - \frac{8}{9\pi})Z = \sum_j E_j(\alpha)\kappa^j +$ (remainders). $E_0 = 0$ exactly ($I_0 = 8/9$), and $E_j = 0$ for j odd by the parity $u \mapsto -u$, i.e. $(s, \tau) \mapsto (-s, -\tau)$. Hence $\Delta_{\mathrm{g}} = \kappa^2\Phi(w)$ with $w = \kappa^2$ and $\Phi(w) = \sum_{k\geq 1} E_{2k} w^{k-1} +$ (Taylor remainder $+ \varepsilon_N Z)/\kappa^2$; the last two terms are increasing in κ and are bounded at $\kappa = K = 59/50$. This is what makes the comparison at $\kappa \to 0$ non-degenerate: no two touching balls are ever compared. [proved]

Since Z is increasing, $\Phi \leq -c_*$ on $[0, K^2]$ implies $D(\kappa, v) \leq \frac{8}{9\pi} - \frac{c_*}{Z(K)}\kappa^2$.

Remark 19 (second order, exact). An independent exact computation (SymPy, harmonic components in $\mathbb{R}^4$; code not included in the archive) gives $E_2(\alpha) = -(1309\cos^2\alpha + 779)/21600$, maximal at $\alpha = \pi/2$. Locally as $\kappa \to 0$ the method cannot give a coefficient better than $779/(21600\pi) \approx 0.0115$ in front of κ^2; along the route actually used ($c_*/Z(K)$, uniform on $[0, K]$), since $c_* \leq -\sup_\alpha \Phi(0) = 779/21600$, the ceiling is $779/(21600\, Z(K)) \approx 0.0097$. The certified 0.0085 is below both. (Status: exact computation; the same E_2 and E_4 are reproduced in exact rationals, by a different method, by the independent check `independent_check_geodesic/`.)

5.2 Computer-assisted part

Interval evaluation. Parameters: series order $N = 16$; Gauss–Legendre with $N_G = 22$ nodes per direction, exact for polynomial degree $\leq 2N_G - 1 = 43$, hence for all $j \leq D = 41$; $\alpha \in [0, \pi]$ cut into 160 boxes of width $\pi/160$, $(\cos\alpha, \sin\alpha)$ enclosed by balls; Arb ball arithmetic (python-flint 0.9.0) with working precision 128 bits and series length cap 200, both set explicitly and read back. Nodes and weights are rigorous balls (`arb.legendre_p_root`). Orders $\kappa^0, \ldots, \kappa^{41}$ are computed; the orders ≥ 42 of $e^{\kappa x}$ are covered by the Lagrange bound $|\mathrm{rem}|/\kappa^2 \leq 4\kappa^{D-1}e^\kappa \sum_i \sup|A_i|/(D + 1 - i)! \leq 1.53 \cdot 10^{-21}$ at $\kappa = K$. On each box, $\Phi(w) \leq \sum_k \max(c_k \cdot 0^k, c_k K^{2k})$ with c_k upper bounds of E_{2k+2} (the constant term absorbing the remainder bounds).

Result. All 160 boxes pass in one piece in w; the worst bound is $\Phi \le -0.03175361$ (box 75, α near $\pi/2$); $Z(59/50) \in [3.721046472 \pm 4.7 \cdot 10^{-10}]$, and $0.03175361/Z(59/50) \in [0.0085335161 \pm 5 \cdot 10^{-12}] > 0.0085$. A second run with $N = 20$, $N_G = 26$, 240 boxes, precision 192 passes with worst bound -0.03333053 (stability check, same code). [interval-certified]

Erratum on an earlier version (important). A first version of this certificate (Lean commit `1f3ace0`) was *invalid*: python-flint's default `ctx.cap` = 10 silently truncated every product of power series at κ^9, so the orders $\kappa^{10}, \dots, \kappa^{41}$ were neither computed nor bounded, and in each of the 160 boxes at least one exported bound (orders E_{12}–E_{40}) was below the true value. Lean accepted it, since Lean only sees the rational inequalities. The error was found by an independent adversarial re-verification, and its numerical effect is small ($\le 4.7 \cdot 10^{-6}$ against a margin 0.0317), but the artefact had to be regenerated. The repaired script makes cap and precision mandatory arguments, refuses cap $< L = 42$ (guard 1) and asserts that every series and product has length exactly L (guard 2, 851 840 assertions per run; the assertion is on `.prec` because `coeffs()` drops trailing zeros). The chordal certificate of §A.3 was written with the same guards from the start.

Independent checks (gates). All logs return code 0 unless a refusal is expected.

gate	outcome
G1: cap $\in \{42, 50, 100, 200\}$	identical results in all 160 boxes; cap = 10 refused by guard 1; cap = 10 and 41 with guard 1 disabled refused by guard 2
G2: explicit Cauchy convolution (no `arb_series`)	all 42 coefficients overlap; $\max \lvert\Delta\text{mid}\rvert = 1.5 \cdot 10^{-39}$
G3: $N \in \{16, 20, 24\}$ vs 32	$\lvert U_N - U_{32}\rvert$ nonzero, decreasing ($1.34 \cdot 10^{-10}$, $1.20 \cdot 10^{-12}$, $1.08 \cdot 10^{-14}$ at $\alpha = 0$, $\kappa = 1.18$), below ε_N
G4: $N_G \in \{21, 22, 26\}$	22 vs 26 overlap for $j \le 41$; with 21 nodes disagreement starts exactly at $j = 40$, as predicted by Gauss exactness
G5: Arb vs float polar grid	gap $\le 1.0 \cdot 10^{-8}$, decreasing as $O(h^2)$ under grid refinement
independent recomputation (Galerkin; code not included in the archive)	Galerkin solver at finite κ and exact E_2, E_4: consistent with every box bound; recomputed data identical byte for byte
independent check (exact rationals; `independent_check_geodesic/`)	$E_j(\alpha) \in \mathbb{Q}[\cos\alpha]$ computed exactly by a different method (no quadrature, no `arb_series`); all 3200 Lean coefficients of `7a211ed` (160 boxes $\times$ $E_2, \dots, E_{40}$) are true upper bounds, via rigorous box majorants with the invariant "majorant $\ge$ exact sample" checked everywhere; the same tool refutes the old commit `1f3ace0` (1634 false bounds, 160/160 boxes); worst `bP` $= -0.03175360$ in rationals; $0.0085335 \ge 0.0085$ checked exactly against a rigorous upper bound of $Z(K)$; U_{16} vs. an independent Galerkin solver at 700 points (κ, α): $\max \lvert\Delta\rvert = 1.34 \cdot 10^{-10} < \varepsilon_{16}$

Floating-point gates are [numerical]; they test the modelling, they are not part of the proof.

Lean. Commit `7a211ed0426e95b36a0ff35f5e896f8594b34fd3` of the author's Lean library (parent `1f3ace0`; only `KevYM/Certif/DobrushinLien/Donnees.lean` changed). The three modules are reproduced byte for byte in the standalone Lean project `lean/` of the accompanying archive (they are unchanged in `1b0f039`). Lean toolchain `v4.34.0`. Main statements (verbatim):

```
theorem sound (cs : List ℚ) (a b : ℚ) (ha : 0 ≤ a) (hneg : bP cs 0 a b < 0)
    (es : List ℝ) (hes : List.Forall₂ (fun (e : ℝ) (c : ℚ) => e ≤ (c : ℝ)) es cs)
    (w : ℝ) (haw : (a : ℝ) ≤ w) (hwb : w ≤ b) : pR es 0 w < 0
theorem certs_ok : certs.all (fun cs => decide (bP cs 0 0 KW < 0)) = true
theorem certificat_boites (cs : List ℚ) (hcs : cs ∈ certs) (es : List ℝ)
    (hes : List.Forall₂ (fun (e : ℝ) (c : ℚ) => e ≤ (c : ℝ)) es cs)
```

```
    (w : ℝ) (hw0 : 0 ≤ w) (hw : w ≤ (KW : ℝ)) : pR es 0 w < 0
theorem somme_dobrushin_lt_one (β : ℝ) (hβ : β < Real.pi / 16) :
    18 * β * (8 / (9 * Real.pi)) < 1
theorem kappa_max_lt : 6 * (Real.pi / 16) < (59 : ℝ) / 50
```

Here `bP` is the monotone upper bound $\sum_k \max(c_k a^k, c_k b^k)$, `pR es 0 w` $= \sum_j e_j w^j$ (`pR_eq_sum`), `KW` $= 3481/2500 = (59/50)^2$, and `certs` is the list of 160 exported rational coefficient vectors (Arb upper bounds rounded up to the grid 2^{-40}, re-checked exactly before export). `certs_ok` is proved by `decide +kernel`. The file also proves $18 \cdot (\pi/16) \cdot 8/(9\pi) = 1$ and the converse for $\beta \geq \pi/16$. Build clean (0 warnings); all 13 theorems depend only on `propext`, `Classical.choice`, `Quot.sound`; the rebuild from `git archive 7a211ed` succeeds, and so does the from-scratch build of the standalone project (`lean/logs/build_from_zero.log`). [Lean-checked]

What Lean covers, exactly. Lean proves `bP cs 0 0 KW < 0` (strict) for each of the 160 exported vectors, and the implication of `certificat_boites`: *if* the true coefficients lie below the exported rationals, *then* $\Phi < 0$ on $[0, K^2]$; plus the threshold arithmetic. Not in Lean: the constant 0.0085 (i.e. $-\max \mathtt{bP}/Z(K) \geq 0.0085$, checked exactly in rationals by the independent check against a rigorous upper bound of $Z(K)$) and the covering of $[0, \pi]$ by the boxes (the association box $\mapsto$ α-interval is a comment in the Lean file); both are [interval-certified].

Trust base. The implication "integrals $\Rightarrow$ exported bounds" rests on Arb and on the reduction of §5.1 (proved on paper; the classical inputs are Kantorovich–Rubinstein duality, quoted from [8, Thm 10.8], the Holley–Stroock perturbation argument, written out in Step 3, and the weak solvability of $-L\varphi = R$, written out in Step 1), and is *not* formalised. It is cross-checked in exact rationals by the independent check (table above), which shares with the certificate only the modelling (reduced ODEs, measure $ds\, d\tau$).

Artefacts (SHA-256). Paths are relative to the accompanying archive.

file	SHA-256
`certificate_geodesic/cert_lien.py` (v2)	`8b76b11125693b8f71b44898e959635066b3804560c2ee314092667a562ea096`
`certificate_geodesic/export_lean.py`	`1885f8bc2e8444bd616efa41c5c18bbfaa3fca78e6bc3e06ae197014160c1bcb`
`certificate_geodesic/cert_N16_G22_B160_cap200_p128.json`	`91700df0e8031e1f6ed4c3d6d9e88c42ddf36f0abeacd58c9f328738b0d9c43e`
`lean/KevYM/Certif/DobrushinLien/Donnees.lean` (= `7a211ed`)	`9d6c06283848a9ef38baecb222e195e9aa528009412e7c8bded6f2555321c37f`

6 Proofs of Theorems 1 and 2, and a corollary

Theorem 1. Let $\beta < 2/9$, and use the chordal metric. By Theorem 6, D_{ch} is decreasing, so $\sup_{\kappa \leq 6\beta} D_{\mathrm{ch}}(\kappa) = D_{\mathrm{ch}}(0) = 1/4$; by Proposition 5 each of the 18 neighbours has influence $\beta/4$ and all other links have influence 0; so (5.2) holds with $d = 9\beta/2 < 1$ (`somme_dobrushin_lt_one`). (5.1) holds for every field. By [1, Thm 4] there is at most one field with the Wilson single-link conditionals, hence at most one Gibbs measure (item 2 of §2); existence is Proposition 3. (Only $\kappa \leq 6\beta < 4/3$ occurs here, the range of the proof of Appendix A, which uses `kappa_max_lt`.) □

Theorem 2. Let $\beta < \pi/16$, and use the geodesic metric. By Theorem 17, $\sup_{\kappa \leq 6\beta} D(\kappa) = D(0) = 8/(9\pi)$, and Proposition 5 gives the sum $16\beta/\pi < 1$ (`somme_dobrushin_lt_one` of the geodesic module). For $\beta \geq \pi/16$ the lower bound of Proposition 5 gives a sum $\geq 16\beta/\pi \geq 1$. □

6.1 A Poincaré inequality for $\beta < 2/9$ (a corollary of Wu's theorem)

The influence coefficients of Proposition 5 are Dobrushin's interdependence coefficients in the sense of Wu [22, (2.3)]. His theorem therefore turns Theorem 6 into a Poincaré inequality, uniform in the volume and in the boundary condition, on the whole range of Theorem 1. This is a corollary, not a new method: the Poincaré inequality under Dobrushin's condition is classical (§8), and we claim only its explicit form for this model. It is a Poincaré inequality only: no log-Sobolev inequality and no decay of correlations is claimed.

Corollary 20 (Poincaré inequality)**.** *Let $0 \le \beta < 2/9$, let $\Lambda \subset E$ be finite, $\bar{U}$ any boundary condition, and $\mu = \mu_\Lambda^{\bar{U}}$ the probability measure (1) on $SU(2)^\Lambda$. For every $f \in C^1(SU(2)^\Lambda)$,*

$$\mathrm{Var}_\mu(f) \;\le\; \frac{1}{\lambda_0\,(1-\frac{9}{2}\beta)} \sum_{\ell\in\Lambda} \mathbb{E}_\mu[|\nabla_\ell f|^2],$$

where ∇_ℓ is the gradient in U_ℓ for the round metric of the unit sphere S^3, and $\lambda_0 > 0$ is the infimum over $\kappa \in [0, 6\beta]$ of the spectral gap of q_κ (its best Poincaré constant) for this metric. One has $\lambda_0 \ge 2 - 6\beta$ (Bakry–Émery) and $\lambda_0 \ge 3$ [17, Lemma 4 of v1, Lemma 122 of v4]. Hence

$$\mathrm{Var}_\mu(f) \;\le\; \frac{1}{3}\Big(1-\frac{9}{2}\beta\Big)^{-1} \sum_{\ell\in\Lambda} \mathbb{E}_\mu|\nabla_\ell f|^2,$$

and the weaker constant $\big[(2-6\beta)(1-\frac{9}{2}\beta)\big]^{-1}$ uses the Bakry–Émery bound only. For the bi-invariant metric $\langle X, Y\rangle = -\mathrm{Tr}(XY)$ (the Frobenius normalisation of Proposition 22: the round S^3 of radius $\sqrt{2}$, in which $|\nabla f|_F^2 = \frac{1}{2}|\nabla f|^2$) the two constants become $\frac{2}{3}(1-\frac{9}{2}\beta)^{-1}$ and $\big[(1-3\beta)(1-\frac{9}{2}\beta)\big]^{-1}$.

Status: [proved], from [22, Thm 2.1] ([read]), the Bakry–Émery criterion [23] (as used in [22, proof of Lemma 5.4]; not re-read) and, for the constant $\frac{1}{3}$, [17, Lemma 4] ([read]; its proof by Barta's method is not re-derived here).

Proof. (1) The Dobrushin matrix. Let $T = \Lambda$. The conditional law of U_ℓ under μ given the other links of Λ is the law q_s of (2) (the links outside Λ are frozen to $\bar{U}$). Let $C = (c_{\ell\ell'})$ be the Dobrushin matrix of μ for the chordal metric, $c_{\ell\ell'} = \sup W_1(\mu_\ell(\cdot|x), \mu_\ell(\cdot|y))/\rho_{\mathrm{ch}}(x_{\ell'}, y_{\ell'})$ over configurations x, y differing only at ℓ'; this is the coefficient (5.4) of [1] and the matrix (2.3) of [22]. The conditions that occur for μ form a subset of those of the full specification, so $c_{\ell\ell'} \le r_{\mathrm{ch}}(\ell', \ell) = \beta/4$ if ℓ' is one of the (at most 18) neighbours of ℓ in Λ, and $c_{\ell\ell'} = 0$ otherwise (Proposition 5, Theorem 6). Hence $r_{\mathrm{sp}}(C) \le \|C\|_\infty \le 9\beta/2 < 1$. *(2) Tensorisation.* Theorem 2.1 of [22] (p. 5; T finite, (E, d) Polish, here S^3 with ρ_{ch}, and a moment condition that is trivial on a compact space) states that if $r_{\mathrm{sp}}(C) < 1$ then $(1 - r_{\mathrm{sp}}(C))\,\mathrm{Var}_\mu(f) \le \mathbb{E}_\mu \sum_\ell \mathrm{Var}_{\mu_\ell}(f)$ for $f \in L^2(\mu)$, where μ_ℓ is the conditional law of the ℓ-th coordinate. The metric d enters only through C and may differ from the metric of the gradient, as in [22, Prop. 5.5] (Euclidean metric for C, Riemannian gradient). *(3) One site.* $\mu_\ell = q_s$ with $\beta|s| \le 6\beta$, i.e. q_s is q_κ up to a rotation, with $\kappa \le 6\beta$; so $\mathrm{Var}_{\mu_\ell}(f) \le \lambda_0^{-1}\mathbb{E}_{\mu_\ell}|\nabla_\ell f|^2$. Summing over ℓ and averaging over the conditions gives the inequality. *(4) The values of λ_0.* On the unit S^3, $\mathrm{Ric} = 2g$ and $\mathrm{Hess}\,\langle x, a\rangle = -\langle x, a\rangle g$, so $q_\kappa \propto e^{-V}$ with $V = -\kappa x_1$ has $\mathrm{Ric} + \mathrm{Hess}\,V = (2+\kappa x_1)g \ge (2-\kappa)g$; the Bakry–Émery criterion gives a gap $\ge 2-\kappa \ge 2-6\beta > 0$. The bound 3 is [17, Lemma 4]. □

Remark 21. (i) *Comparison.* Shen–Zhu–Zhu [2] prove a log-Sobolev inequality, hence a Poincaré inequality, for $\beta < 1/12$. Theorem B of [17] (v1, p. 2) gives the Bakry–Émery condition, with log-Sobolev and Poincaré inequalities uniform in the volume and the boundary condition, for $|\beta| < 1/(8d)$ in their normalisation, i.e. $\beta_W < 1/8$ for $SU(2)$ in $d = 4$ (without gauge fixing; in a line-field gauge, version 4.45.0 of [18] obtains the Bakry–Émery method up to $\beta_W < 4/29$, §8). Corollary 20 extends the range of the Poincaré inequality alone to $2/9$ (a

factor 8/3 over [2]); it says nothing about the log-Sobolev inequality beyond 1/8. The authors of [17] write that Dobrushin's condition, its Wasserstein form and its consequences for decay and for transport and Poincaré inequalities (citing, among others, [22]) "are classical" (v1, p. 19, footnote), without writing out the Poincaré inequality at $\beta_W < 2/9$ for this model, to our knowledge (literature search not exhaustive). (ii) *Infinite volume.* For f local and C^1, both sides of the inequality are continuous under weak convergence of μ, and the limit points of the finite-volume measures are Gibbs measures (proof of Proposition 3), hence equal the unique Gibbs measure of Theorem 1. That measure therefore satisfies the same inequality for local C^1 functions f (the sum over $\ell \in E$ has finitely many non-zero terms). (iii) *Numerical controls* ([numerical]; they check conventions and are not part of the proof). The one-site gap of q_κ, computed by Rayleigh–Ritz in the sectors $l = 0, 1, 2$ of spherical harmonics on S^2 and, independently, by a grid discretisation, is 3 at $\kappa = 0$ and 3.0083, 3.2337, 3.5164, 10.580 at $\kappa = \frac{1}{4}, \frac{4}{3}, 2, 10$. A Monte Carlo test on the 4^4 torus (1024 links, exact heat bath, four observables, $\beta \in \{0.05, 0.1, 0.15, 0.2\}$, $2 \cdot 10^4$ measurements each) satisfied both $(1 - \frac{9}{2}\beta) \operatorname{Var} f \le \mathbb{E} \sum_\ell \operatorname{Var}_\ell f$ and the Poincaré inequality with both constants in all 16 cases, within 3σ and with a wide margin.

7 Application: Wilson's area law for $\beta < 2/3$

The coefficient of Theorem 6 is not specific to uniqueness. It is applied here to the three-dimensional σ-models into which Durhuus and Fröhlich [24] decompose the four-dimensional theory, and gives Wilson's area law for $\beta = \beta_W < 2/3$. *This range is not new*: Borgs and Seiler [25] obtained confinement for $\beta_W < 12/17 \approx 0.706$ by a different argument (§7.2), and *we claim no priority on any threshold.* What this section offers is an independent proof by Dobrushin's comparison, valid on tori uniformly in the volume, in boxes with free boundary conditions and for every Gibbs measure on $\mathbb{Z}^4$, with explicit constants for every rectangular loop and the explicit lower bound $\operatorname{arcosh}(2/\beta - 2)$ for the string tension.

7.1 Statement

Measures. For $L \ge 2$ let $\mathbb{T}_L = (\mathbb{Z}/L\mathbb{Z})^4$ be the periodic torus and $\mathbb{B} = \prod_{\mu=1}^4 \{0, \dots, L_\mu\}$ a box. Links, plaquettes $p = (z; \mu, \nu)$, $\mu < \nu$, and $U_p = U_{z,\mu} U_{z+e_\mu,\nu} U^{-1}_{z+e_\nu,\mu} U^{-1}_{z,\nu}$ are as in §1; on $\mathbb{B}$ only the links and plaquettes inside the box are present (free boundary conditions). For $\Lambda \in \{\mathbb{T}_L, \mathbb{B}\}$,

$$\mu_\Lambda(dU) = Z^{-1} \exp\Big(\beta \sum_p \tfrac{1}{2} \operatorname{Re} \operatorname{Tr} U_p\Big) \prod_\ell dU_\ell, \tag{11}$$

the weight of (1) up to a constant factor. A Gibbs measure on $\mathbb{Z}^4$ is a probability measure μ with $\mu\gamma_\Delta = \mu$ for every finite set Δ of links (§1); one exists for every β (Proposition 3), and uniqueness is established in this note only for $\beta < 2/9$ (Theorem 1).

Wilson loops. For a closed lattice path γ, U_γ is the ordered product of its link variables and $W_\gamma = \frac{1}{2} \operatorname{Tr} U_\gamma \in [-1, 1]$. A *rectangular loop with sides R (along e_μ) and T (along e_ν)*, $\mu \ne \nu$, $R, T \ge 1$, is the boundary of a planar $R \times T$ rectangle of lattice squares; its area is RT.

Theorem 7.1 (Area law)**.** *Let $0 \le \beta < 2/3$ and $\lambda_*(\beta) := \operatorname{arcosh}(2/\beta - 2)$ for $\beta > 0$, $\lambda_*(0) := +\infty$. For $0 \le \lambda < \lambda_*(\beta)$ put*

$$\theta_\lambda := \frac{\beta}{4}\,(2\cosh\lambda + 4) \in [0, 1), \qquad C_\lambda := \frac{8\,\theta_\lambda}{1 - \theta_\lambda}. \tag{12}$$

Let γ be a rectangular loop with sides R along e_μ and T along e_ν. Then, for every $\lambda \in [0, \lambda_*(\beta))$,

$$|\langle W_\gamma \rangle| \;\le\; \min\Big\{ C_\lambda^T\, e^{-\lambda RT},\; \big(1 - e^{-12\beta}\big)^T \Big\} \tag{13}$$

in each of the following cases: (i) $\langle\cdot\rangle = \mu_{\mathbb{T}_L}$, $L \geq 2$, *provided* $R \leq L/2$ *and* $T \leq L-1$; (ii) $\langle\cdot\rangle = \mu_{\mathbb{B}}$ *for a box* $\mathbb{B}$ *containing* γ; (iii) $\langle\cdot\rangle = \mu$ *for any Gibbs measure* μ *on* $\mathbb{Z}^4$. *The two sides may be exchanged. The constants depend on* β *and* λ *only.*

Status: [proved] (§§7.3–7.5 and Appendix B), from Theorem 6 (through Corollary 14(b)); not formalised in Lean.

Corollary 7.2 (String tension and uniform area law)**.** *Let* $0 < \beta < 2/3$. *In each case of Theorem 7.1:* (a) $\liminf_{R,T\to\infty} \dfrac{-\log|\langle W_{R\times T}\rangle|}{RT} \geq \lambda_*(\beta) = \operatorname{arcosh}\left(\dfrac{2}{\beta} - 2\right)$, *and on tori the bound is uniform in* $L \geq \max(2R, T+1)$; (b) *with* $\lambda_\beta = \lambda_*(\beta)/2$, $R_\beta = \max\{1, \lceil 2\log^+ C_{\lambda_\beta}/\lambda_\beta\rceil\}$ *and* $c(\beta) = \min\{\lambda_\beta/2,\ |\log(1-e^{-12\beta})|/R_\beta\} > 0$, *one has* $|\langle W_\gamma\rangle| \leq e^{-c(\beta)RT}$ *for every rectangular loop* γ.

Proof. (a) By (13), $-\log|\langle W\rangle|/(RT) \geq \lambda - \log C_\lambda/R \to \lambda$ for every $\lambda < \lambda_*(\beta)$. (b) If $R \geq R_\beta$, then $\lambda_\beta R/2 \geq \log^+ C_{\lambda_\beta}$ and the first bound in (13) is at most $e^{-\lambda_\beta RT/2}$. If $R < R_\beta$, the second bound is $e^{-|\log(1-e^{-12\beta})|T} \leq e^{-|\log(1-e^{-12\beta})|RT/R_\beta}$. □

Remark 7.3 (Behaviour of the bound). $\lambda_*(\beta) = \log(4/\beta) - \beta + O(\beta^2)$ as $\beta \to 0$, and $\lambda_*(\beta) \sim 3\sqrt{2/3-\beta}$ as $\beta \uparrow 2/3$. The strong-coupling expansion of the string tension is $\sigma = -\log u - 4u^4 + O(u^8)$ with $u = I_2(\beta)/I_1(\beta) = \beta/4 + O(\beta^3)$ [32, Table 1] [read], so the lower bound has the exact leading behaviour $\log(4/\beta)$ at strong coupling, and $\sigma - \lambda_*(\beta) = \beta + O(\beta^2)$.

7.2 Relation to earlier work

Ranges are in β_W (Table 1); they are collected in Table 5.

Borgs and Seiler. Borgs and Seiler [25] use the slab decomposition of [24] to reduce confinement to the requirement that "the u-spins show exponential clustering uniformly in the random coupling field v" (preprint p. 20), and verify it, in Appendix B of the preprint version, by a random-walk expansion together with operator inequalities for the transfer matrix; no Dobrushin criterion is involved. They state that "irrespective of J_M and temperature we are sure to have confinement for $J_E < \tilde{J}_c$", with $\tilde{J}_c = (d - \frac{1}{6})^{-1}$ for $SU(2)$ (preprint (II.55), p. 21; proof pp. B1–B6), d being the space dimension and J_E the coefficient of $\mathrm{Re}\,\mathrm{Tr}$ in the action (their (II.2)–(II.5), pp. 4–5). In the normalisation of this note $\beta_W = 2J_E$, so for $d = 3$ their range is $\beta_W < 12/17 \approx 0.706$, which is larger than $2/3$ [read]. Thus [25, Appendix B of the preprint version] proves confinement in a larger range than Theorem 7.1: *the threshold* $2/3$ *is not new, and no priority is claimed for it or for any other threshold.* We read the preprint MPI-PAE/PTh 16/83 (March 1983); in v1.6 we compared the journal version, Comm. Math. Phys. 91 (1983) 329–380, with it on the passages cited ([read] on page images): same text and equation numbers; (II.55) is on p. 343 and Appendix B on pp. 373–377 of the journal version, and Crossref lists no erratum (as of 11 November 2026; Crossref lists only errata that were deposited). We did not find in the preprint a statement for individual Wilson loops, nor one for arbitrary Gibbs measures.

Durhuus and Fröhlich. They introduced the slab decomposition (their (1.17)–(1.19)) and the clustering criterion [24]. They prove confinement for $\beta < 1/(2(\nu-1))$ and state, in a parenthesis closing the appendix to their §5, that "a more refined estimate extends this result to $\beta < 1/(\nu-1)$" (preprint p. 80), without giving it; in their normalisation (1.13) (plaquettes counted once, $\beta_W = 2\beta$) the announced range is $\beta_W < 2/3$ for $\nu = 4$ [read]. One identity of their derivation, $\vec{S}_1 \cdot \vec{S}_2 = \mathrm{tr}(\varphi(\vec{S}_1)^{-1}\varphi(\vec{S}_2))$ (preprint p. 48; their φ is our ι), holds only up to a factor 2 (see (14)); with this correction the range covered by their written argument is $\beta_W < 1/6$ in the normalisation of this note.

Other ranges. Osterwalder and Seiler [3, Thm. 5.1] prove the area law for planar loops in the region of convergence of the cluster expansion (small β, no explicit range) [read]. Fröhlich [26, (21)–(25)] compares by Ginibre inequalities the $SU(2)$ theory with the $\mathbb{Z}_2$ gauge theory and the latter with decoupled Ising slabs [read]; with the sharpness of the Ising transition [27] this should give an area law up to $\beta_W \approx K_c = 0.221654626(5)$, the critical coupling of the three-dimensional Ising model [28] (up to $K_c/2 \approx 0.1108$ with the constant of [26, (23)] as printed). We have not seen this combination written out and have not checked its uniformity in the volume [not checked]. Cao, Nissim and Sheffield [5] verify the criterion of [24] by the Bakry–Émery condition and obtain the area law on tori, uniformly in the volume, for $\beta_{\mathrm{tH}} < 1/(8(d-1))$, i.e. $\beta_W < 1/6$ [read]. Brydges, Giffen, Durhuus and Fröhlich [30] (surface representations of Wilson loop expectations; only the abstract and the metadata were seen) and Seiler's monograph [4] were not consulted.

What is claimed. Theorem 7.1 is an independent proof, by Dobrushin's comparison, of the area law in a range already covered by [25]. It holds on tori uniformly in the volume, with free boundary conditions and for every Gibbs measure on $\mathbb{Z}^4$, with explicit constants for every rectangular loop, and gives the string-tension bound $\operatorname{arcosh}(2/\beta - 2)$. Its ingredients are the slab decomposition [24, 5], the exact one-link coefficient (Theorem 6, whose bound $\frac{1}{4}$ for all fields is also [17, Theorem 7]), Dobrushin's comparison, a count of walks and an elementary Doeblin bound; no technical novelty beyond their combination is claimed.

7.3 The slab σ-models

Quaternions. As in §3, $SU(2)$ is identified with $S^3 \subset \mathbb{R}^4$, here through the map of Durhuus and Fröhlich [24, (5.1)]

$$\iota(q) = \begin{pmatrix} q^0 + iq^3 & -q^1 + iq^2 \\ q^1 + iq^2 & q^0 - iq^3 \end{pmatrix}, \qquad q \in \mathbb{R}^4,$$

which maps S^3 onto $SU(2)$ and the normalised surface measure $\bar\sigma$ onto Haar measure. For $q, r \in S^3$ and $h \in \mathbb{R}^4$,

$$\operatorname{Tr}\left(\iota(q)^{-1}\iota(r)\right) = 2\langle q, r\rangle, \qquad \|\iota(h)\|_{\mathrm{op}} = |h| \tag{14}$$

(since $\iota(h)^*\iota(h) = |h|^2 I$), so that $\rho_{\mathrm{ch}}(q,r) = \|\iota(q) - \iota(r)\|_{\mathrm{op}}$; and for $A, B \in SU(2)$ the map $r \mapsto \iota^{-1}(A\,\iota(r)B^{-1})$ is the restriction to S^3 of a matrix $O_{A,B} \in SO(4)$ (left and right multiplications by unit quaternions). [proved]; checked numerically (`s1_normalisations.py`, items (a)–(c), folder `verif_loi_aire/`).

Twisted $O(4)$ models. A *bond* is a triple $b = (b_-, b_+, O_b)$ with $b_\pm$ sites and $O_b \in SO(4)$; several bonds may join the same two sites. Let V (free sites) and F (frozen sites) be finite disjoint sets, $\mathfrak{B}$ a finite family of bonds with $b_\pm \in V \cup F$ and at least one end in V, $\beta \geq 0$ and $\xi \in (S^3)^F$. The *twisted $O(4)$ model* is the probability measure on $(S^3)^V$

$$\mu^\xi(dq) = \frac{1}{Z} \exp\Big(\beta \sum_{b \in \mathfrak{B}} \langle q_{b_-}, O_b q_{b_+}\rangle\Big) \prod_{x \in V} \bar\sigma(dq_x), \qquad q_z := \xi_z \ (z \in F). \tag{15}$$

Let $\deg(x)$ be the number of bonds with an end at x and m_{xz} the number of bonds joining x and z. The substitution $(q, \xi) \mapsto (-q, -\xi)$ leaves every term of the exponent and $\bar\sigma$ unchanged (this is the action of the centre $\{\pm I\}$ of $SU(2)$), hence

$$\text{the image of } \mu^\xi \text{ under } q \mapsto -q \text{ is } \mu^{-\xi}; \quad \text{if } F = \emptyset,\ \mathbb{E}_{\mu^\emptyset}[\psi(q_x)] = 0 \text{ for every bounded odd } \psi. \tag{16}$$

Lemma 7.4 (Slab decomposition; [24, (1.17)–(1.19)]). *Let $\mu = \mu_\Lambda$ with $\Lambda = \mathbb{T}_L$ or a box $\mathbb{B}$, and let $\mathcal{H}$ be the σ-algebra generated by the* horizontal *links, those in the directions $1, 2, 3$. For a slab k ($x_4 \in [k, k+1]$) write $Q^k_x = U_{(x,k),4}$ and $q^k_x = \iota^{-1}(Q^k_x)$, x in the spatial lattice $\Lambda^{(3)}$ ($(\mathbb{Z}/L\mathbb{Z})^3$, or a box of $\mathbb{Z}^3$). Conditionally on $\mathcal{H}$ the families $(q^k_x)_x$, one for each slab, are independent, and $(q^k_x)_x$ has the law (15) with $V = \Lambda^{(3)}$, $F = \emptyset$ and one bond $(x, x + e_\mu, O_{A,B})$ for each vertical plaquette $p = ((x,k);\mu,4)$, $\mu \le 3$, where $A = U_{(x,k),\mu}$ and $B = U_{(x,k+1),\mu}$. On the torus $\deg(x) = 6$ for every x; in a box $\deg(x) \le 6$.*

Proof. For $p = ((x,k);\mu,4)$, $U_p = A\, Q^k_{x+e_\mu} B^{-1} (Q^k_x)^{-1}$, so by (14)

$$\tfrac{1}{2} \operatorname{Re} \operatorname{Tr} U_p = \tfrac{1}{2} \operatorname{Tr} \left(\iota(q^k_x)^{-1} \iota(O_{A,B} q^k_{x+e_\mu})\right) = \langle q^k_x, O_{A,B}\, q^k_{x+e_\mu} \rangle .$$

A horizontal plaquette contains no vertical link, and a vertical plaquette contains two vertical links of the same slab. So the density of μ_Λ is $e^{\beta h(\mathcal{H})} \prod_k e^{\beta S_k}$, h collecting the horizontal plaquettes and S_k being the exponent of (15) for slab k; Fubini gives the claim. On the torus x lies in the six bonds $(x, x \pm e_\mu)$, $\mu \le 3$ (for $L = 2$ two of them join the same sites). □

Single-site laws and influence. For $x \in V$ and a configuration η of the other sites (frozen ones included), the conditional law of q_x under (15) is

$$\gamma_x(du \mid \eta) = \nu_{\beta s_x(\eta)}(du), \qquad s_x(\eta) = \sum_{b:\, b_- = x} O_b \eta_{b_+} + \sum_{b:\, b_+ = x} O_b^{\mathsf{T}} \eta_{b_-}, \quad |s_x(\eta)| \le \deg(x), \tag{17}$$

where $\nu_h(du) \propto e^{\langle u, h\rangle} \bar\sigma(du)$ is the law of Corollary 14(b). It is the same law as the single-link conditional law (2) of the Wilson measure: the six plaquettes through a vertical link all lie in its slab. Let Lip_1 be the set of real functions on S^3 that are 1-Lipschitz for ρ_{ch}. For $z \in (V \cup F) \setminus \{x\}$ put

$$c_{xz} = \sup \Big\{ \frac{\left| \int f\, d\gamma_x(\cdot \mid \eta) - \int f\, d\gamma_x(\cdot \mid \eta') \right|}{\rho_{\mathrm{ch}}(\eta_z, \eta'_z)} : \ f \in \mathrm{Lip}_1,\ \eta, \eta' \text{ equal off } z,\ \eta_z \ne \eta'_z \Big\}; \tag{18}$$

by Kantorovich–Rubinstein duality this is Dobrushin's coefficient (5.4) of [1] for the chordal metric (as in Proposition 5), but only the form (18) is used below.

Lemma 7.5 (Influence). *For the model (15), $c_{xz} \le \frac{\beta}{4}\, m_{xz}$; hence $\sum_z c_{xz} \le \frac{\beta}{4} \deg(x) \le \frac{3\beta}{2}$.*

Proof. Let η, η' differ only at z, $s = s_x(\eta)$, $s' = s_x(\eta')$. Then $s' - s = \sum_b \tilde O_b(\eta'_z - \eta_z)$, the sum over the m_{xz} bonds joining x and z, with $\tilde O_b \in \{O_b, O_b^{\mathsf{T}}\}$ orthogonal, so $|s' - s| \le m_{xz}\, \rho_{\mathrm{ch}}(\eta_z, \eta'_z)$. In the proof of Corollary 14(b), Theorem 6 gives $\int f\, d(\nu_{b'} - \nu_b) \le \frac{1}{4}|b' - b|$ for $f \in \mathrm{Lip}_1$ and all $b, b' \in \mathbb{R}^4$ (before the duality is used). With $b = \beta s$, $b' = \beta s'$ and $f \mapsto -f$ this gives $\left| \int f\, d\gamma_x(\cdot \mid \eta') - \int f\, d\gamma_x(\cdot \mid \eta) \right| \le \frac{\beta}{4}|s' - s|$. □

Remark 7.6 (The threshold $2/3$ is the limit of the criterion). Let $L \ge 3$ (or work on $\mathbb{Z}^3$), fix the twists, a site x, a neighbour z and a unit vector v. Choose η_z with $w := \tilde O \eta_z \perp v$ ($\tilde O$ the orthogonal map of the bond xz) and the five other neighbours with $\sum \tilde O_b \eta_b = -w$ (five unit vectors can sum to any vector of norm ≤ 5), so that $s_x = 0$. The lower-bound argument of Proposition 5 gives $c_{xz} \ge \beta D_{\mathrm{ch}}(0) = \beta/4$. So the Dobrushin sum of the slab is exactly $6 \cdot \beta/4 = 3\beta/2$, and the single-site criterion holds if and only if $\beta < 2/3$. Moreover, for trivial horizontal links the slab is the untwisted $O(4)$ model on $\mathbb{Z}^3$ with coupling $J = \beta$, whose critical coupling is $J_c = 0.9360(1)$ [31] [read]: no decay in the slab that is uniform in the twists can hold beyond it, and $2/3$ is the mean-field point $Jz/N = 1$ ($z = 6$, $N = 4$).

7.4 Decay of correlations in a slab

Let μ^ξ be a twisted model, $C = (c_{xx'})_{x,x' \in V}$ with $c_{xx} := 0$, and assume Dobrushin's condition

$$\alpha := \max_{x \in V} \sum_{z \in (V \cup F) \setminus \{x\}} c_{xz} < 1, \qquad D := \sum_{k \geq 0} C^k \tag{19}$$

(D converges: the row sums of C^k are $\leq \alpha^k$). For $G : (S^3)^V \to \mathbb{R}$ let $\delta_x(G) = \sup\{|G(q) - G(q')|/\rho_{\mathrm{ch}}(q_x, q'_x) : \ q, q' \text{ equal off } x, \ q_x \neq q'_x\}$, and let $\mathcal{L}$ be the set of G with $\delta_x(G) < \infty$ for all $x \in V$ (such G are continuous).

Proposition 7.7 (Comparison)**.** *Under* (19)*, for $G \in \mathcal{L}$ and $\xi, \xi' \in (S^3)^F$,*

$$|\mu^\xi(G) - \mu^{\xi'}(G)| \ \leq \ \sum_{x,x' \in V} \ \sum_{z \in F} \delta_x(G)\, D_{xx'}\, c_{x'z}\, \rho_{\mathrm{ch}}(\xi_z, \xi'_z). \tag{20}$$

This is a Wasserstein form of Dobrushin's comparison theorem (see [29] for a covariance estimate under Dobrushin's condition; the statements there were not checked). A complete proof, in the spirit of [22, Prop. 2.5] [read], is given in Appendix B. [proved]

Corollary 7.8 (Two-point bounds)**.** *Assume* (19)*. For $x \neq y$ in V put $G_{xy} = \sum_{k \geq 1} (C^k)_{xy}$. Let $\psi : S^3 \to \mathbb{C}$ be bounded and measurable, and $\chi : S^3 \to \mathbb{C}$ with $|\chi(u) - \chi(u')| \leq \rho_{\mathrm{ch}}(u, u')$. Then*

$$|\operatorname{Cov}_{\mu^\xi}(\chi(q_x), \psi(q_y))| \leq 2\, \|\psi\|_\infty\, G_{xy}, \tag{21}$$

$$|\mathbb{E}_{\mu^\xi} \chi(q_x) - \mathbb{E}_{\mu^{\xi'}} \chi(q_x)| \leq \sum_{z \in F} \rho_{\mathrm{ch}}(\xi_z, \xi'_z) \sum_{k \geq 1} (\bar{C}^k)_{xz}, \tag{22}$$

where $\bar{C} = (c_{uv})$ is indexed by $V \cup F$, with zero rows at the frozen sites.

Proof. Let χ be real. Given $q_y = a$, the law of $q_{V \setminus \{y\}}$ under μ^ξ is the model with free set $V' = V \setminus \{y\}$ and frozen data (ξ, a), with the same coefficients c (so (19) holds). Proposition 7.7 with $G = \chi(q_x)$ gives $|\mathbb{E}[\chi(q_x) \mid q_y = a] - \mathbb{E}[\chi(q_x) \mid q_y = a']| \leq \rho_{\mathrm{ch}}(a, a') \sum_{k \geq 0} (C^k_{V'} C_{V',y})_x \leq \rho_{\mathrm{ch}}(a, a')\, G_{xy}$, since every product of coefficients along a path in V' followed by one step to y is a term of $(C^{k+1})_{xy}$. As $\rho_{\mathrm{ch}} \leq 2$, $\operatorname{Cov}(\chi(q_x), \psi(q_y)) = \mathbb{E}[(\mathbb{E}[\chi(q_x) \mid q_y] - \mathbb{E}\chi(q_x))\psi(q_y)]$ is bounded by $2G_{xy}\|\psi\|_\infty$. Inequality (22) is (20) for $G = \chi(q_x)$, because $\sum_{x'} D_{xx'} c_{x'z} = \sum_{k \geq 1} (\bar{C}^k)_{xz}$. A complex χ is handled through $\mathrm{Re}(e^{-i\vartheta}\chi)$, ϑ real. □

Lemma 7.9 (Tilted walk count)**.** *Let $x \in V$ and $y = x + Re_1$, $R \geq 1$, in a model of Lemma 7.4 on $(\mathbb{Z}/L\mathbb{Z})^3$, with $d = \min_{j \in \mathbb{Z}} |R + jL|$, or in a model on a subset $V \subset \mathbb{Z}^3$ with nearest-neighbour bonds, with $d = R$. Let $N_k(x, y)$ be the number of walks of k bonds from x to y through sites of V. Then, for every $\lambda \geq 0$ and $k \geq 1$, $N_k(x, y) \leq 2e^{-\lambda d}(2\cosh\lambda + 4)^k$. Consequently, by Lemma 7.5 ($(C^k)_{xy} \leq (\beta/4)^k N_k(x, y)$), if $\theta_\lambda < 1$,*

$$G_{xy} \ \leq \ \sum_{k \geq 1} \left(\tfrac{\beta}{4}\right)^k N_k(x, y) \ \leq \ 2e^{-\lambda d}\, \frac{\theta_\lambda}{1 - \theta_\lambda}.$$

Proof. A walk is a sequence of steps $\varepsilon_1, \dots, \varepsilon_k \in \{\pm e_1, \pm e_2, \pm e_3\}$ (on the torus with $L = 2$ the two bonds joining x and $x + e_\mu$ are the steps $\pm e_\mu$). Let $D = \sum_s \langle \varepsilon_s, e_1 \rangle \in \mathbb{Z}$. A walk from x to y has $D \in R + L\mathbb{Z}$ in the torus case and $D = R$ in the other; in both cases $|D| \geq d$, so its indicator is at most $e^{\lambda(|D| - d)} \leq e^{-\lambda d}(e^{\lambda D} + e^{-\lambda D})$. Summing over the 6^k step sequences, $\sum e^{\pm \lambda D} = (e^\lambda + e^{-\lambda} + 4)^k$. □

7.5 Proof of Theorem 7.1

By the symmetry of the Wilson specification under permutations of the axes we may take $\nu = 4$ (the side of length T is vertical) and $\mu = 1$. Let the loop start at (x, k_0) and put $y = x + Re_1$. Then

$$U_\gamma = H_0\, Q_y^{k_0}\, Q_y^{k_0+1} \cdots Q_y^{k_0+T-1}\, H_T^{-1}\, (Q_x^{k_0+T-1})^{-1} \cdots (Q_x^{k_0})^{-1}, \tag{23}$$

where H_0 (resp. H_T) is the product of the R horizontal links from (x, k_0) to (y, k_0) (resp. at time $k_0 + T$). On the torus, $T \le L - 1$ makes the slabs $k_0, \dots, k_0 + T - 1$ distinct, and $R \le L/2$ gives $d = R$ in Lemma 7.9.

Factorisation. For a twisted model ν_k define the linear map $\mathcal{M}_k(X) = \mathbb{E}_{\nu_k}[\iota(q_y)\, X\, \iota(q_x)^{-1}]$ on 2×2 complex matrices. If the slabs are independent with laws ν_k and H_0, H_T are fixed, (23) gives, by expanding the products entrywise and using independence,

$$\mathbb{E}\big[\tfrac{1}{2}\operatorname{Tr} U_\gamma\big] = \tfrac{1}{2}\operatorname{Tr}\Big(H_0\, \mathcal{M}_{k_0}\big(\mathcal{M}_{k_0+1}(\cdots \mathcal{M}_{k_0+T-1}(H_T^{-1})\cdots)\big)\Big). \tag{24}$$

Lemma 7.10. *Let ν be a twisted model satisfying (19), $x \neq y$ in V, and suppose $|\mathbb{E}_\nu \psi(q_x)| \le \epsilon$ and $|\mathbb{E}_\nu \psi(q_y)| \le \epsilon$ for all functions $\psi(u) = a^* \iota(u) b$ and $\psi(u) = a^* \iota(u)^{-1} b$, $a, b \in \mathbb{C}^2$ unit vectors. Then $\|\mathcal{M}(X)\|_{\mathrm{op}} \le 2(2G_{xy} + \epsilon^2)\|X\|_{\mathrm{op}}$.*

Proof. Write $X = \sum_{r=1}^{2} s_r p_r w_r^*$ (singular values), so $\sum_r s_r \le 2\|X\|_{\mathrm{op}}$. For unit a, b, $a^* \mathcal{M}(X) b = \sum_r s_r \mathbb{E}[\psi_r(q_y)\chi_r(q_x)]$ with $\psi_r(u) = a^* \iota(u) p_r$, $\chi_r(u) = w_r^* \iota(u)^{-1} b$. Both are bounded by 1 and 1-Lipschitz for ρ_{ch}, because $\|\iota(u) - \iota(u')\|_{\mathrm{op}} = |u - u'|$ and $\iota(u)^{-1} = \iota(\bar{u})$ with $u \mapsto \bar{u}$ an isometry. By (21), $|\mathbb{E}[\psi_r(q_y)\chi_r(q_x)]| \le |\operatorname{Cov}(\chi_r(q_x), \psi_r(q_y))| + \epsilon^2 \le 2G_{xy} + \epsilon^2$. Hence $|a^* \mathcal{M}(X) b| \le (2G_{xy} + \epsilon^2) \sum_r s_r$. □

Case (i): torus. By Lemma 7.4, conditionally on $\mathcal{H}$ the slabs are independent twisted models on $(\mathbb{Z}/L\mathbb{Z})^3$ with $F = \emptyset$; by Lemma 7.5, (19) holds with $\alpha \le 3\beta/2 < 1$ whatever the twists; by (16), $\epsilon = 0$ in Lemma 7.10 (the functions ψ are odd). Hence, by (24), Lemma 7.10 and Lemma 7.9,

$$|\mathbb{E}[W_\gamma \mid \mathcal{H}]| \le \big\|\mathcal{M}_{k_0} \circ \cdots \circ \mathcal{M}_{k_0+T-1}(H_T^{-1})\big\|_{\mathrm{op}} \le (4G_{xy})^T \le \Big(\frac{8\theta_\lambda}{1-\theta_\lambda}\Big)^T e^{-\lambda R T},$$

using $|\frac{1}{2}\operatorname{Tr}(H_0 Y)| \le \|Y\|_{\mathrm{op}}$ for $H_0 \in SU(2)$. Taking expectations gives the first bound in (13).

Case (ii): free boundary conditions. Identical, with Lemma 7.9 on a box of $\mathbb{Z}^3$, $F = \emptyset$ and $\deg \le 6$.

Case (iii): Gibbs measures. Let μ be a Gibbs measure, $n \ge R$, $C_n = \{x' \in \mathbb{Z}^3 : \|x' - x\|_\infty \le n\}$ and Δ_n the set of vertical links $((x', k), 4)$ with $x' \in C_n$, $k_0 \le k \le k_0 + T - 1$. Then $\mu(W_\gamma) = \mu(\gamma_{\Delta_n} W_\gamma)$. The plaquettes meeting Δ_n are vertical plaquettes of the slabs $k_0, \dots, k_0 + T - 1$, so $\gamma_{\Delta_n}(\cdot \mid \omega)$ is the product over these slabs of twisted models (15) with $V = C_n$ and $F = \partial C_n$ (the outer boundary), with twists and frozen spins given by ω. For each slab, (19) holds with $\alpha \le 3\beta/2$ and $G_{xy} \le 2e^{-\lambda R}\theta_\lambda/(1-\theta_\lambda)$ (Lemmas 7.5 and 7.9). For ψ as in Lemma 7.10, (16) gives $\mathbb{E}_{\mu^\xi}\psi(q_x) = \frac{1}{2}(\mathbb{E}_{\mu^\xi}\psi(q_x) - \mathbb{E}_{\mu^{-\xi}}\psi(q_x))$ and, by (22) with $\xi' = -\xi$ and $\rho_{\mathrm{ch}}(\xi_z, -\xi_z) = 2$,

$$|\mathbb{E}_{\mu^\xi}\psi(q_x)| \le \sum_{z \in F}\sum_{k \ge 1} (\bar{C}^k)_{xz} \le \sum_{k \ge n+1-R} \Big(\frac{3\beta}{2}\Big)^k =: \epsilon_n,$$

because a walk from x (or from y) to ∂C_n has at least $n + 1 - R$ bonds and the row sums of $\bar{C}$ are $\le 3\beta/2$; the same bound holds for $\psi(q_y)$. By (24) and Lemma 7.10, $|\gamma_{\Delta_n} W_\gamma(\omega)| \le \big(2(2G_{xy} + \epsilon_n^2)\big)^T$ for every ω. Letting $n \to \infty$ ($\epsilon_n \to 0$) gives the first bound in (13). No uniqueness of μ is used.

The second bound in (13). In all three cases the links Q_y^k, $k_0 \le k \le k_0 + T - 1$, share no plaquette (a plaquette contains two vertical links of one slab, at different spatial sites). Under the Wilson specification, their conditional law given all the other links is therefore the product of single-link laws (17) with $|s| \le 6$, whose densities with respect to $\bar\sigma$ are $\ge e^{-6\beta}/e^{6\beta} = e^{-12\beta}$. Write such a law as $e^{-12\beta}\bar\sigma + (1-e^{-12\beta})\nu'$, with ν' a probability measure; since $\int \iota(u)\,\bar\sigma(du) = 0$ and $\|\iota(u)\|_{\mathrm{op}} = 1$, $\|\mathbb{E}[Q_y^k \mid \mathrm{rest}]\|_{\mathrm{op}} \le 1 - e^{-12\beta}$. Conditioning (23) on all links but the Q_y^k gives $|\mathbb{E}[W_\gamma \mid \mathrm{rest}]| \le \|\prod_k \mathbb{E}[Q_y^k \mid \mathrm{rest}]\|_{\mathrm{op}} \le (1-e^{-12\beta})^T$. (For the torus and the box this conditional law follows from (11) by Fubini; for Gibbs measures it is the DLR equation for the finite set of these T links.) □

7.6 Scope and checks

Only rectangular planar loops are treated. For $2/9 \le \beta < 2/3$ this note does not prove uniqueness of the Gibbs measure (Rajabu [20, Theorem 9.1(i)] states that the Wilson specification on $\mathbb{Z}^4$ has exactly one Gibbs measure for $0 < \beta_W \le 0.511$; we have not checked that proof, §8); Theorem 7.1(iii) holds for each Gibbs measure. The walk count of Lemma 7.9 is sharp for the walk sum: a Fourier computation of $\sum_k (\beta/4)^k N_k$ on $(\mathbb{Z}/48\mathbb{Z})^3$ gives the decay rate $\lambda_*(\beta)$ within 1% at $\beta \in \{0.5, 0.6, 0.65\}$ [numerical] (`s3_marches.py`, folder `verif_loi_aire/`, where the 840 exact walk counts of Lemma 7.9 on $(\mathbb{Z}/L\mathbb{Z})^3$, $L \le 5$, $k \le 12$, are also checked), so any improvement of the rate would have to come from the coefficients c_{xz}, not from the counting. The identity $\frac{1}{2}\operatorname{Re}\operatorname{Tr} U_p = \langle q_x, O_{A,B} q_{x+e_\mu}\rangle$ and the count of 6 neighbours (Lemma 7.4) are checked numerically on a 3^4 torus (`s1_normalisations.py`, items (c), (e)). These checks are not part of any proof.

8 Relation to the literature

Bhattacharjee–Bhattacharya(–Lockwood) [17] (added in v1.4, updated here). After v1.3 of this note (28 September 2026, not publicly released) was completed, we learned of the public preprint [17], dated 18 October 2026. *Versions and numbering.* The Hexagon record has four versions (dates of the record: 18, 22, 28 and 30 October 2026; the PDFs are dated 18, 20, 27 and 30 October); its web pages show the title of the latest one for all four, and the titles below are those of the PDFs. Version v1 (29 pages, two authors) is *Mass Gap of Lattice Yang–Mills Theory up to the Dobrushin Ceiling*; v2 (28 pages) is *Mass Gap of Four-Dimensional Lattice Yang–Mills Theory in a Forest Gauge*, which cites the all-field estimate below as Theorem 7 of the Dobrushin-ceiling paper (its reference [9], GitHub tag v2.0.2, Zenodo record doi:10.5281/zenodo.23449466, whose version string is 2.0.0); v3 (54 pages) and v4 (70 pages), with a third author, J. M. Lockwood, contain its proof in an appendix, as Theorem 91 (v3, p. 47) and Theorem 125 (v4, p. 64, Appendix I). The Zenodo record 10.5281/zenodo.23786523 (30 October; all versions: 10.5281/zenodo.23444338) contains the v1 paper (Theorem 7, p. 9) and the v4 paper (Theorem 125). Below, "Theorem 7 (= 125)" means Theorem 7 of v1 and Theorem 125 of v4; likewise Lemmas 4, 5, 6 of v1 are Lemmas 122, 123, 124 of v4 (p. 63).

What [17] proves. Its Theorem E (v1, p. 2) reads (text of the PDF): "Let $G = SU(2)$ and $|\beta| < 1/(6(d-1))$. The Wilson theory on $\mathbb{Z}^d$ has exactly one infinite-volume Gibbs measure, and its truncated correlations of local Lipschitz functions decay exponentially with every rate $\lambda < -\log 6(d-1)|\beta|$. At $d = 4$ the range is $\beta_W < 2/9$. No single-link Wasserstein–Dobrushin argument, in the chordal or the geodesic metric, reaches beyond this range." with $\beta_W = N^2\beta$ (their footnote 4, the normalisation of [2]); its Theorem F adds a spectral gap of the transfer matrix in this range, and its Theorem 7 (p. 9) states that for every $c \in \mathbb{R}^4$ and unit $h \in \mathbb{R}^4$, "$L(c,h) \le \frac{1}{4}$, with equality at $c = 0$. Equivalently, $W_1(\nu_b, \nu_{b'}) \le \frac{1}{4}|b - b'|$ for all $b, b' \in \mathbb{R}^4$, and the constant $\frac{1}{4}$ is the best possible." [read]. (Here $L(c,h) = \sup_g \operatorname{Cov}_{\nu_c}(g, \langle h, x\rangle)$ over chordally 1-Lipschitz g, i.e. $L(\kappa e_1, h) = D_{\mathrm{ch}}(\kappa, h)$.) To our knowledge [17] is the first public statement

of the range of our Theorem 1, with stronger conclusions, and of the maximum-at-zero-field property of Levin's Remark 3.7 for $N = 4$ and *all* fields.

What this note does not claim. The following are in [17] (numbers of v1, with those of v4 after "=" when they differ): the uniqueness range $\beta_W < 2/9$ (Theorem E); the bound $D_{\mathrm{ch}}(\kappa) \le \frac{1}{4}$ for every κ (Theorem 7 = 125), hence Levin's maximum-at-zero-field property for $N = 4$; the reflection computation $L(c,h) = \mathbb{E}x_2^2 = A/\kappa$ for $h \perp c$, with $A = I_2/I_1$ (Lemma 6(ii) = 124(ii); our Lemma 7); the monotonicity $A' \le A/\kappa$ of the transverse moment (Lemma 5(iii) = 123(iii)), i.e. the non-strict decrease of $\kappa \mapsto I_2/(\kappa I_1)$; the exactness of the influence coefficient, $\sup C_{ee'} = |\beta|$ for every β, i.e. $\beta_W/4$ in our normalisation (Proposition 13 with Theorem 7; our Corollary 14(a)); and the limit $\beta_W = 1/8$ of the Bakry–Émery curvature method (Theorem C, p. 2; our Proposition 22, which we had written before seeing [17]). We make *no priority claim* on any of these.

What is new here, relative to [17] (for $\kappa \le 4$ see the comparison with Rajabu below). (i) The exact value of the supremum over directions, $\sup_v D_{\mathrm{ch}}(\kappa, v) = I_2(\kappa)/(\kappa I_1(\kappa))$ for every $\kappa \ge 0$: [17] proves "$\le \frac{1}{4}$"; for $v \perp \hat{s}$ it computes the value $I_2/(\kappa I_1)$ by reflection, but does not show that no other direction does better. (ii) The maximisers: for $\kappa > 0$, exactly $v \perp \hat{s}$. (iii) The two-sided estimate (3) for oblique directions. (iv) The strict decrease of $\kappa \mapsto I_2/(\kappa I_1)$ and, with (i), of D_{ch}. (v) The proof: a divergence-free flux field with a slice estimate and an exact sum-of-squares identity, in place of harmonic extension, Cauchy–Schwarz and the positivity of Bessel-series coefficients. For oblique directions the majorant that the proof of [17] gives is close to $\frac{1}{4}$ without being exact: at $\kappa = 4/3$ we recompute it from their formulas as 0.2498, against the exact $D_{\mathrm{ch}}(4/3) = 0.2333$ [numerical]. Independently of [17], this note also provides: the exact geodesic coefficient $8\beta/(9\pi)$ for $\kappa \le 3\pi/8$ (Theorem 17) and the range $\pi/16$; two interval-arithmetic certificates, each recertified by an independent implementation; the Lean-checked final steps of Appendix A and §5.2; and the transfers of the published constants of Levin, Royer, Gross, Simon and Shen–Zhu–Zhu to the Wilson link (Table 5).

Concerning the geodesic metric, [17] notes (footnote to Theorem 7, v1, p. 9) that the quarter fails in that metric already at $c = 0$, because $\arcsin x_2$ is geodesically 1-Lipschitz with covariance $8/(9\pi) \approx 0.283 > \frac{1}{4}$ with x_2 under the uniform law, and (Proposition 13) that no single-link argument in either metric goes beyond 2/9. That covariance is the lower bound of our Lemma 18; the field-uniform upper bound of Theorem 17 and the exact value $8\beta/(9\pi)$ of the geodesic influence coefficient for $\kappa \le 3\pi/8$ are not in [17], to our knowledge. We have not checked the proofs of [17] in full: we recomputed the steps of the proof of Theorem 7 and confirmed its statement numerically by optimal transport up to $\kappa = 50$ ([numerical]), but nothing here depends on that check. Versions v2–v4 of the preprint treat a different object (gauge-invariant observables and free-boundary limits in a gauge-fixed theory, with the ranges $\beta_W < 1/3$ and $\beta_W \le 0.345$ for $SU(2)$ in the abstract of v4), which we do not compare with Theorem 1, whose statement is DLR uniqueness for the Wilson specification. Nissim [21] treats $U(N)$ at sufficiently small 't Hooft coupling; no numerical $SU(2)$ threshold is stated there.

Later versions of [17] (added in v1.6). The dates below are the `created` timestamps of the Zenodo records (UTC); they do not establish first public availability. The 18 October 2026 date for Hexagon preprint `2610.00341v1` is the date cited in [17], separately from these Zenodo timestamps. The metadata for record 10.5281/zenodo.23449141 (version string 2.0.0, created 18 October 2026, 08:35:38 UTC) describe the Dobrushin-ceiling paper with the 2/9 uniqueness range and the bound $\frac{1}{4}$; those for record 10.5281/zenodo.23484043 (version 3.0.0, created 19 October 2026, 20:51:57 UTC) describe a forest gauge with the range 1/3. The latest version we have read is 4.45.0 (Zenodo 10.5281/zenodo.24111624, 10 November 2026, 14:11:39 UTC, title *Yang–Mills Theory and the Mass Gap*, five authors; [18]), read through its 22-page short account. Its Theorem A (gauge fixing by a field of straight lines, gauge-invariant observables, free boundary) gives, for $SU(2)$, a lattice mass gap for $|\beta_W| < \frac{1}{3}$ with $m \ge -\log(3\beta_W)$, and,

by the Bakry–Émery method, a decay rate for $N|\beta| < \frac{2}{29}\kappa_G$, i.e. $\beta_W < 4/29$ for $SU(2)$ ([read]; proofs not checked here). Its Theorem B reduces a continuum mass gap to a cube-resampling spectral-gap condition (D) along the weak-coupling tuning and to the existence of a nontrivial continuum limit (E); the authors state that (D) and (E) are not proved. Theorem A concerns gauge-invariant observables and free-boundary limits; Theorem B assumes a cube condition for Wilson measures on tori. We keep this distinction from the DLR uniqueness statement of Theorem 1. A further version 4.46.0 (Zenodo 10.5281/zenodo.24114232, 10 November 2026, 15:36:28 UTC, with a sixth author) has an archive listed in its metadata, `creelie/4Y-M-v4.46.0.zip`; we have read only those metadata and have not opened that archive. Two later records exist: 10.5281/zenodo.24117469 (version string 4.46.0, archive named `creelie/4Y-M-v4.47.0.zip`, 10 November 2026, 18:26:47 UTC) and 10.5281/zenodo.24117822 (version 4.47.1, 10 November 2026, 18:49:04 UTC). The Zenodo abstract of the former is that of 4.46.0; the abstract of 4.47.1 differs from that of 4.45.0 (cube gap for $SU(2)$ up to $\beta_W = 7/20$, and no mention of the Bakry–Émery bound 29/2); we have read only those metadata, and we have not checked whether the statements attributed here to version 4.45.0 are kept in version 4.47.1. A further record, 10.5281/zenodo.24147967 (version 5.0.0, created on 11 November 2026 at 23:29:13 UTC, publication date 12 November 2026), has a new title, *The Yang–Mills Mass Gap from One Bound at One Scale*, and a paper of 157 pages in the archive `creelie/4Y-M-v5.0.0.zip`. We have read its abstract and its introduction only: its Theorem A states a gap for $N|\beta| < \frac{2}{29}\kappa_G$ and, for $SU(2)$, for $|\beta_W| < 1/3$ with $m \geq -\log(3\beta_W)$; its Theorem C(iii) states, for $SU(2)$, a cube condition for $|\beta_W| \leq 7/20$ and a transfer-matrix gap for $0 \leq \beta_W \leq 9/25$, in the normalisation of that paper. We have not checked these proofs, and the comparisons made in this note are with version 4.45.0.

Rajabu [19, 20] (added in v1.6). The Zenodo record 10.5281/zenodo.23691339 (27 October 2026, 13:09:49 UTC) contains a preprint and companion working notes of H. T. Rajabu, among them "Part I" and "Part II" (both dated 26 October 2026 in the notes); the record 10.5281/zenodo.23729328 (version 2.2; created 29 October 2026, 00:48:34 UTC, its publication-date field reads 28 October) is a new version of it. All numbers below are the statements of the authors, taken from the texts we read ([read]); we have not checked the proofs, which are partly computer-assisted. Concept DOI: 10.5281/zenodo.23691338. *Part I*: in a line gauge, uniqueness of the Gibbs measure of the Wilson specification and a transfer-matrix gap for $\beta_W < 1/(2\sqrt{3}) \approx 0.2887$ (Theorem 6.1, Corollary 6.3), and Wilson's area law on tori for $\beta_W < 1/\sqrt{3} \approx 0.577$ (Corollary 6.5). *Part II*: the chordal one-link bound $\|D_v\|_{\mathrm{ch}} \leq q(|s|)|v| \leq \frac{1}{4}|v|$ for $|s| \leq 4$, with equality for $v \perp s$, where $q(\kappa) = I_2(\kappa)/(\kappa I_1(\kappa))$ (Theorem 7.6, with a computer-assisted certificate, Theorem 7.5); uniqueness of the Gibbs measure for the Wilson specification on all links of $\mathbb{Z}^4$ for $\beta_W \leq 0.393$ (Corollary 5.2), obtained by fixing one quarter of the links to the identity, integrating out one further quarter exactly, and applying Dobrushin contraction in the geodesic metric to the remaining specification (Theorem 5.1); and Wilson's area law on tori for $\beta_W < 2/3$ (Corollary 7.7(iii)). *Preprint of 27 October*: Rajabu states uniqueness of the Wilson Gibbs measure, exponential decay of gauge-invariant correlations and a transfer-matrix gap for $\beta_W \leq 0.419$ (abstract). Theorem 5.1 of version 2.2 also states these conclusions, including uniqueness on all links of $\mathbb{Z}^4$. In addition, version 2.2 states uniqueness in every half-space for every boundary condition for $0 < \beta_W \leq 0.419$ (Section 11). *Version 2.2* of the preprint (dated 29 October 2026): Rajabu states in Theorem 9.1(i): "For $0 < \beta \leq 0.511$ the Wilson specification on $\mathbb{Z}^4$ has exactly one Gibbs measure ν_β. It is invariant under lattice symmetries and gauge transformations, and the free-boundary box measures converge to it." Here his β is β_W. Theorem 9.1(ii)–(iii) states exponential clustering for gauge-invariant local chordal-Lipschitz observables and a gap for the physical (gauge-invariant) Osterwalder–Schrader transfer matrix for $0 < \beta_W \leq 0.51$. These are the author's statements; we have not checked their proofs. A second preprint in the record

10.5281/zenodo.23729328 ("Multiscale covariance transfer and a conditional continuum mass gap for four-dimensional SU(2) lattice Yang–Mills theory", version 2.1, 29 October 2026) is a conditional continuum statement, which we do not use here. *Relation to this note.* The record of 27 October precedes v1.4 of this note (Zenodo 10.5281/zenodo.23712954, 27 October 2026, 23:03:42 UTC) by about ten hours, and v1.5 (1 November 2026, 10:24:30 UTC) by about five days. In the range $\kappa \le 4$, the bound of Theorem 6 on the supremum over directions, with equality for directions orthogonal to the field, is Theorem 7.6 of Part II; we make no priority claim on it for $\kappa \le 4$. Theorem 7.6 of Part II is stated on $|s| \le 4$; the present note's all-field statement, its maximisers and the two-sided oblique estimate are given in Theorem 6 and (3). We have not made an exhaustive comparison of the series. Part II of [19], dated 26 October 2026 and distributed in the Zenodo record created on 27 October 2026 at 13:09:49 UTC, states Wilson's area law on tori for $\beta_W < 2/3$ (Corollary 7.7(iii)). Borgs and Seiler [25] (1983) state a confinement condition $J_E < (d - \frac{1}{6})^{-1}$; in $d = 3$ and with $\beta_W = 2J_E$ this reads $\beta_W < 12/17$ (§7.2). We report these statements without a priority ranking. The uniqueness ranges stated by Rajabu exceed 2/9; their use of gauge fixing, decimation and, for Theorem 9.1, a fiber–base cluster expansion differs from the one-site criterion for the undecimated specification in Theorem 1.

Shen–Zhu–Zhu [2]. Their action is (verbatim LaTeX from the arXiv source, eq. (1.2), p. 4 of the arXiv PDF)

```
\cS (Q) \eqdef N\beta  \Re \sum_{p\in \CP^+_{\Lambda_{L}}} \Tr(Q_p),
```

i.e. $\mathcal{S}(Q) = N\beta \operatorname{Re} \sum_{p\in\mathcal{P}^+} \operatorname{Tr}(Q_p)$, one term per plaquette; their Assumption 1.1 is equivalent to (1.3), p. 4: $|\beta| < \frac{1}{16(d-1)}$ for $G = SU(N)$. For $SU(2)$, $2\beta_{\mathrm{tH}} \operatorname{Re}\operatorname{Tr} = 4\beta_{\mathrm{tH}} \cdot \frac{1}{2} \operatorname{Re}\operatorname{Tr}$, so $\beta_W = 4\beta_{\mathrm{tH}}$ and their range is $\beta_W < 1/12 \approx 0.0833$ in $d = 4$. Under it they prove (Theorem 1.2, p. 5) uniqueness of the invariant measure of the Langevin dynamics and convergence of all tight limits of the finite-volume measures, together with a log-Sobolev inequality (Theorem 1.4) and exponential decay of correlations (Corollary 1.6). On p. 6 (arXiv v1, the only arXiv version; the published text was not compared) they write (verbatim): "We remark that uniqueness for small β could possibly also be proven using the method of Dobrushin, see e.g. [Dob70]. To this end one would also need to consider the related Wasserstein metric with respect to the Riemannian distance similarly as we do in this paper. However as we understand such an argument has not been carried out in detail for lattice Yang Mills in the literature." The present note carries out this programme for the gauge model, for uniqueness only (with the Riemannian distance in Theorem 2, and with the Euclidean distance of $\mathbb{R}^4$ in Theorem 1). The single-site ingredient (W_1 on a sphere, derivative along a path of fields written as a covariance) is not new: it is the method of Levin [11] (Euclidean metric) and Royer [10] (geodesic metric) for rotator models, and Levin's many-body version already covers the Wilson model (next paragraphs). Neither paper mentions gauge theories. Our uniqueness statement is directly of DLR type; we do *not* claim that [2] lacks DLR uniqueness (their statement is formulated for the Langevin invariant measure and limits with periodic, Dirichlet "or other" boundary conditions).

Levin: the Euclidean W_1 criterion on spheres, many-body interactions. Levin [11] (read on the page images of the Project Euclid scan) considers "the lattice model on $\mathbb{Z}^\nu$ with single spin space S^{N-1}, $N \geqslant 2$" with "a priori measure μ_0 the invariant probability measure on S^{N-1}" (p. 65) and the "Vaserstein distance" $V(\mu,\nu) = \sup_{\|f\|_L=1} |\mu(f) - \nu(f)|$ over functions of Lipschitz norm 1 for "$\|\cdot\|_2$ [...] Euclidean norm on $\mathbb{R}^N$" ((3.3)–(3.4), p. 67), i.e. the *chordal* metric of $S^{N-1} \subset \mathbb{R}^N$. His Theorem 3 (p. 67) reads: "For $\mathbf{h}, \mathbf{h}' \in \mathbb{R}^N$, let $\mu_{\mathbf{h}}, \mu_{\mathbf{h}'}$ be probability measures on S^{N-1} given by $d\mu_{\mathbf{h}}(\mathbf{x}) = \frac{1}{Z_{\mathbf{h}}} e^{\mathbf{h}\cdot\mathbf{x}} d\mu_0(\mathbf{x})$ [...] Then, $V(\mu_{\mathbf{h}}, \mu_{\mathbf{h}'}) \leqslant \frac{\sqrt{5}}{N} \|\mathbf{h}' - \mathbf{h}\|_2$. (3.8)", proved by writing the distance "as the variation of an expectation along

an appropriate path in **h**-space" (p. 68), with two lemmas controlling "separately variation parallel to the 'magnetic field', and perpendicular to the 'magnetic field'," the perpendicular one being Lemma 3.2 (p. 68, quoted in §4.1). Remark 3.6 (p. 73) extends the uniqueness theorem to "many-body interactions of the form $\sum J_X^{(i_a)} \prod_{a\in X} \sigma_a^{i_a}$" with the condition "$T > T_C^N \equiv (\sup_a \sum_{X\ni a}(|X|-1)\mathscr{J}_X)\frac{\sqrt{5}}{N}$, (3.49)", where $\mathscr{J}_X$ is the supremum of the interaction over configurations; the details are "[left] to the reader". Remark 3.7 (p. 73) conjectures that "the numerical factor of $\sqrt{5}$ appearing in Theorem 1 can be replaced by 1" and notes that "the conjecture is true if one shows $\sup_{\|f\|_L=1} |\mu_{\mathbf{g}}[(\mathbf{x}\cdot\hat{\mathbf{c}} - \langle \mathbf{x}\cdot\hat{\mathbf{c}}\rangle_{\mathbf{g}})f]|\,[\dots]$ attains its maximum value at $\mathbf{g}=0$". [read]. The paper does not mention gauge theories.

Transfer to the Wilson link (our computation from the published statement, not a published result). As noted in §3, the Wilson weight $e^{(\beta/2)\operatorname{Re}\operatorname{Tr} U_p} = e^{\beta\langle u_1 u_2 \bar{u}_3 \bar{u}_4, \mathbf{1}\rangle}$ is of Levin's form (3.48) with $N=4$, $|X|=4$ and $\mathscr{J}_X = \beta$; each link lies in 6 plaquettes, so (3.49) at unit temperature reads $6\cdot 3\cdot\beta\cdot\sqrt{5}/4 < 1$, i.e. $\beta_W < 2\sqrt{5}/45 \approx 0.09938$. [proved] from [read] (exact arithmetic; log `calc/levin_transfer.log`). This already exceeds 1/12.

What Theorem 6 adds. The supremum in Levin's Remark 3.7 is our $D_{\mathrm{ch}}(|\mathbf{g}|)$ for $N=4$. Theorem 6 shows that it attains its maximum at $\mathbf{g}=0$, with value $1/N = 1/4$, and is strictly decreasing in $|\mathbf{g}|$, for every $\mathbf{g}$; it identifies it exactly ($\mathbb{E}_{\mathbf{g}}[x_\perp^2]$, attained by the coordinate orthogonal to the field). It therefore confirms, for $N=4$, the maximum-at-$\mathbf{g}=0$ property conjectured in [11, Rem. 3.7]; for all fields this had been proved by [17, Theorem 7], which Theorem 6 recovers independently. It does not prove the conjecture, which is stated for all N, all $\mathbf{g}$ and general interactions. Since the fields that occur for $\beta < 2/9$ satisfy $|\mathbf{g}| \le 6\beta < 4/3$, the Wilson threshold is Levin's transferred bound with $\sqrt{5}$ replaced by 1: $2\sqrt{5}/45\cdot\sqrt{5} = 2/9$ exactly.

Royer: the geodesic W_1 criterion on spheres. Royer [10] (read on the Numdam page images) works with "the unit sphere S^n with its natural (geodesic) distance r" and Dobrushin's distance "$\mathrm{R}(\nu_1,\nu_2) = \sup\{\int f\,d\nu_1 - \int f\,d\nu_2;\ \mathscr{L}(f)\leqslant 1\}$" (p. 30). He defines $C(n,h)$ as "the best constant in the inequality $\operatorname{var}(f,\nu(h)) \leqslant C(n,h)(\mathscr{L}(f))^2$" and $C(n) = \sup_h C(n,h)$ (Lemma 1.2, p. 31), proves (3)–(4) quoted in §3, and bounds $C(n)$: the proof of Corollary 2.3 (p. 34) gives "$C(n,h) \leqslant \chi(n,h) < (n-2)^{-1}$; thus $C(n) \leqslant (n-2)^{-1}$", Corollary 3.5 (p. 37) "$C(n,0) = v_n$", and Theorem 3.6 (p. 37) "For $n>1$, $C(n) < v_n + v_{n-1}$", with $v_n + v_{n-1} = 2\sum_{k\geqslant n} k^{-2}$ (p. 37). He states that his estimates "improve on that given by Levin [10], namely $N/\sqrt{5}$, for $N\geqslant 3$" (p. 30), and on p. 38 he conjectures the geodesic analogue of Levin's Remark 3.7: "one might think that $C(n,h)\leqslant C(n,0)$". [read].

Transfer to the Wilson link (our computation from the published statements, not a published result). By (3) with $n=3$, $K=\beta$ and z the staple moved by the neighbour, each neighbour has influence at most $\beta(C(3)/4)^{1/2}$, hence uniqueness when $18\beta(C(3)/4)^{1/2} < 1$. With $C(3)\le 1$ (Cor. 2.3) this gives $\beta_W < 1/9 \approx 0.1111$; with $C(3) < v_3 + v_2 = \pi^2/3 - 5/2$ (Thm 3.6) it gives $\beta_W < \left(9\sqrt{\pi^2/3-5/2}\right)^{-1} \approx 0.12502$. [proved] (exact arithmetic; SymPy logs `calc/royer_transfer.log` and `calc/chiffres_v1.log`). At zero field, Royer's route gives $\beta(v_3/4)^{1/2}$ with $(v_3/4)^{1/2} = \frac{1}{2}(\pi^2/12 - \frac{1}{2})^{1/2} \approx 0.28393$, only 0.35% above our exact $D(0) = 8/(9\pi) \approx 0.28294$; but this is not uniform in the field and gives no uniqueness statement.

Where the gain comes from. Our main threshold 2/9 exceeds the transferred Royer bound 0.12502 by a factor 1.777, Levin's transferred bound by exactly $\sqrt{5}\approx 2.236$, and 1/12 by 8/3; the geodesic threshold $\pi/16$ exceeds 0.12502 by 1.5705, and 2/9 exceeds $\pi/16$ by 13.2%. The gain is not in the constants at $h=0$: it comes from computing the exact single-link coefficient as a function of $\kappa = \beta|s|$ and showing that it is maximal at $\kappa = 0$ on $[0, 6\beta]$ (Theorems 6 and 17), which replaces the field-uniform constants $\sqrt{5}/N$ (Levin) and $C(n)$ (Royer). This maximality is the property conjectured by Levin (Euclidean metric, all N, all fields) and, for

the variance constant, by Royer. For $N = 4$ we establish it for the Euclidean W_1 coefficient at every field (Theorem 6; for all fields it had been proved before in [17], Theorem 7) and for the geodesic one only for $\kappa \le 3\pi/8$, by a computer-assisted proof. To our knowledge [17] does not contain the geodesic coefficient of Theorem 17, and we found no other published proof of either property (literature search not exhaustive).

Total variation. The classical total-variation form of Dobrushin's criterion for many-body interactions on a compact single-spin space is due to Gross [13] and Simon [14], neither of whom mentions gauge theories. Gross [13] (read on the page images) treats "X a compact metric space", with "the cases of principle current interest [...] X = finite set, S^m, compact group, or a finite product of these" (p. 10), coefficients $\varrho_{a,b} = \frac{1}{2}\sup\|\mu_b(\cdot|s) - \mu_b(\cdot|t)\|_{\mathrm{Var}}$ ((2.3), p. 11), and proves (Theorem 3, p. 22): "If $\alpha_0 < 1$ then φ has a unique Gibbs state σ" with the decay bound (4.9), where $\alpha_0 = 2e^{4\|\varphi\|_1}\sup_a\sum_{b\neq a}\zeta(a,b)e^{d(a,b)}$ (4.8), and "$\|\mu_a(\cdot|t) - \mu_a(\cdot|s)\|_{\mathrm{Var}} \leqslant e^{4\zeta(a,b)} - 1$" ((4.11), Lemma 4.1). [read]. Simon [14] (read on the page images) proves: "If $\sum_{0\in X}(|X|-1)\|\Phi(X)\|_\infty < 1$, then (2) holds" (Theorem, p. 184), with "$\|\Phi(X)\|_\infty$ [replaceable] by $\frac{1}{2}[\max(\Phi(X)) - \min(\Phi(X))]$" (Remark 1, p. 185), via the Lemma "$\|\mu_h - \mu_g\| \leqslant \|h - g\|_\infty$" whose proof writes $\frac{d}{d\theta}\nu_\theta(f)$ as a covariance ((3)–(4), p. 184); in Remark 5 (p. 185) he observes that a one-site quantity "takes its maximum at $a = 0$ by the GHS inequality" for spins in $[-1, 1]$. [read]. *Transfer to the Wilson link* (ours): with $\Phi(p) = -\frac{\beta}{2}\operatorname{Re}\operatorname{Tr} U_p$, Simon's condition with his Remark 1 reads $18\beta < 1$, i.e. $\beta_W < 1/18 \approx 0.056$; Gross's Theorem 3 gives $36\beta e^{24\beta} < 1$, i.e. $\beta_W < 0.0180$, and his (4.11) in his Theorem 1 gives $\beta_W < \ln(10/9)/4 \approx 0.0263$ [proved] from [read] (log `calc/gross_transfer.log`). Chatterjee [12] (read in arXiv:1811.09770v3) applies the total-variation criterion to $\mathbb{Z}_2$ (Ising) lattice gauge theory in $d = 4$: "The easiest way to prove exponential decay of correlations at strong coupling is by using Dobrushin's condition" (p. 21) and, in the proof of Theorem 5.2 (p. 24), "There are at most 18 such neighbors", with a total-variation bound $L_0\lambda$ per neighbour, L_0 a universal constant that is not made explicit. [read]. For the $SU(2)$ single-link conditionals the exact one-link total-variation coefficient is $c_{\mathrm{TV}}(\beta) = \sup\|q_s - q_{s'}\|_{\mathrm{TV}} = \frac{4\beta}{3\pi} + O(\beta^2)$, the supremum being over $|s|, |s'| \le 6$ differing by one staple, so $|s - s'| \le 2$. [proved]: *upper bound* (this is the route (3) of Simon's Lemma, without the Schwarz step), along the segment $s(t) = s + t\delta$, $\delta = s' - s$, $|s(t)| \le 6$, the density h_t of $q_{s(t)}$ w.r.t. Haar measure satisfies $e^{-12\beta} \le h_t \le e^{12\beta}$, and $\frac{d}{dt}q_{s(t)} = \beta(\langle u, \delta\rangle - \mathbb{E}_{q_{s(t)}}\langle u, \delta\rangle)q_{s(t)}$, so $\|q_s - q_{s'}\|_{\mathrm{TV}} \le \frac{\beta}{2}\sup_t(\mathbb{E}_{q_{s(t)}}|\langle u, \delta\rangle| + |\mathbb{E}_{q_{s(t)}}\langle u, \delta\rangle|) \le \beta(e^{12\beta}\frac{4}{3\pi} + e^{12\beta} - 1)$, using $\mathbb{E}_{\mathrm{Haar}}|\langle u, e\rangle| = 4/(3\pi)$ and $\mathbb{E}_{\mathrm{Haar}}\langle u, \delta\rangle = 0$; *lower bound*, take $s = 0$ and $s' = -2w$ (flip the staple $w \mapsto -w$, the other five summing to $-w$), then $\|q_0 - q_{s'}\|_{\mathrm{TV}} = \frac{1}{2}\mathbb{E}_{\mathrm{Haar}}|e^{\beta\langle u, s'\rangle}/Z - 1| = \frac{\beta}{2}\mathbb{E}_{\mathrm{Haar}}|\langle u, s'\rangle| + O(\beta^2) = \frac{4\beta}{3\pi} + O(\beta^2)$. The resulting sum $24\beta/\pi$ gives the threshold $\approx \pi/24 \approx 0.131$ *at leading order only*; the finite-β threshold, from a floating-point evaluation of $c_{\mathrm{TV}}(\beta)$, is ≈ 0.1310 [numerical]. The ratio of the first-order coefficients is $\frac{12/(9\pi)}{8/(9\pi)} = \frac{3}{2}$ for the geodesic metric, and $\frac{4/(3\pi)}{1/4} = \frac{16}{3\pi} \approx 1.70$ for the chordal one; the "gain of W_1 over total variation" therefore depends on the metric.

Summary: five levels. (i) Total variation: transferred published constants give $\beta_W < 1/18$ (Simon) [proved] from [read]; with the exact one-link coefficient, $\beta_W \approx \pi/24 \approx 0.131$ (leading order [proved], value [numerical]). (ii) Euclidean W_1 with Levin's published constant transferred to the Wilson link: $\beta_W < 0.09938$ [proved] from [read]. (iii) Geodesic W_1 with Royer's published constants transferred: $\beta_W < 0.12502$ (Thm 3.6) or $< 1/9$ (Cor. 2.3) [proved] from [read]. (iv) Geodesic W_1 with the exact coefficient, uniform in $|h| \le 6\beta$ (Theorem 2): $\beta_W < \pi/16 \approx 0.19635$. (v) Euclidean W_1 with the exact coefficient, uniform in $|h| \le 6\beta$ (Theorem 1): $\beta_W < 2/9 \approx 0.2222$. Levels (ii)–(v) exceed $1/12$ rigorously; the exact-coefficient TV level does so only at leading order and numerically, since the explicit upper bound on c_{TV} written above proves the total-variation criterion only for $\beta < 0.0414$.

Bakry–Émery on link variables, without gauge fixing, cannot pass $\beta_W = 1/8$.

Proposition 22. *Equip each factor $SU(2)$ with a bi-invariant metric (any constant scale per link). At the π-flux configuration ($U_\ell = \pm\mathbf{1}$ with every $U_p = -\mathbf{1}$; it exists on $\mathbb{Z}^4$ and on periodic boxes of even side) the quadratic form $\mathrm{Ric} + \beta\,\mathrm{Hess}\,S_W$, $S_W = \sum_p(1 - \frac{1}{2}\,\mathrm{Re}\,\mathrm{Tr}\,U_p)$, is not positive definite for $\beta > 1/8$ (and, on even periodic boxes, has the eigenvalue 0 at $\beta = 1/8$, so it is not positive definite there either). Hence the global Bakry–Émery condition $\mathrm{Ric} + \beta\,\mathrm{Hess}\,S_W \geq K > 0$ on link variables fails for $\beta_W \geq 1/8$.* [proved]

Proof. In the Frobenius normalisation ($X_a = i\sigma_a/\sqrt{2}$), $\exp(sX_a)$ has rotation angle $s/\sqrt{2}$, so $SU(2)$ is the round S^3 of radius $\sqrt{2}$ and $\mathrm{Ric} = g$. Both Ric and $\mathrm{Hess}\,S_W$ are $(0,2)$-tensors unchanged by constant rescalings of the factors, so the sign question is scale-free. Perturb $U_\ell \mapsto e^{X_\ell}U_\ell$. At a configuration with all U_ℓ central ($\pm\mathbf{1}$), $U_p \mapsto U_p\,e^{X_1}e^{X_2}e^{-X_3}e^{-X_4}$, so the second-order expansion of $1 - \frac{1}{2}\,\mathrm{Re}\,\mathrm{Tr}\,U_p$ at $U_p = -\mathbf{1}$ is minus the one at $U_p = \mathbf{1}$, and both configurations are critical. Therefore $\mathrm{Hess}\,S_W(\pi\text{-flux}) = -\mathrm{Hess}\,S_W(\text{flat}) = -\frac{1}{2}\,d^*d \otimes I_3$, where $d^*d = B^\mathsf{T}B$ with B the lattice curl (since $1 - \frac{1}{2}\,\mathrm{Re}\,\mathrm{Tr}\,e^Y = |Y|^2/4 + O(|Y|^3)$ and the commutator terms of Y are traceless). By Fourier analysis the largest eigenvalue of d^*d is $4d = 16$, attained at momentum (π,π,π,π) with $d - 1 = 3$ transverse polarisations. So $\lambda_{\min}(\mathrm{Hess}\,S_W) = -8$ with multiplicity 9 on even periodic boxes (on $\mathbb{Z}^4$: infimum of the quadratic form, via Weyl sequences), and $\mathrm{Ric} + \beta\,\mathrm{Hess}\,S_W$ has the eigenvalue $1 - 8\beta$. □

Checks [numerical]: an independent arbiter script (finite differences, $L = 2$) finds $\lambda_{\min} = -8.000000$, multiplicity 9, $\max|H(\pi) + H(\text{flat})| = 7.5 \cdot 10^{-8}$; a second script written for this note finds $\max\mathrm{eig}(B^\mathsf{T}B) = 16$ with multiplicity 3 on $L = 2$ and $L = 4$, and $Q_\pi(v) = -Q_{\text{flat}}(v) = -\frac{1}{2}v^\mathsf{T}(B^\mathsf{T}B \otimes I_3)v$ to $2 \cdot 10^{-7}$ along random directions on $L = 4$. Whether $\sup_U(-\lambda_{\min}\mathrm{Hess}\,S_W) = 8$ (so that global Bakry–Émery holds up to exactly 1/8) is [conjecture]. Consistently, [2] bound |Hess| by $12\beta_W$ and obtain 1/12. Since $2/9 > \pi/16 > 0.12502 > 1/8$ (the transferred Royer bound exceeds 1/8 by a margin of $2 \cdot 10^{-5}$; and, at leading order only, $\pi/24 > 1/8$), Dobrushin-type criteria, which need no convexity, go beyond the ceiling of the global Bakry–Émery route on link variables, without gauge fixing, with bi-invariant product metrics (and $2/9 > 4/29 \approx 0.138$, the range of the gauge-fixed Bakry–Émery method of [18, Theorem A], version 4.45.0, where one link of every plaquette is set to the identity by a field of straight lines and the Hessian bound is 29/2 instead of 16; Proposition 22 concerns all link variables and says nothing about that gauge-fixed measure). This does not contradict [5], which obtains an area law up to $\beta_W < 1/6$ by applying Bakry–Émery to a different (vertex) spin system. The same limit, for every $SU(N)$ and every d, is Theorem C of [17] (v1, p. 2): $\inf_Q \lambda_{\min}(\mathrm{Ric} - N\beta\,\mathrm{Hess}\,S_0(Q)) = N/2 - 4dN|\beta|$ on every torus of even side, which vanishes at $\beta_W = N^2|\beta| = 1/8$ for $SU(2)$ in $d = 4$; Proposition 22 was written before we saw that preprint.

Other works. Osterwalder–Seiler [3] prove, by cluster expansion, decay, a mass gap and uniqueness of the infinite-volume limit for "g_0 sufficiently large", without an explicit constant [read]. Chatterjee [6] (arXiv:2006.16229v5, p. 7, after Definition 2.3; the published version was not compared) states that exponential decay under arbitrary boundary conditions "can be shown to hold for essentially any lattice gauge theory when β is small enough, using the methods of Dobrushin and Shlosman" [7], without a constant [read]. Cao–Nissim–Sheffield [5] (arXiv preprint; "Submitted" in the arXiv comment on 11 November 2026) prove an area law on the torus for $\beta_{\text{tH}} < \frac{1}{8(d-1)}$, i.e. $\beta_W < 1/6$; this is a statement on Wilson loops, not on uniqueness, and its relation to §7 is discussed in §7.2. Seiler's monograph [4] could not be obtained and was *not consulted*; for Georgii's book [16] only the table of contents was read (Chapter 8 "Uniqueness", §8.1 "Dobrushin's condition of weak dependence", p. 140), and its content was *not consulted*; Simon's book [15] was *not consulted*. No claim of this note rests on any of them (see §9, item 4).

source	statement	range in β_W ($SU(2)$, $d = 4$)	status
[3]	decay, gap, unique limit	small, not explicit	[read]
[12]	TV-Dobrushin, decay, $\mathbb{Z}_2$ gauge, $d = 4$	not explicit (L_0)	[read]; other group
[13] Thm 3, transferred (here)	uniqueness, decay	< 0.0180	[proved] from [read]
[14] Thm + Rem. 1, transferred (here)	uniqueness	$< 1/18 \approx 0.056$	[proved] from [read]
[2]	uniqueness, LSI, decay	$< 1/12 \approx 0.083$	[read]
[11] Rem. 3.6, transferred (here)	uniqueness	$< 2\sqrt{5}/45 \approx 0.0994$	[proved] from [read]
[10] Cor. 2.3, transferred (here)	uniqueness	$< 1/9 \approx 0.111$	[proved] from [read]
Prop. 22	global BE on links fails (no gauge fixing)	$\geq 1/8 = 0.125$	[proved]
[17] Thm B (preprint)	Bakry–Émery, log-Sobolev and Poincaré inequalities, decay	$< 1/8 = 0.125$	[read]; proofs not checked here
[10] Thm 3.6, transferred (here)	uniqueness	< 0.12502	[proved] from [read]
TV-Dobrushin, exact one-link coefficient (here)	uniqueness	$\approx \pi/24 \approx 0.131$	leading order [proved], value [numerical]
[18, Thm A] (v4.45.0)	Bakry–Émery, field of straight lines; decay, gap (gauge-invariant observables, free boundary)	$< 4/29 \approx 0.138$	[read]; different object; proofs not checked here
[5]	area law on torus (§7.2)	$< 1/6 \approx 0.167$	[read], preprint
Theorem 2 (geodesic W_1, exact coefficient)	uniqueness (DLR) only	$< \pi/16 \approx 0.196$	see Thm 2
Theorem 1 (Euclidean W_1, exact coefficient)	uniqueness (DLR) only	$< 2/9 \approx 0.222$	see Thm 1
Cor. 20 (via [22])	Poincaré inequality, uniform in volume and b.c.	$< 2/9 \approx 0.222$	[proved] from [read]
[17] Thms E, F (preprint, v1, 18 Oct. 2026)	uniqueness, decay, transfer-matrix gap	$< 2/9 \approx 0.222$	[read]; proofs not checked here
[19] Part I, Thm 6.1, Cor. 6.3	uniqueness (Wilson specification), gap; line gauge	$< 1/(2\sqrt{3}) \approx 0.2887$	[read]; proofs not checked here
[18, Thm A] (v4.45.0)	lattice gap, field of straight lines (gauge-invariant observables, free boundary)	$< 1/3$	[read]; different object; proofs not checked here
[19] Part II, Cor. 5.2	stated Wilson uniqueness; exact integration, geodesic Dobrushin contraction	≤ 0.393	[read]; proofs not checked here
[19] preprint (27 Oct.), abstract	stated Wilson uniqueness, gauge-invariant decay, gap	≤ 0.419	[read]; proofs not checked here
[20] v2.2, Thm 9.1(i)–(iii)	stated Wilson uniqueness on $\mathbb{Z}^4$; gauge-invariant chordal-Lipschitz clustering and physical gap up to 0.51	$0 < \beta_W \leq 0.511$	[read]; proofs not checked here
[24], written argument (corrected, §7.2)	area law (slab criterion)	$< 1/6 \approx 0.167$	[read]
[24], announced (literal reading)	area law, "more refined estimate", no proof	$< 2/3 \approx 0.667$	[read]
[26] + [27]	area law via Ising slabs	≈ 0.2217	[read] for [26]; [not checked]
[19] Part I, Cor. 6.5	area law on tori (line gauge, geodesic constant)	$< 1/\sqrt{3} \approx 0.577$	[read]; proofs not checked here
Theorem 7.1 (Dobrushin comparison on the slabs)	area law; tori, free b.c., every Gibbs measure	$< 2/3 \approx 0.667$	[proved] from Theorem 6
[19] Part II, Cor. 7.7(iii)	area law on tori (chordal	$< 2/3 \approx 0.667$	[read]; proofs not

9 Limitations

1. *Scope.* The note proves uniqueness of the Gibbs measure and, as a corollary of Wu's theorem [22], the Poincaré inequality of Corollary 20 for $\beta < 2/9$; and, as an application (§7), Wilson's area law for rectangular loops for $\beta < 2/3$, a range that is not new ([25]). No decay of correlations of the Wilson measure, spectral gap of the transfer matrix or log-Sobolev inequality is claimed at $\beta < 2/9$ in this note. The preprint [17] states exponential decay and a transfer-matrix gap in the same range (proofs not checked here); among refereed publications, the best explicit range we know for these conclusions remains 1/12 [2]. Rajabu [20, Theorem 9.1(i)–(iii)] states exactly one Gibbs measure for the Wilson specification on $\mathbb{Z}^4$ for $0 < \beta_W \leq 0.511$, and exponential clustering of gauge-invariant local chordal-Lipschitz observables and a physical transfer-matrix gap for $0 < \beta_W \leq 0.51$ (proofs not checked here). Explicit decay statements covering four-body plaquette interactions do exist in total variation: Gross's Theorem 3 [13], transferred, gives uniqueness and exponential decay for $\beta_W < 0.0180$, far below 1/12. Gross's Theorem 1 [13, p. 11] turns any total-variation Dobrushin sum below 1 into exponential decay (take $d(a,b) = r|a-b|$ with $r > 0$ small); with a certified bound on c_{TV} it would give decay up to the TV threshold ≈ 0.131, hence beyond 1/12 [conjecture] (not certified, not written out). For the W_1 criterion, Levin [11, Remark 3.5, p. 72] states that, "parallel to the discussion of Gross", the truncated spin–spin correlations decay exponentially under the W_1 condition with a semi-metric weight (his (3.46)–(3.47)); applied to our influence matrices this would plausibly give exponential decay of correlations of Lipschitz observables for $\beta < 2/9$ [conjecture]. Nothing is claimed.

2. *Formalisation boundary.* The analytic proof of Theorem 6 (all fields, §4) is not formalised in Lean; Lean covers none of it. In the second, computer-assisted proof of Appendix A (chordal, $\kappa \leq 4/3$), the flux bound, the explicit field and the elimination of the direction (§§4.3, A.2) are proved on paper; the box bounds, the positivity of m and the constant 0.0084 are interval-certified; Lean checks only the final rational implication, two algebraic steps and the threshold arithmetic. For the geodesic certificate, the chain "integrals → coefficient bounds" (Arb, reduction of §5.1) is outside Lean, and so are the constant 0.0085 and the covering of $[0,\pi]$ by the boxes (§5.2). The differential identity $R\,E\,\mathcal{Q}[\phi] = (R\,E\,\eta\,\phi^2)'$, the sum-of-squares algebra, the moment and Riccati identities and the Chebyshev argument of §4 are natural candidates for formalisation; none was formalised. Kantorovich–Rubinstein duality is quoted from [8, Thm 10.8, p. 53]; McShane's extension, the divergence theorem for Lipschitz fields, the Holley–Stroock perturbation argument [9] and the weak solvability of $-L\varphi = R$ (Lax–Milgram + Poincaré) are standard and written out here; none of them is formalised. Wu's theorem, used in Corollary 20, is quoted, not reproved. The area law of §7 and Appendix B (slab decomposition, Wasserstein comparison, walk count, factorisation of the loop) is proved on paper only; Lean covers none of it.

3. *History and audit of the certificates and proofs.* The first geodesic certificate was wrong (see the erratum); the present one passed guards designed to catch that failure mode and two independent recomputations. The chordal certificate of Appendix A was written with the same guards and is confirmed by an independent implementation with different arithmetic, tail bounds and quadrature (§A.3); its mathematical reduction was re-derived line by line in that independent verification. The analytic proof of §4, new in this version, was checked as follows: every algebraic identity symbolically (SymPy); the key identity $\Xi + 3\Delta + \mathcal{S} = 0$ as a formal power series in exact rational arithmetic up to κ^{39}, and numerically at 120 values of $\kappa \in [0.01, 400]$ (Arb, 90 digits) and by independent mpmath codes (30 and 50 digits); the resulting bounds against exact optimal-transport computa-

tions of the supremum (§4.9); and it was read step by step by one independent reviewer, a language model of a different family from the one that assisted the writing (Kimi K3.1 Pro, Moonshot AI), which found no error and recomputed the identities with its own code. The area-law section (§7 and Appendix B) was read by the same reviewer, first as a separate working note and then in its present condensed form (no objection; the reviewer recomputed the quaternionic identities and exact walk counts with its own code and compared the statements on [25, 24] and on the other sources with the page images). This is not a refereeing: neither certificate, and none of the proofs, has been audited or refereed by a party outside the project, and the new proofs (§4, §7) have not been read by a human referee. For version 1.6, which changes the literature only, the passages on the literature were revised on 11 November 2026 with the assistance of GPT-6.1 Sol (OpenAI), after a review of the draft of 10 November 2026 by Qwen3.8 (Qwen team, run locally); the final text was then read against the sources by Claude (Anthropic) agents on 11 November 2026; no statement or proof was modified since version 1.5.

4. *Novelty and prior art.* The single-site W_1 machinery on spheres is due to Levin [11] (Euclidean metric, many-body multilinear interactions, which cover the Wilson model; transferred, 0.0994) and Royer [10] (geodesic metric; transferred, 0.12502). The field-uniformity of the one-site coefficient (maximum at zero field) was conjectured by both. For $N = 4$ and the Euclidean metric it was proved, for all fields, by [17] (Theorem 7), before the identity of Theorem 6, which adds the exact value of the supremum over directions, its maximisers, the estimate for oblique directions and a different proof (§8). The total-variation criterion for many-body interactions is due to Gross [13] and Simon [14], and the Poincaré inequality under Dobrushin's condition to Wu [22] ("classical" for [17]). [17] contains $\beta < 2/9$ (Theorem E); Rajabu [20, Theorem 9.1(i)] states uniqueness of the Wilson specification on $\mathbb{Z}^4$ for $0 < \beta_W \le 0.511$, and Part II of [19] states $\|D_v\|_{\mathrm{ch}} \le q(|s|)|v| \le \frac{1}{4}|v|$ for $|s| \le 4$, with equality in the first inequality for $v \perp s$ (§8). For the area law of §7, the range $\beta < 2/3$ is not new: Wilson's area law for $\beta < 2/3$ is Corollary 7.7 of Part II of [19], and confinement for $\beta < 12/17$ is in Appendix B of [25], which we read in the preprint version and compared with the journal version on the passages cited; Durhuus and Fröhlich [24] announced $\beta < 2/3$ without proof (literal reading of their normalisation), and the ranges of [26, 27] and [5] are smaller (§7.2). The literature search for the area law is not exhaustive: MathSciNet was not queried, and [30] and [4] were not consulted. The contribution of this note is the exact field-uniform single-link coefficient for the Euclidean metric (all fields, with an analytic proof) and for the geodesic metric ($\kappa \le 3\pi/8$), the certification of the latter, the resulting thresholds $2/9$ (also in [17]) and $\pi/16$, and the transfers of published constants. *Limitation of the literature search:* Seiler's monograph [4] (not obtainable), the content of Georgii's book [16] (table of contents only) and Simon's book [15] were not consulted; any of them may contain a Dobrushin-type statement for plaquette interactions or for $SU(2)$ gauge fields. No claim of this note rests on them, but a priority claim for the gauge application should be read with this limitation in mind.

5. *Range of fields.* The chordal identity is proved for all $\kappa \ge 0$; the computer-assisted proof of v1.4 (Appendix A) covers $\kappa \le 4/3$ only, which is what $\beta < 2/9$ requires. Levin's conjecture for $N = 4$ (maximum at zero field, Remark 3.7 of [11]) was decided by [17], Theorem 7, which gives $D_{\mathrm{ch}}(\kappa) \le \frac{1}{4}$ for every κ; Theorem 6 recovers it with the exact value of the supremum over directions. Levin's conjecture for $N \ne 4$, and for interactions other than the Wilson one, is not addressed. Royer's analogous conjecture for the variance constant, "one might think that $C(n,h) \leqslant C(n,0)$" [10, p. 38], is not addressed either, and the geodesic W_1 bound is certified only for $\kappa \le 59/50$.

6. *The value* $\pi/24$ is exact only at leading order in β; the finite-β TV threshold is a floating-point estimate, and has not been recertified in interval arithmetic.

7. *Physics.* The results are strong-coupling statements ($g^2 > 18$ for uniqueness, $g^2 > 6$ for the area law, with $\beta_W = 4/g^2$). They say nothing about the continuum limit or the weak-coupling regime relevant to the Yang–Mills mass gap problem.

A The computer-assisted proof of v1.4 on $0 \le \kappa \le 4/3$

This appendix reproduces, with its mathematics unchanged, the argument by which v1.4 proved the chordal identity on $0 \le \kappa \le 4/3$, the range that $\beta < 2/9$ needs. No statement of the main text depends on it: the analytic proof of §4 covers all κ. It is kept because it is an independent second proof on $[0, 4/3]$ whose final rational steps are checked in Lean, and because the certificates, their independent check and the Lean files are part of the archive. The notation is that of §4, except that $m(y_1)$ below is the function of Proposition 24, not the mean m_1; Lemmas 7 and 9 and the explicit field (6) are those of §4.

Proposition 23 (v1.4)**.** *For $\kappa \in [0, 4/3]$ and all α, $D_{\mathrm{ch}}(\kappa, \alpha) \le \mathbb{E}_\kappa[x_2^2]$; with Lemma 7 this is the identity $D_{\mathrm{ch}}(\kappa) = \mathbb{E}_\kappa[x_2^2]$ on $[0, 4/3]$.*

Status: the upper bound is [proved] (§4.3, §A.2) up to the one-variable inequality (25), which is [interval-certified] (§A.3); its last rational step is [Lean-checked].

A.1 Series for A and regularity for $\kappa \le 4/3$

Series and regularity. $A = \sum_{j \ge 0} \kappa^j A_j$ where A_j is the unique polynomial solution of $L[A_j] = q_j$, $L[p] = tp - \frac{1}{3}(1-t^2)p'$, q_j the coefficient of κ^j in q: L maps degree d to degree $d+1$ with leading factor $1 + d/3$, the homogeneous solutions $c(1-t^2)^{-3/2}$ are unbounded, and the solvability condition $\int q_j \sqrt{1-t^2} = 0$ holds order by order; so A_j has degree j ($A_0 = 1$, $A_1 = \frac{3}{4}t$, $A_2 = \frac{3}{10}t^2 - \frac{7}{40}$). A is holomorphic in κ for $|\kappa| \le 2$ (where $|\bar{Z}| \ge 2 - \bar{Z}(2) > 0.4$) and bounded there, so Cauchy's estimates and Markov's inequality $\|p'\|_\infty \le (\deg p)^2 \|p\|_\infty$ make the series and its first two t-derivatives converge uniformly on $[-1, 1]$ for $\kappa \le 4/3$; hence $V \in C^1(\bar{B})$ [proved]. The explicit bounds are those of the certificate (§A.3).

A.2 Elimination of the direction

Proposition 24. *Let $m(y_1) = \min(A, E) - |A'|R/6$ and assume $m > 0$ on $[-1, 1]$. With*

$$J_0 = \frac{2}{5\pi} \int_{-1}^{1} \Big(\frac{A'}{3}\Big)^2 \frac{R^5}{2m}\, dy_1, \qquad J_1' = \frac{2}{3\pi} \int_{-1}^{1} (E - A)^2 \frac{R^3}{2m}\, dy_1, \qquad J_1 = J_1' + \tfrac{2}{3} J_0,$$

one has, for all α, $D_{\mathrm{ch}}(\kappa, \alpha) \le \cos^2\alpha \,(\operatorname{Var}_\kappa x_1 + \max(J_0, J_1)) + \sin^2\alpha\, \mathbb{E}_\kappa x_2^2$. [proved]

Proof. By Lemma 9 applied to (6), $D_{\mathrm{ch}}(\kappa, \alpha) \le \frac{1}{2\pi^2} \int_B |V|$. Pointwise, $\langle v, V\rangle = \cos^2\alpha\, A + \sin^2\alpha\, E - \cos\alpha \sin\alpha\, \frac{A'}{3} y_2 \ge m(y_1) > 0$ (as $|\cos\alpha \sin\alpha| \le \frac{1}{2}$ and $|y_2| \le R$), and $\sqrt{a^2 + b^2} \le a + b^2/(2a)$ for $a > 0$ gives $|V| \le \langle v, V\rangle + |V_\perp|^2/(2m)$, where, in the orthonormal frame (v, n, e_3, e_4) with $n = -\sin\alpha\, e_1 + \cos\alpha\, e_2$, $|V_\perp|^2 = \cos^2\alpha \big[\big(\sin\alpha(E - A) - \cos\alpha \frac{A'}{3} y_2\big)^2 + \big(\frac{A'}{3}\big)^2 (y_3^2 + y_4^2)\big]$. The integral of $\langle v, V\rangle$ is $2\pi^2 \operatorname{Var}_\kappa(u) = 2\pi^2(\cos^2\alpha \operatorname{Var}_\kappa x_1 + \sin^2\alpha\, \mathbb{E}_\kappa x_2^2)$ (end of §4.3; $\operatorname{Cov}_\kappa(x_1, x_2) = 0$ and $\mathbb{E}_\kappa x_2 = 0$ by the symmetry $x_2 \mapsto -x_2$). On the section $\{|z| \le R\}$ of the ball, the cross term is odd in y_2, and $\int 1 = \frac{4}{3}\pi R^3$, $\int y_2^2 = \frac{4}{15}\pi R^5$, $\int (y_3^2 + y_4^2) = \frac{8}{15}\pi R^5$. Dividing by $2\pi^2$, with $s = \sin^2\alpha$, the excess is at most $\cos^2\alpha \,[s J_1' + (1 - \frac{s}{3}) J_0]$, affine in $s \in [0, 1]$, hence at most $\cos^2\alpha \max(J_0, J_1)$. □

Reduction to one variable. If, for a given κ, $m > 0$ on $[-1,1]$ and $\mathrm{Var}_\kappa\, x_1 + \max(J_0, J_1) \le \mathbb{E}_\kappa x_2^2$, Proposition 24 gives $D_{\mathrm{ch}}(\kappa,\alpha) \le \mathbb{E}_\kappa x_2^2$ for all α (the convex combination `etape_alpha`, [Lean-checked]), and Lemma 7 gives equality at $\alpha = \pi/2$. Now $N := \bar{Z}^2(\mathbb{E}_\kappa x_2^2 - \mathrm{Var}_\kappa\, x_1) = \frac{1}{3}\bar{Z}^2 - \frac{4}{3}\bar{Z}''\bar{Z} + \bar{Z}'^2 = \kappa^2\tilde{N}(\kappa)$ with $\tilde{N}$ a power series, and $A' = \kappa\tilde{A}$, $E - A = \kappa\tilde{D}$ with power series $\tilde{A}, \tilde{D}$ (since $A_0 = E_0 = 1$). Writing $\tilde{J}_0, \tilde{J}_1$ for J_0, J_1 with $A', E - A$ replaced by $\tilde{A}, \tilde{D}$ (so $J_i = \kappa^2\tilde{J}_i$), it suffices to prove

$$\frac{\tilde{N}(\kappa)}{\bar{Z}(\kappa)^2} > \max\left(\tilde{J}_0(\kappa), \tilde{J}_1(\kappa)\right) \quad \text{and} \quad m > 0 \text{ on } [-1,1], \qquad \text{for all } \kappa \in [0, 4/3] \tag{25}$$

(the step from (25) to the inequality before it is `etape_kappa2`, [Lean-checked]). At $\kappa = 0$, exactly, $\tilde{N}/\bar{Z}^2 = 1/48$, $\tilde{A} = \frac{3}{4}$, $\tilde{D} = \frac{t}{4}$, $m = 1$, and $\tilde{J}_0 = \tilde{J}_1 = 1/256$: the margin is $13/768$, so (25) is *not* degenerate at zero field, unlike the geodesic inequality of §5. [proved]

A.3 Computer-assisted part

Interval evaluation. Script `certificate_chordal/cert_ch.py`, Arb ball arithmetic (python-flint 0.9.0), working precision 128 bits and series length cap 200, both mandatory arguments, set explicitly and read back. All series in κ ($\bar{Z}, \bar{Z}', \bar{Z}'', 1/\bar{Z}$, m_1, $m_1/\bar{Z}$, $\bar{Z}^2$, $\bar{Z}''\bar{Z}$, $\bar{Z}'^2$, N) are computed to length $L = D + 5 = 85$ ($D = 80$); guard 1 refuses cap $< L$, guard 2 asserts `.prec` $= L$ on each of these ten series and products (trailing zeros are padded explicitly, since `coeffs()` drops them). The A_j are obtained by exact downward recursion; the solvability residual contains 0 for all $j \le 80$. *Tails:* all orders > 80 are bounded, not neglected: Cauchy's estimates on $|\kappa| \le 2$ ($|\bar{Z}| \ge 0.409363$, $|m_1| \le 1.6830$, $|E| \le 18.050$, $|A| \le 228.21$, using $\int_t^1 \sqrt{1-s^2}\,ds \le \frac{\pi}{2}(1-t^2)^{3/2}$) and on $|\kappa| \le 4$ for N, and Markov's inequality for A_j'. At $\kappa = 4/3$ the tails are $1.9 \cdot 10^{-8}$ ($\tilde{A}$), $3.0 \cdot 10^{-12}$ ($\tilde{D}$) and $7.3 \cdot 10^{-39}$ (N). *Boxes:* $[0, 4/3]$ is cut into 256 boxes $[i/192, (i+1)/192]$, κ being a ball in each; the series are folded into polynomials in $t = y_1$, and $[-1,1]$ is cut into 1024 dyadic sub-intervals, on which the polynomials are evaluated in centred form $p(t_c) + p'(T)(T - t_c)$. On each sub-interval $m > 0$ is *asserted*, and $\tilde{J}_0$, $\tilde{J}_1$ are bounded above by upper Riemann sums (with exact upper bounds for R^3, R^5). $\tilde{N}$ is bounded below on the box (minus its tail), and $\bar{Z}$ above by its value at the right end (nonnegative coefficients).

Result. All 256 boxes pass; the worst margin is $1.142 \cdot 10^{-2}$, on the last box $[85/64, 4/3]$ (lower bound $1.6883 \cdot 10^{-2}$ for $\tilde{N}/\bar{Z}^2$ against $\tilde{J}_1 \le 5.463 \cdot 10^{-3}$, $\tilde{J}_0 \le 3.975 \cdot 10^{-3}$, $m \ge 0.192$); at $\kappa \in [0, 1/192]$ the margin is $1.690 \cdot 10^{-2}$. The right-hand side can be multiplied by about 3 before the inequality fails. [interval-certified]

Gates. All logs return code 0 unless a refusal is expected (logs in the archive).

gate	outcome
cap $\in \{85, 100, 200, 300\}$	identical results in all 256 boxes; cap 10 and $84 = L - 1$ refused by guard 1, and by guard 2 when guard 1 is disabled
order $D \in \{40, 60, 80, 100\}$	the bound moves ($\tilde{J}_0$ of the last box $4.8604 \cdot 10^{-3}$, $3.97576 \cdot 10^{-3}$, $3.975228508 \cdot 10^{-3}$, $3.975228233 \cdot 10^{-3}$) and stabilises; all four runs pass
sub-intervals $\in \{256, \dots, 2048\}$	upper Riemann sums strictly decreasing, step halving ($O(1/N_T)$)
precision 128 vs 192	$\max \lvert\Delta \tilde{J}_1\rvert = 6.7 \cdot 10^{-14}$; all boxes pass
boxes $\in \{64, 128, 512\}$	all pass; worst margin 1.106, 1.130, $1.148 \cdot 10^{-2}$
exact gate (fractions, no Arb)	solvability residuals exactly 0 for $j \le 80$; $\tilde{N}_0 = 1/48$, odd orders 0; all 6891 exact coefficients lie in the Arb balls
numerical gate ([numerical])	series A vs direct quadrature $1.3 \cdot 10^{-15}$; $\lvert \operatorname{div} V\rvert \le 4 \cdot 10^{-11}$; $\lvert V \cdot x - w\rvert \le 7 \cdot 10^{-16}$; full flux bound $\frac{1}{2\pi^2} \int_B \lvert V\rvert$ by 3D quadrature within $1.7 \cdot 10^{-4}$ of the linear programs of `calc/lp2d_m14.log` (e.g. 0.20892, 0.22140, 0.23332 vs 0.20887, 0.22145, 0.23324 at $\kappa = 4/3$, $\alpha = 0, \pi/4, \pi/2$)

Independent check. A second implementation (`independent_check_chordal/`), written without reading or importing the certificate code, recomputes everything with different tools: exact rational coefficients up to order 100 (own downward solver; solvability residuals exactly 0 and $L[A_j] = q_j$ checked exactly), `mpmath.iv` interval arithmetic instead of Arb, elementary tail bounds instead of complex Cauchy estimates ($\lvert[\kappa^n]1/\bar{Z}\rvert \le 2^{-n}$ by induction, and $(1 - t^2)^{-3/2} \int_t^1 \sqrt{1 - s^2}\, ds \le 1$ on $[0, 1)$ instead of $\pi/2$), a uniform partition in θ ($t = \cos\theta$) with exact weights $\int \sin^4$, $\int \sin^6$, and naive Horner evaluation. On its own partition (200 boxes of width 1/150) it certifies (25) on all of $[0, 4/3]$ (worst margin $1.1395 \cdot 10^{-2}$, $m \ge 0.188$); it also recertifies the 256 boxes of the Lean data file and checks, two-sidedly, that none of the exported rationals contradicts its own enclosures. Its negative gates fail as they should (precision sabotaged to 53 bits: refused; series shorter than $D + 2$: refused; left side of (25) scaled by 0.30: fails on the last box, while 0.35 passes). Exact checks: $\mathbb{E}_{\bar{\sigma}} u^2 = 1/4$ and the cost 1/4 of the reflection coupling (SymPy), the coefficient $3k/(k + 2)$ symbolically for all k, $c_2 = 1/96$; a Monte-Carlo evaluation of the full flux bound at $\kappa = 4/3$ (0.20884, 0.22131, 0.23322) agrees with the linear programs [numerical]. [interval-certified]

Lean. Commit `1b0f039c53c1e2a11c496cc5a1cb7774ab5a010e` of the author's Lean library (parent `7a211ed`, which holds the geodesic modules of §5.2; three new files, no existing file changed), module `KevYM/Certif/DobrushinCordale/`. The three modules are reproduced byte for byte in the standalone project `lean/` of the accompanying archive. Lean toolchain `v4.34.0`. Main statements (verbatim):

```
theorem sound (L : List Boite) (a c : ℚ) (hch : chaine L a c = true) (hok : L.all Boite.ok = true)
    (Nt J0 J1 : ℝ → ℝ)
    (hN : ∀ b ∈ L, ∀ κ : ℝ, (b.k0 : ℝ) ≤ κ → κ ≤ (b.k1 : ℝ) → (b.lb : ℝ) ≤ Nt κ)
    (hJ0 : ∀ b ∈ L, ∀ κ : ℝ, (b.k0 : ℝ) ≤ κ → κ ≤ (b.k1 : ℝ) → J0 κ ≤ (b.h0 : ℝ))
    (hJ1 : ∀ b ∈ L, ∀ κ : ℝ, (b.k0 : ℝ) ≤ κ → κ ≤ (b.k1 : ℝ) → J1 κ ≤ (b.h1 : ℝ))
    (κ : ℝ) (ha : (a : ℝ) ≤ κ) (hc : κ ≤ (c : ℝ)) : max (J0 κ) (J1 κ) < Nt κ
theorem etape_kappa2 (k Nt Jt V1 E2 : ℝ) (hE : E2 - V1 = k ^ 2 * Nt) (h : Jt < Nt) :
    V1 + k ^ 2 * Jt ≤ E2
theorem etape_alpha (V1 E2 J c s : ℝ) (hcs : c ^ 2 + s ^ 2 = 1) (h : V1 + J ≤ E2) :
    c ^ 2 * (V1 + J) + s ^ 2 * E2 ≤ E2
theorem somme_dobrushin_lt_one (β : ℝ) (hβ : β < 2 / 9) : 18 * β * (1 / 4) < 1
theorem kappa_max_lt (β : ℝ) (hβ : β < 2 / 9) : 6 * β < 4 / 3
theorem pi_div_16_lt : Real.pi / 16 < 2 / 9
theorem certificat_cordal (Nt J0 J1 : ℝ → ℝ)
    (hN : ∀ b ∈ boites, ∀ κ : ℝ, (b.k0 : ℝ) ≤ κ → κ ≤ (b.k1 : ℝ) → (b.lb : ℝ) ≤ Nt κ)
    (hJ0 : ∀ b ∈ boites, ∀ κ : ℝ, (b.k0 : ℝ) ≤ κ → κ ≤ (b.k1 : ℝ) → J0 κ ≤ (b.h0 : ℝ))
    (hJ1 : ∀ b ∈ boites, ∀ κ : ℝ, (b.k0 : ℝ) ≤ κ → κ ≤ (b.k1 : ℝ) → J1 κ ≤ (b.h1 : ℝ))
    (κ : ℝ) (h0 : 0 ≤ κ) (hK : κ ≤ 4 / 3) : max (J0 κ) (J1 κ) < Nt κ
```

Here `Boite.ok b` is `h0 < lb && h1 < lb && k0 ≤ k1`, `chaine` checks that the boxes are

chained from a to c, and `boites` is the list of the 256 exported boxes $\langle k_0, k_1, lb, h_0, h_1\rangle$ (Arb bounds rounded outward to the grid 2^{-40}, re-checked exactly before export); `boites_ok` and `boites_chaine` are proved by `decide +kernel`. The file also proves $18 \cdot (2/9) \cdot \frac{1}{4} = 1$, the converse for $\beta \geq 2/9$, and $6 \cdot (2/9) = 4/3$. Build clean (0 warnings); all 16 theorems of the module depend only on `propext`, `Classical.choice`, `Quot.sound`; the rebuild from `git archive 1b0f039` succeeds, and the standalone project `lean/` of the archive builds from an empty `.lake` with the same axiom report (`lean/logs/build_from_zero.log`). [Lean-checked]

What Lean covers, exactly. `certificat_cordal` is a conditional implication: *if* real functions `Nt`, `J0`, `J1` (standing for $\tilde{N}/\bar{Z}^2$, $\tilde{J}_0$, $\tilde{J}_1$) are bounded box by box by the exported rationals, *then* $\max(\tilde{J}_0, \tilde{J}_1) < \tilde{N}/\bar{Z}^2$ on $[0, 4/3]$. Lean also checks the two algebraic steps `etape_alpha`, `etape_kappa2` and the threshold arithmetic. Not in Lean: the flux bound, the explicit field and Proposition 24 (proved on paper, §§4.3–A.2), the bounds of each box and the positivity of m ([interval-certified]), and the constant 0.0084.

Artefacts (SHA-256). Paths are relative to the accompanying archive.

file	SHA-256
`certificate_chordal/cert_ch.py`	`d83aa23b2fd20707b08076c87053aa8c9bc9e277ca15d78be95e5afa7dab14c0`
`certificate_chordal/export_lean.py`	`f9f59cbeb0a0ac0c609ce7471b97ea3c69ea7f7330ac4c5975c152416b374ac4`
`certificate_chordal/cert_D80_NT1024_NK256_cap200_p128.json`	`aa5da0f05c1ee230782c30bd451981df2100ca75a9fc46ca73a6b7e54c765bb5`
`lean/KevYM/Certif/DobrushinCordale/Donnees.lean` (= `1b0f039`)	`72a0901472475b3a6d98033692b899e32c5f9b2a1daab7b9ed5c22416dc462fe`

B Proof of Proposition 7.7 (Dobrushin's comparison)

Reduction. Ordering $F = \{z_1, \dots, z_r\}$ and replacing ξ by ξ' one frozen site at a time, it suffices to treat ξ, ξ' differing at a single $z \in F$: the right side of (20) is additive over z, and C does not depend on ξ.

Step 1 (one-site kernels). For $x \in V$ let $(\gamma_x^\xi G)(q) = \int G(q_{V\setminus x}, u)\, \gamma_x(du \mid q_{V\setminus x}, \xi)$. Clearly $\delta_x(\gamma_x^\xi G) = 0$. If q, q' differ only at $x' \neq x$,

$$\gamma_x^\xi G(q) - \gamma_x^\xi G(q') = \int [G(q_{V\setminus x}, u) - G(q'_{V\setminus x}, u)]\gamma_x(du \mid q, \xi) + \int G(q'_{V\setminus x}, u)[\gamma_x(du \mid q, \xi) - \gamma_x(du \mid q', \xi)];$$

the first term is at most $\delta_{x'}(G)\, \rho_{\mathrm{ch}}(q_{x'}, q'_{x'})$ and, $u \mapsto G(q'_{V\setminus x}, u)$ being $\delta_x(G)$-Lipschitz, the second is at most $\delta_x(G)\, c_{xx'} \rho_{\mathrm{ch}}(q_{x'}, q'_{x'})$ by (18). So $\delta_{x'}(\gamma_x^\xi G) \leq \delta_{x'}(G) + \delta_x(G)\, c_{xx'}$. In the same way, for every q,

$$|\gamma_x^\xi G(q) - \gamma_x^{\xi'} G(q)| \leq \delta_x(G)\, c_{xz}\, \rho_{\mathrm{ch}}(\xi_z, \xi'_z). \tag{26}$$

Step 2 (random scan). Let $n = |V|$ and $P^\xi = n^{-1} \sum_{x\in V} \gamma_x^\xi$. Since $\gamma_x(\cdot \mid q_{V\setminus x}, \xi)$ is the conditional law of q_x under μ^ξ, $\mu^\xi \gamma_x^\xi = \mu^\xi$ and so $\mu^\xi P^\xi = \mu^\xi$. By Step 1, writing $\delta(G)$ for the row vector $(\delta_x(G))_{x\in V}$,

$$\delta(P^\xi G) \leq \delta(G)\, M \quad \text{entrywise}, \qquad M := (1 - \tfrac{1}{n})I + \tfrac{1}{n} C \;\geq 0,$$

and the row sums of M are $\leq \vartheta := 1 - (1-\alpha)/n < 1$. In particular $P^\xi G \in \mathcal{L}$.

Step 3 (convergence). Put $G_t = (P^\xi)^t G$. Then $\delta(G_t) \le \delta(G) M^t$, so

$$\sum_x \delta_x(G_t) \le \vartheta^t \sum_x \delta_x(G) \to 0.$$

Changing one coordinate at a time, $\sup G_t - \inf G_t \le 2 \sum_x \delta_x(G_t) \to 0$; as $\mu^\xi(G_t) = \mu^\xi(G)$, $G_t \to \mu^\xi(G)$ uniformly.

Step 4 (telescoping). Using $\mu^{\xi'} P^{\xi'} = \mu^{\xi'}$ and (26),

$$\begin{aligned} |\mu^{\xi'}(G_t) - \mu^{\xi'}(G_{t+1})| &= |\mu^{\xi'}((P^{\xi'} - P^\xi) G_t)| \\ &\le \frac{\rho_{\mathrm{ch}}(\xi_z, \xi'_z)}{n} \sum_x \delta_x(G_t)\, c_{xz} \le \frac{\rho_{\mathrm{ch}}(\xi_z, \xi'_z)}{n} \, (\delta(G) M^t) c_{\cdot z}. \end{aligned}$$

Summing over $t \ge 0$ and using Step 3, $|\mu^{\xi'}(G) - \mu^\xi(G)| \le n^{-1} \rho_{\mathrm{ch}}(\xi_z, \xi'_z)\, \delta(G) (\sum_t M^t) c_{\cdot z}$, and $\sum_t M^t = (I - M)^{-1} = n(I - C)^{-1} = nD$. □

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